

CMPT 732-G200 Practices for Visual Computing

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2D Animations

- For a long while, all cartoons were 2D



3D Objects

- Pixar made the first 3D animated movie: Toy Story 1



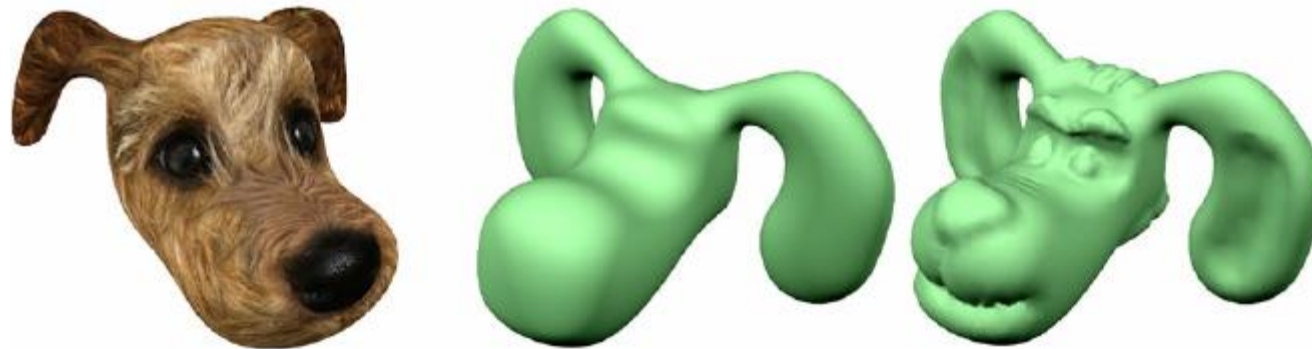
3D Objects

- A piece of Steve Jobs's key note speech at SIGGRAPH 1995.



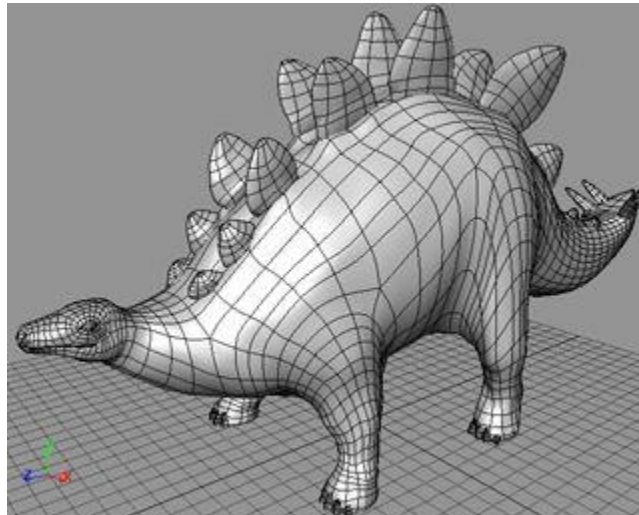
3D objects

- How can we represent 3D objects?
 - Implicit Modelling



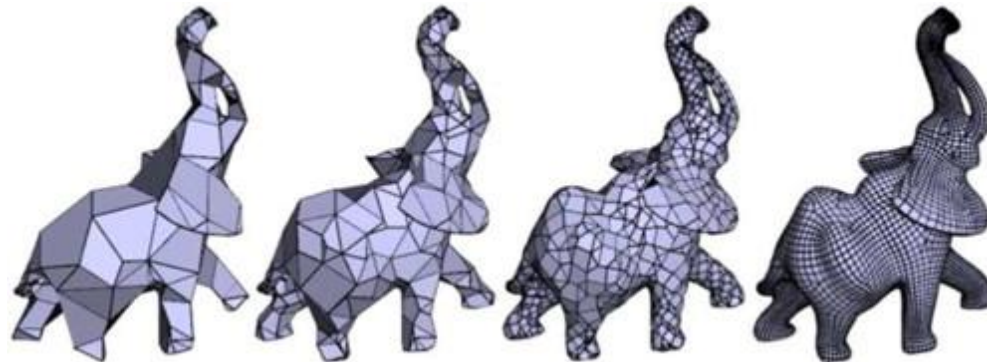
3D objects

- How can we represent 3D objects?
 - Implicit Modelling
 - Parametric Modelling



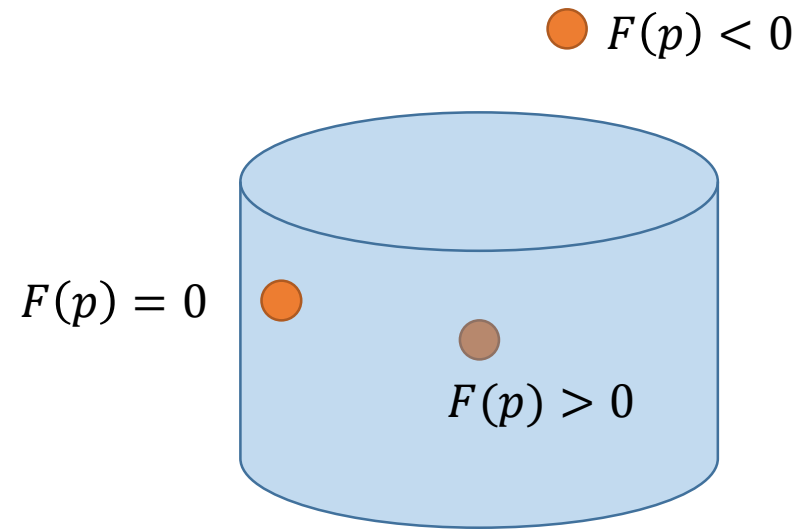
3D objects

- How can we represent 3D objects?
 - Implicit Modelling
 - Parametric Modelling
 - Polygonal Mesh Processing



Implicit Modelling

- To generate a shape, We define a function F , if point p is inside the object if
 - $F(p) < 0$ then p is outside the shape
 - $F(p) > 0$ then p is inside the shape
 - $F(p) = 0$ then p is on the shape



Implicit Modelling

- Example:

- $F(x, y, z) = 1 - x^2 - y^2 - z^2$ What is this shape?

Implicit Modelling

- Example:
 - $F(x, y, z) = 1 - x^2 - y^2 - z^2$
 - $F(0,0,0) = 1 > 0 \rightarrow (0,0,0)$ is inside



Implicit Modelling

- Example:
 - $F(x, y, z) = 1 - x^2 - y^2 - z^2$
 - $F(0,0,0) = 1 > 0 \rightarrow (0,0,0)$ is inside
 - $F(1,0,0) = 0 \rightarrow (1,0,0)$ is on the surface

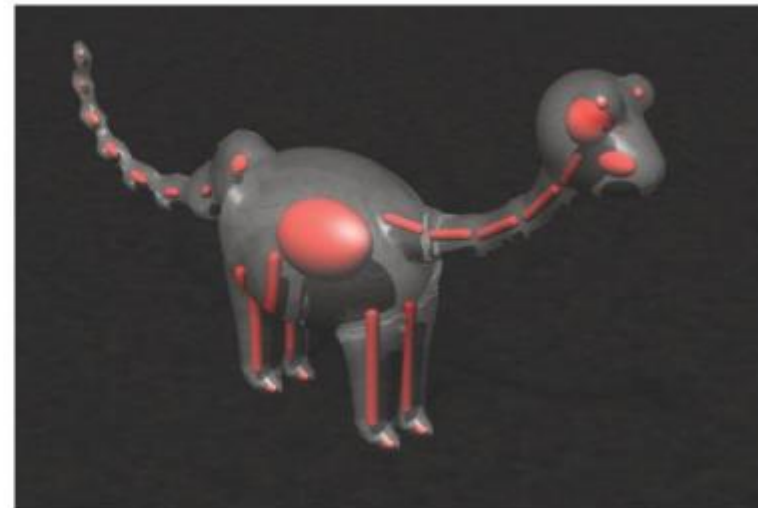
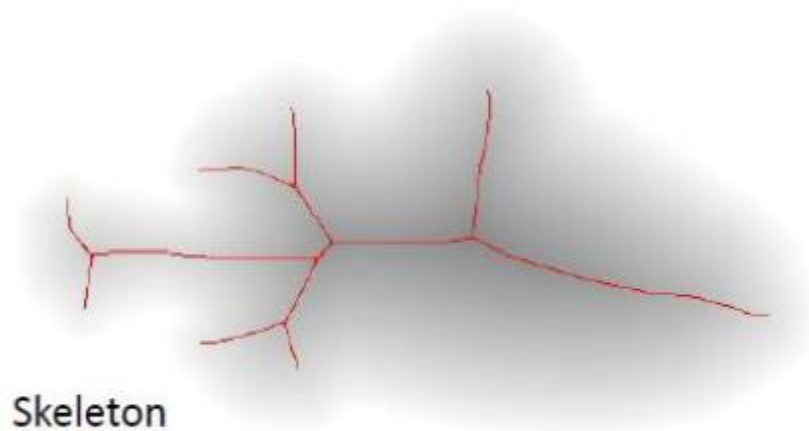


Implicit Modelling

- Sphere is too easy. How to make general surfaces?

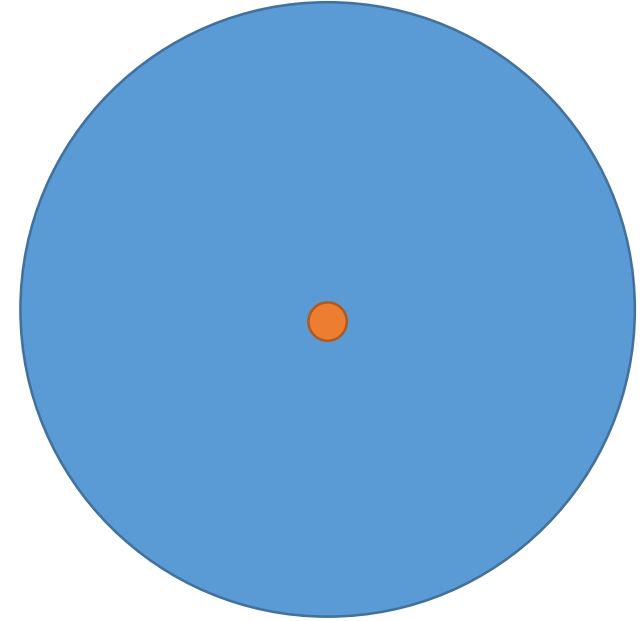
Implicit Modelling

- Sphere is too easy. How to make general surfaces?
- Basic idea is to define a **function based on distance**, make a skeletal structure of primitives and **combine primitives**.



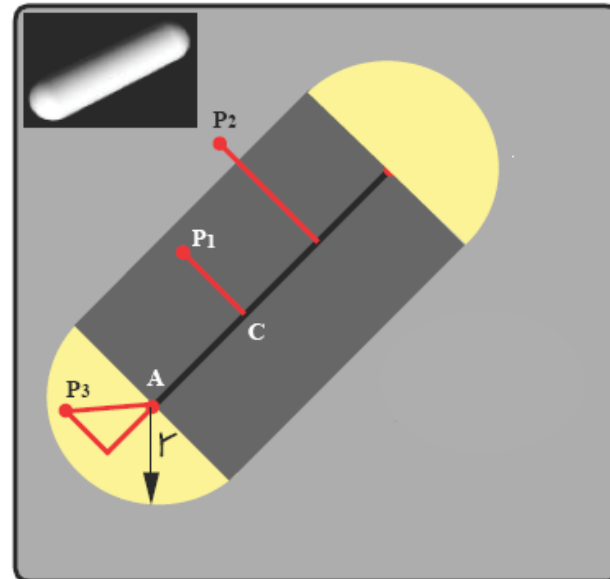
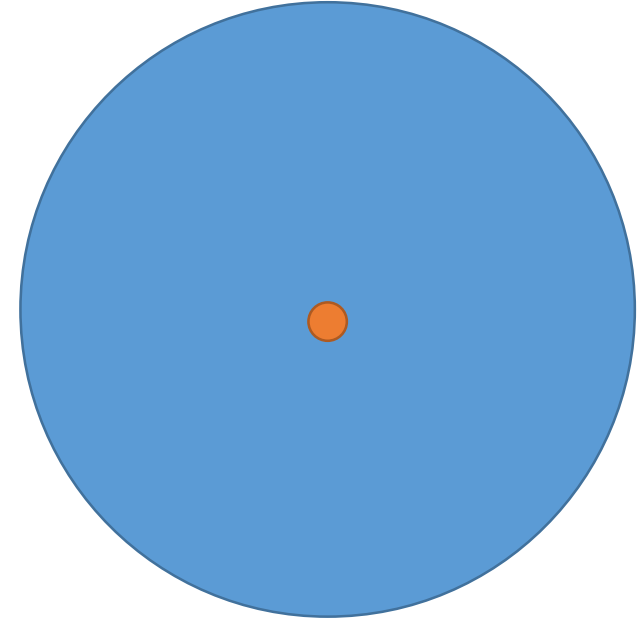
Primitives

- Sphere (dot)
 - All the points within a certain distance (radius) are inside.



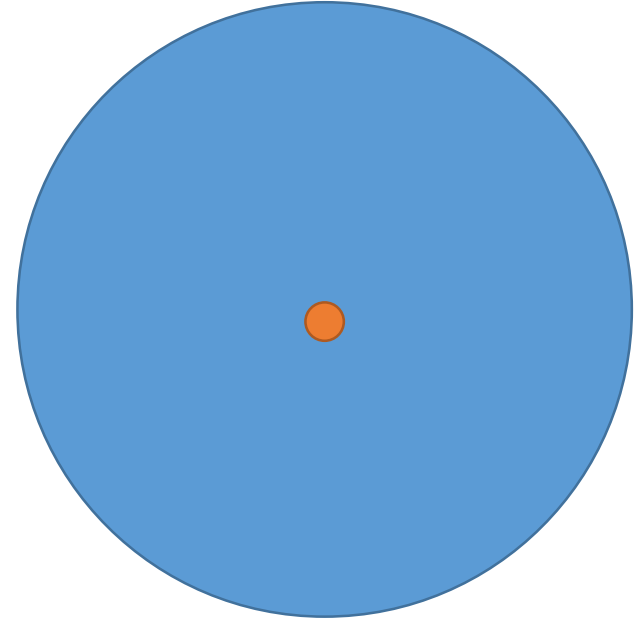
Primitives

- Sphere (dot)
 - All the points within a certain distance (radius) are inside.
- Cylinders with spherical caps (line)



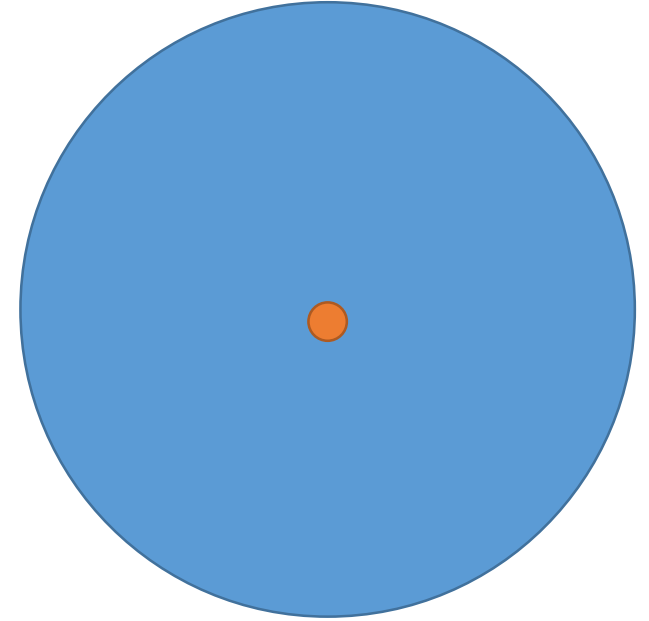
Primitives

- Function $F(p) = \frac{1}{r^2}$



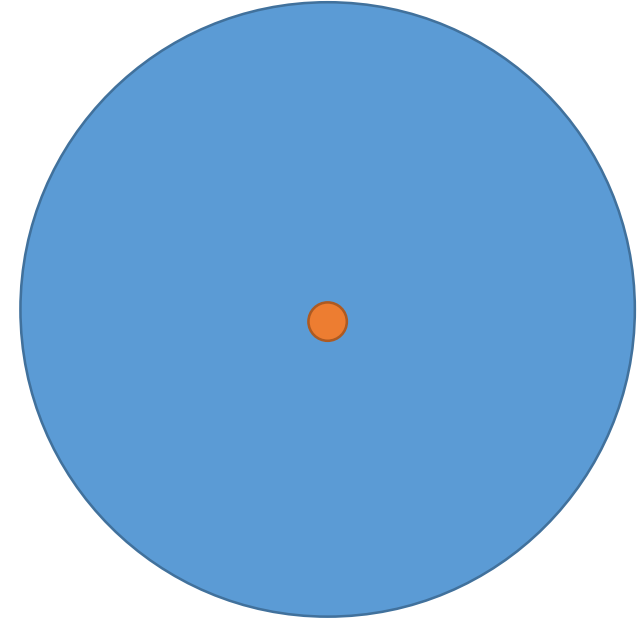
Primitives

- Function $F(p) = \frac{1}{r^2}$
- F is always positive



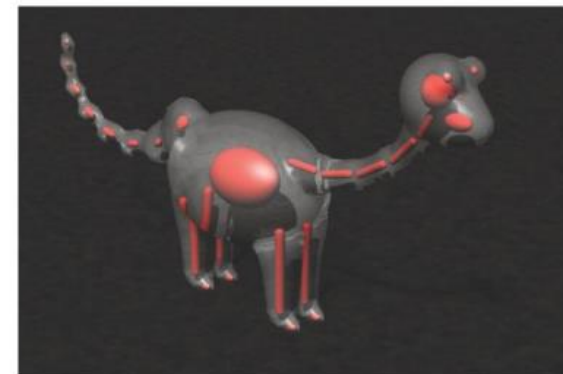
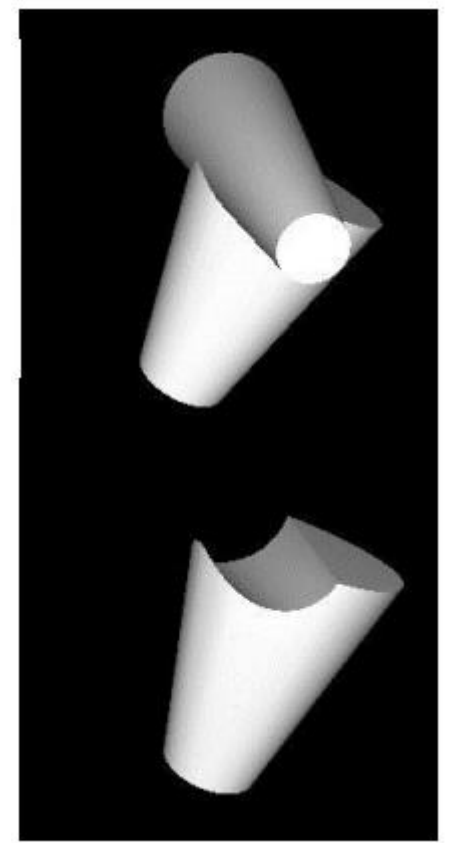
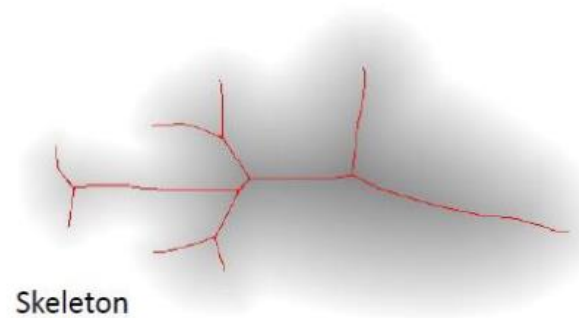
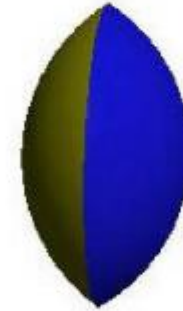
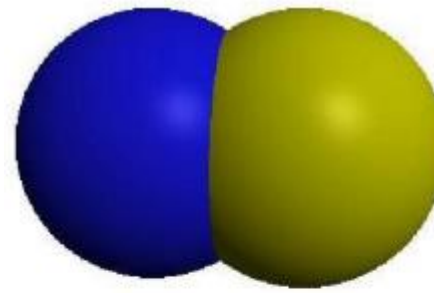
Primitives

- Function $F(p) = \frac{1}{r^2}$
- F is always positive
- We consider a threshold T
 - $F(p) = \frac{1}{r^2} - T$



Primitives

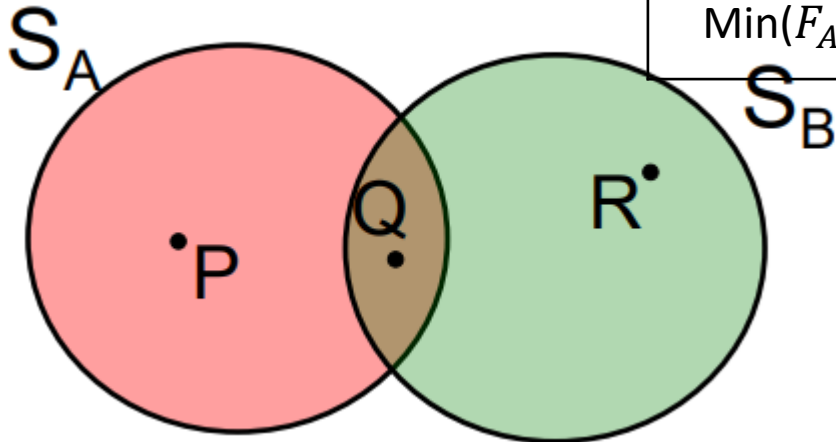
- Boolean Operations
- Union
- Intersection
- Difference



Primitives

- Functions for Boolean operations

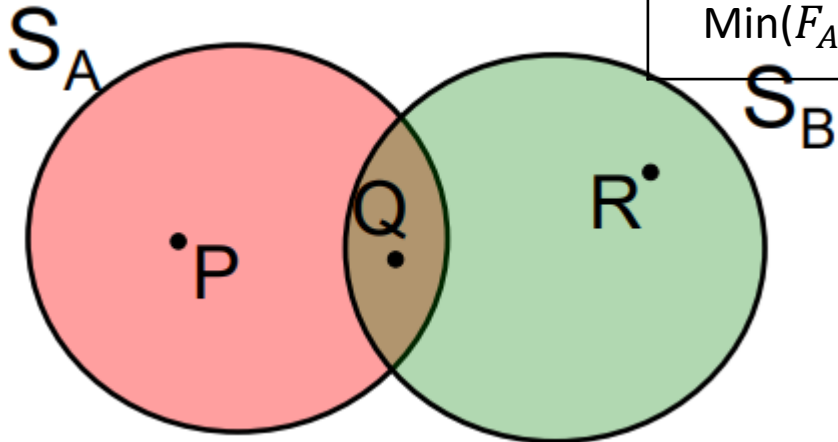
Function	P	Q	R	Solid
F_A				S_A
F_B				S_B
$\text{Max}(F_A, F_B)$				$S_{A \cup B}$
$\text{Min}(F_A, F_B)$				$S_{A \cap B}$
$\text{Min}(F_A, -F_B)$				S_{A-B}



Primitives

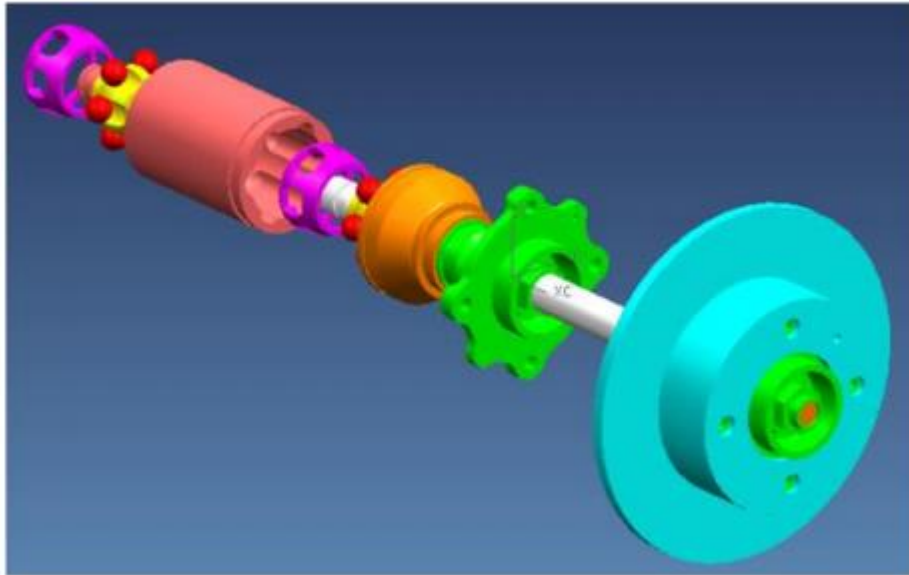
- Functions for Boolean operations

Function	P	Q	R	Solid
F_A	+	+	-	S_A
F_B	-	+	+	S_B
$\text{Max}(F_A, F_B)$	+	+	+	$S_{A \cup B}$
$\text{Min}(F_A, F_B)$	-	+	-	$S_{A \cap B}$
$\text{Min}(F_A, -F_B)$	+	-	-	S_{A-B}



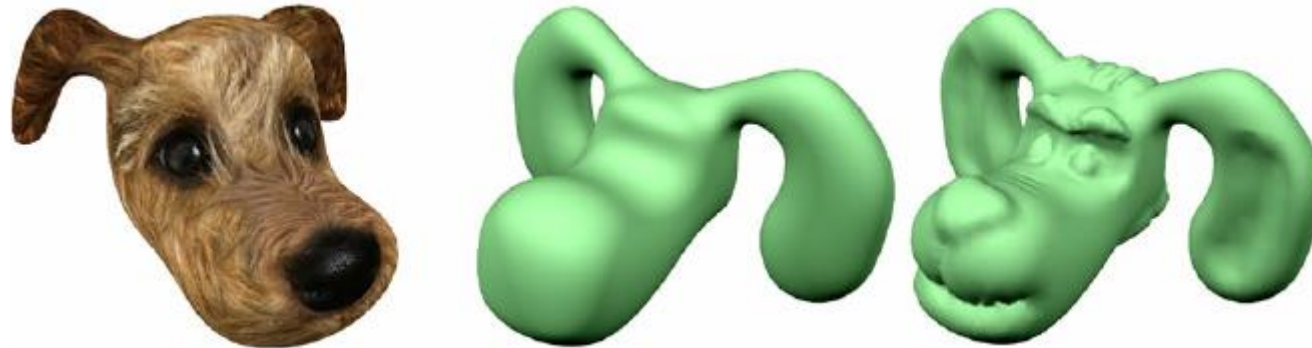
Implicit Modelling

- It is very common in CAD but it is not as common in Computer graphics as objects are too primitive.



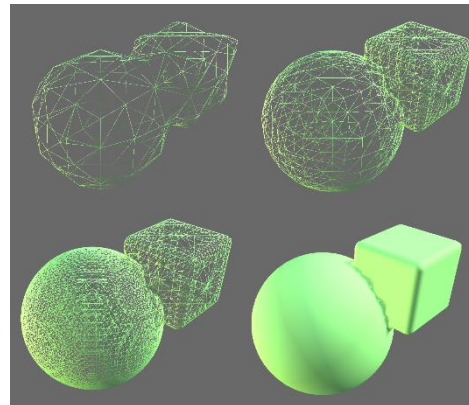
Implicit Modelling

- It is very common in CAD but it is not as common in Computer graphics as objects are too primitive.
- It is used for overall shapes of a model. (sketch based systems)



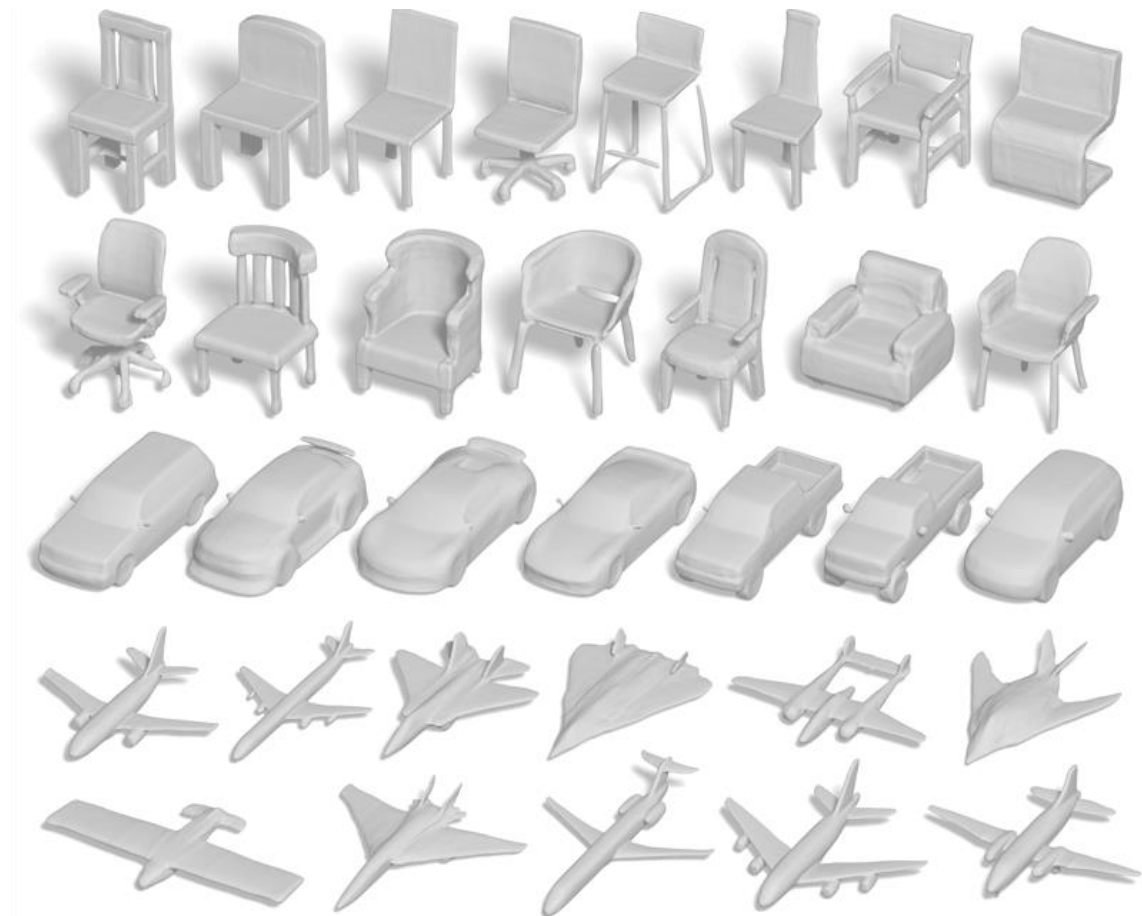
Implicit Modelling

- It is very common in CAD but it is as common in Computer graphics as objects are too primitive.
- It is used for overall shapes of a model. (sketch based systems)
- Needs to be triangulated as standard approaches work with polygonal shapes. (Marching Cube)



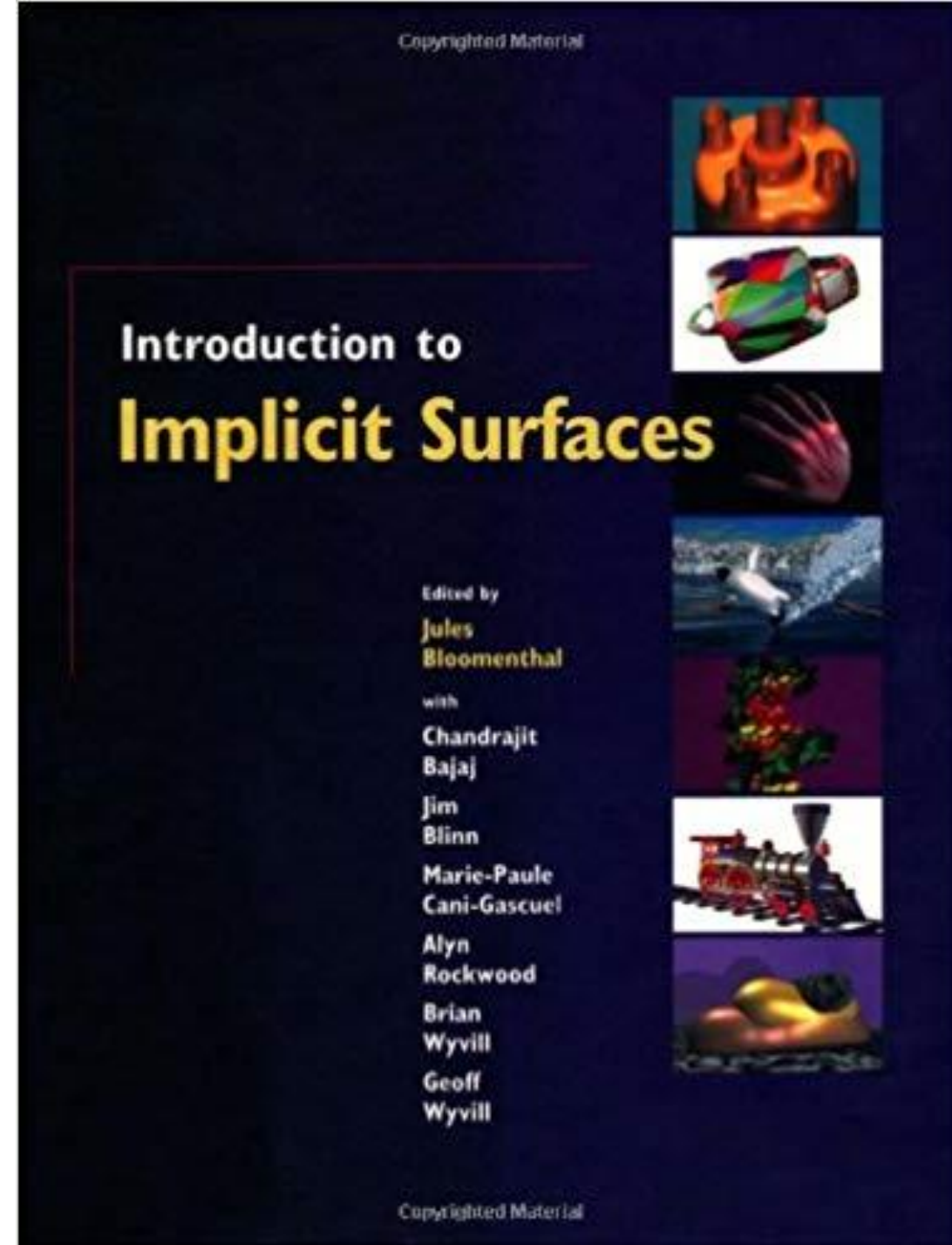
Implicit Modelling

- Currently, implicit modeling is one of the most popular representation for deep geometric processing.



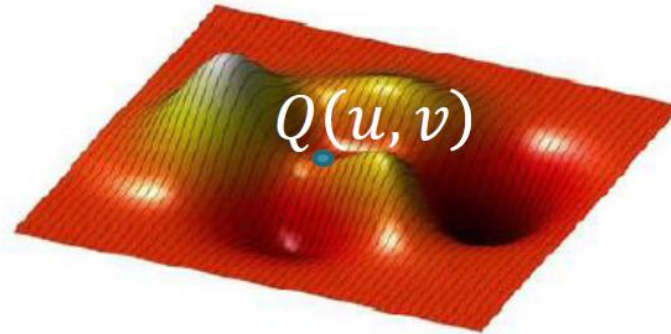
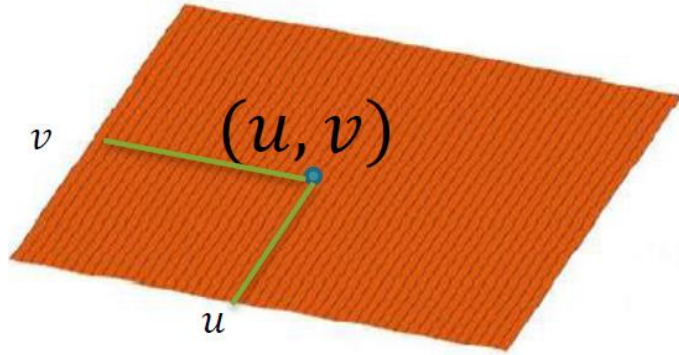
Implicit Modelling

- Reference
 - Bajaj
 - Blinn
 - Cani
 - Rockwood
 - B. Wyvill
 - G. Wyvill



Parametric Surfaces

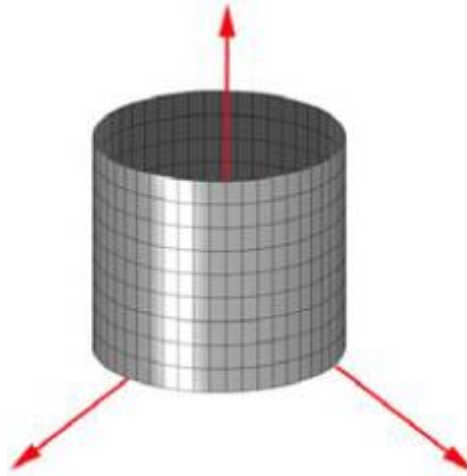
- $U - V$ parameter domain, we need a function $Q(u, v)$ to map the parameter space to the surface



$$Q(u, v) = \begin{bmatrix} X(u, v) \\ Y(u, v) \\ Z(u, v) \end{bmatrix} \quad \begin{array}{l} Q(u, v): A \rightarrow \mathbb{R}^3 \\ (u, v) \in A \end{array}$$

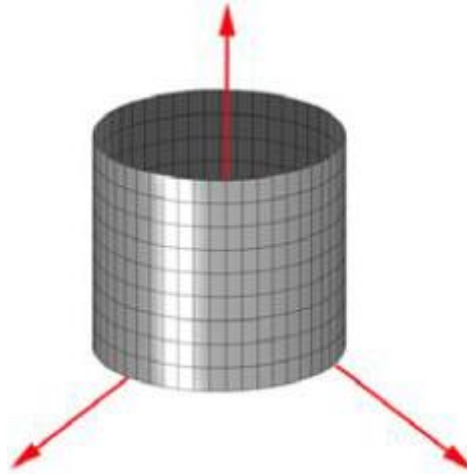
Parametric Surfaces

- What is a cylinder?



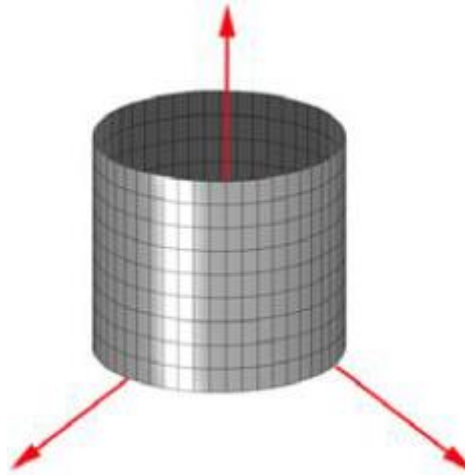
Parametric Surfaces

- What is a cylinder?
 - A set of circles stacked on top of each other.



Parametric Surfaces

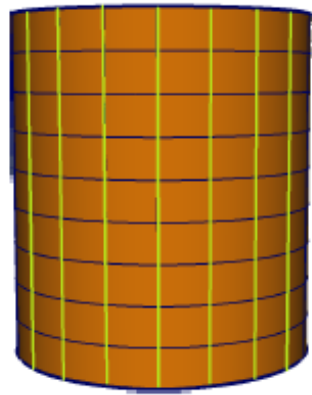
- What is a cylinder?
 - A set of circles stacked on top of each other.



$$Q(u, v) = \begin{bmatrix} \cos u \\ \sin u \\ v \end{bmatrix}, \quad \begin{matrix} 0 \leq u < 2\pi \\ 0 \leq v < 1 \end{matrix}$$

Parametric Surfaces

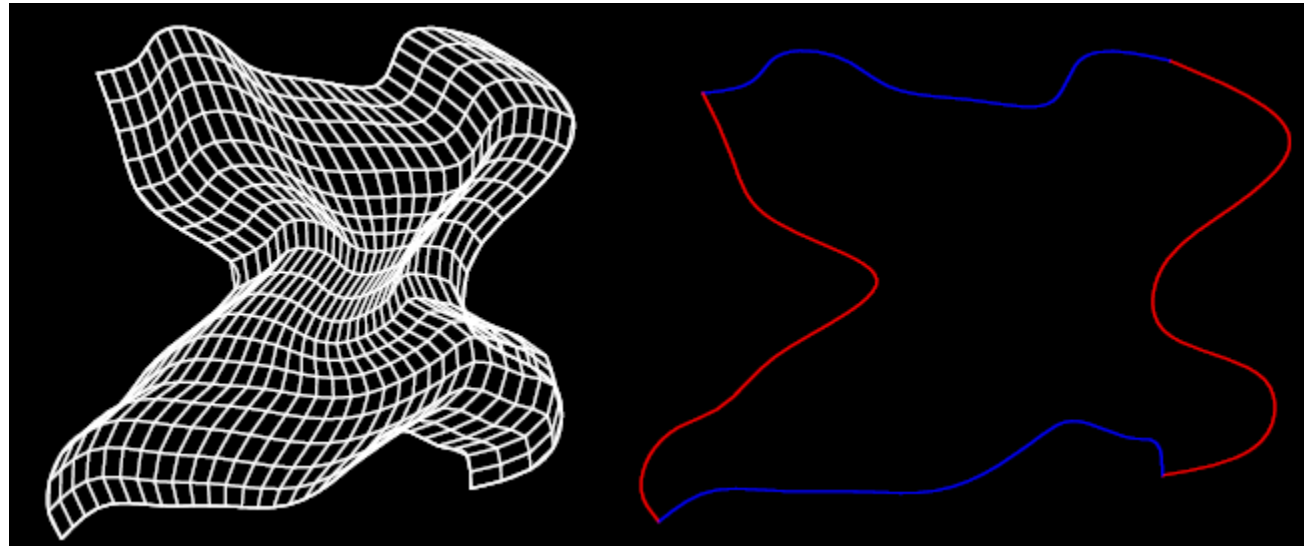
- What is a cylinder?
 - A set of circles stacked on top of each other.
- What are U-Curves and V-Curves



$$Q(u, v) = \begin{bmatrix} \cos u \\ \sin u \\ v \end{bmatrix}, \quad \begin{matrix} 0 \leq u < 2\pi \\ 0 \leq v < 1 \end{matrix}$$

Parametric Surfaces

- One common way to find parametric surfaces is to combine curves along U and V .



Ruled Surfaces

- One possibility is to sweep a line between two parametric curves.



Ruled Surfaces

- One possibility is to sweep a line between two parametric curves.

$$Q(u, v) = (1 - v)P_0(u) + vP_1(u)$$

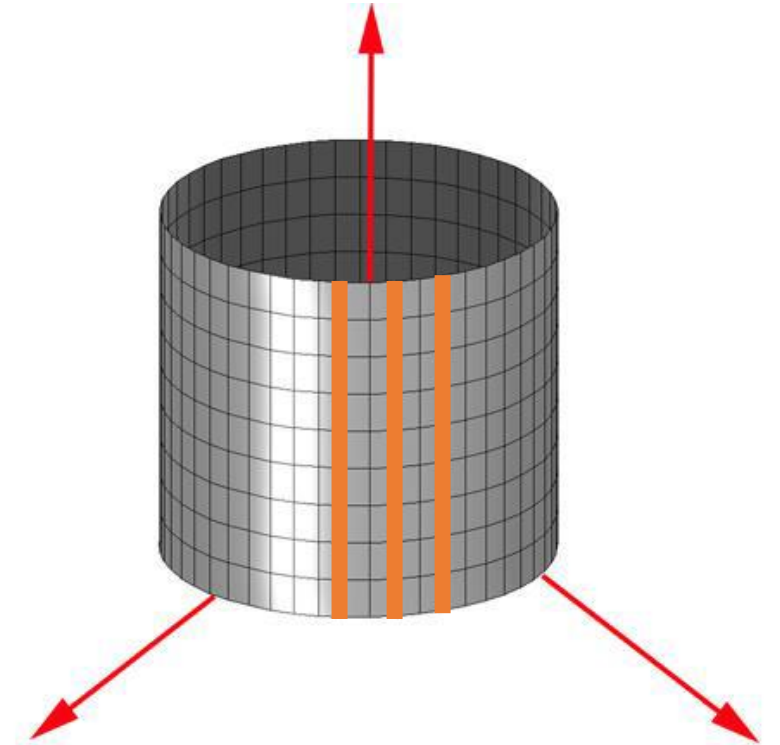


Ruled Surfaces

- One possibility is to sweep a line between two parametric curves.
 - Cylinder

$$Q(u, v) = (1 - v)P_0(u) + vP_1(u)$$

Circles



Ruled Surfaces

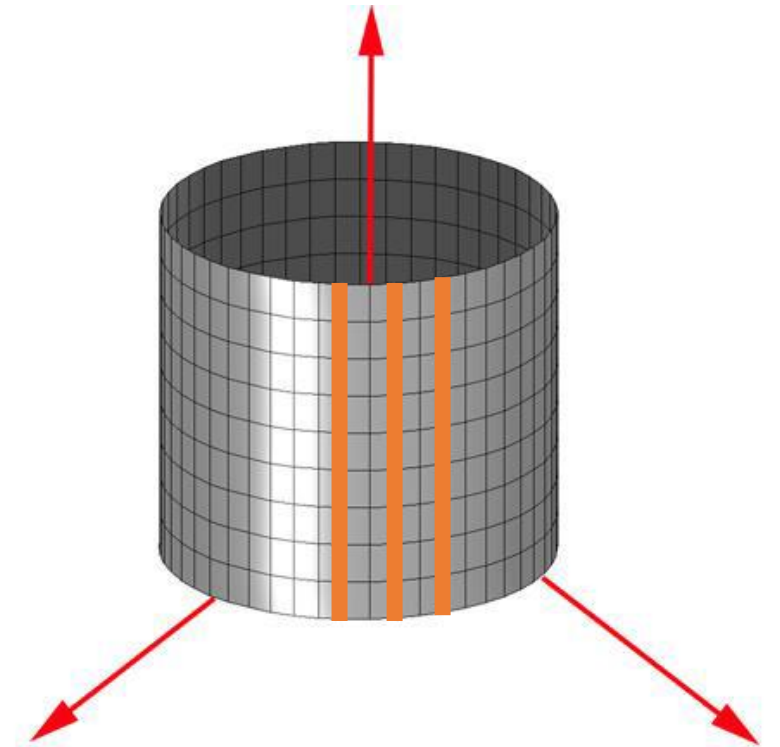
- One possibility is to sweep a line between two parametric curves.
 - Cylinder

$$Q(u, v) = (1 - v)P_0(u) + vP_1(u)$$

$$P_0(u) = (\cos u, \sin u, 1)$$

$$P_1(u) = (\cos u, \sin u, 0)$$

Circles

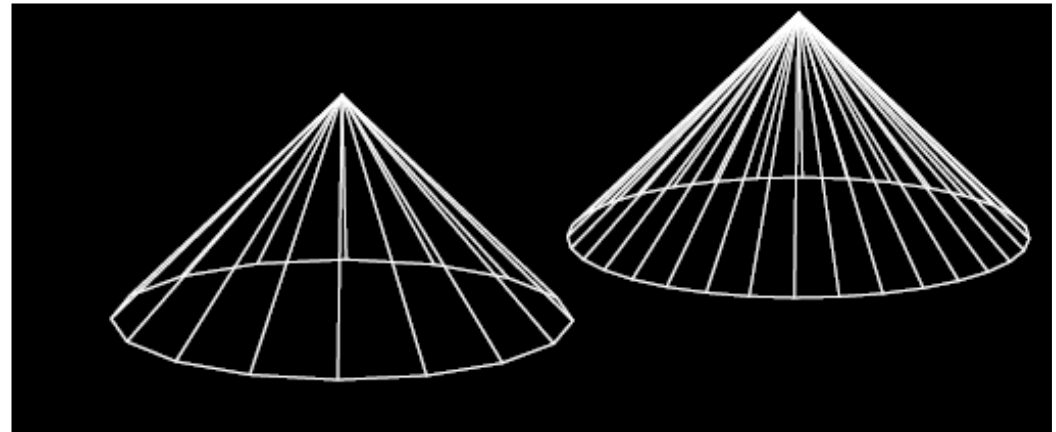
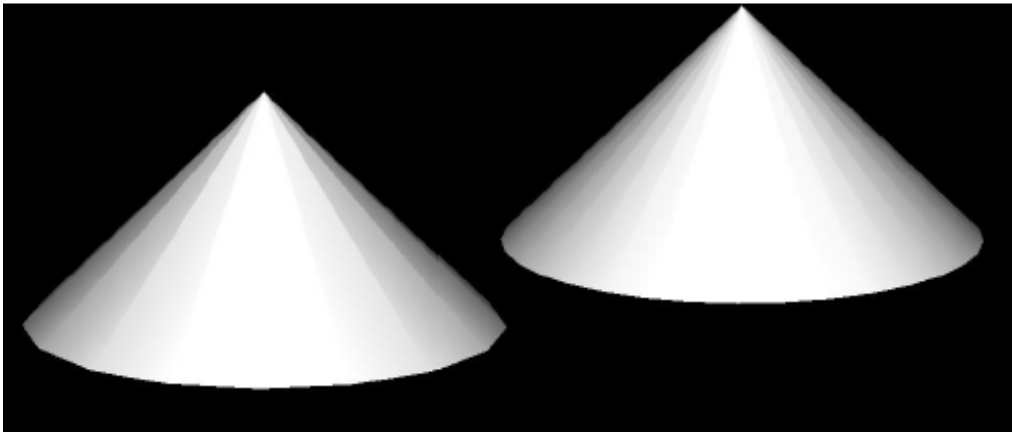


Ruled Surfaces

- One possibility is to sweep a line between two parametric curves.
 - Cylinder
 - Cone

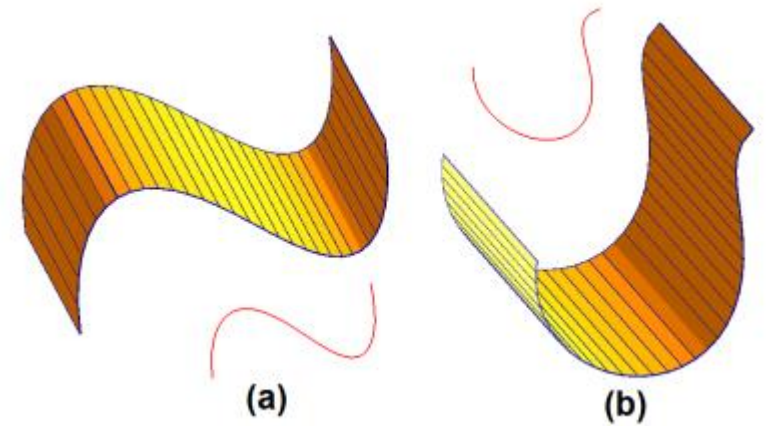
$$P_0(u) = (\cos u, \sin u, 1)$$

$$Q(u, v) = (1 - v)P_0(u) + v \boxed{P_1} \text{ Single point}$$



Ruled Surfaces

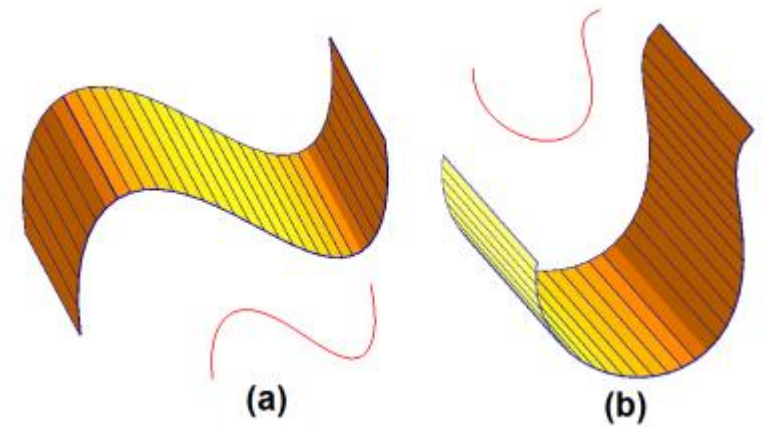
- One possibility is to sweep a line between two parametric curves.
 - Cylinder
 - Cone
 - General Cylinder
 - Two translated same curves



Ruled Surfaces

- One possibility is to sweep a line between two parametric curves.
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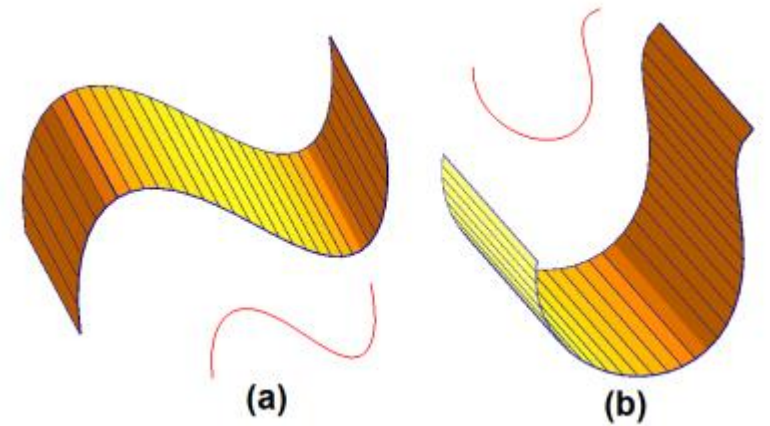
$$Q(u, v) = (1 - v)P_0(u) + v[P_0(u) + d]$$



Ruled Surfaces

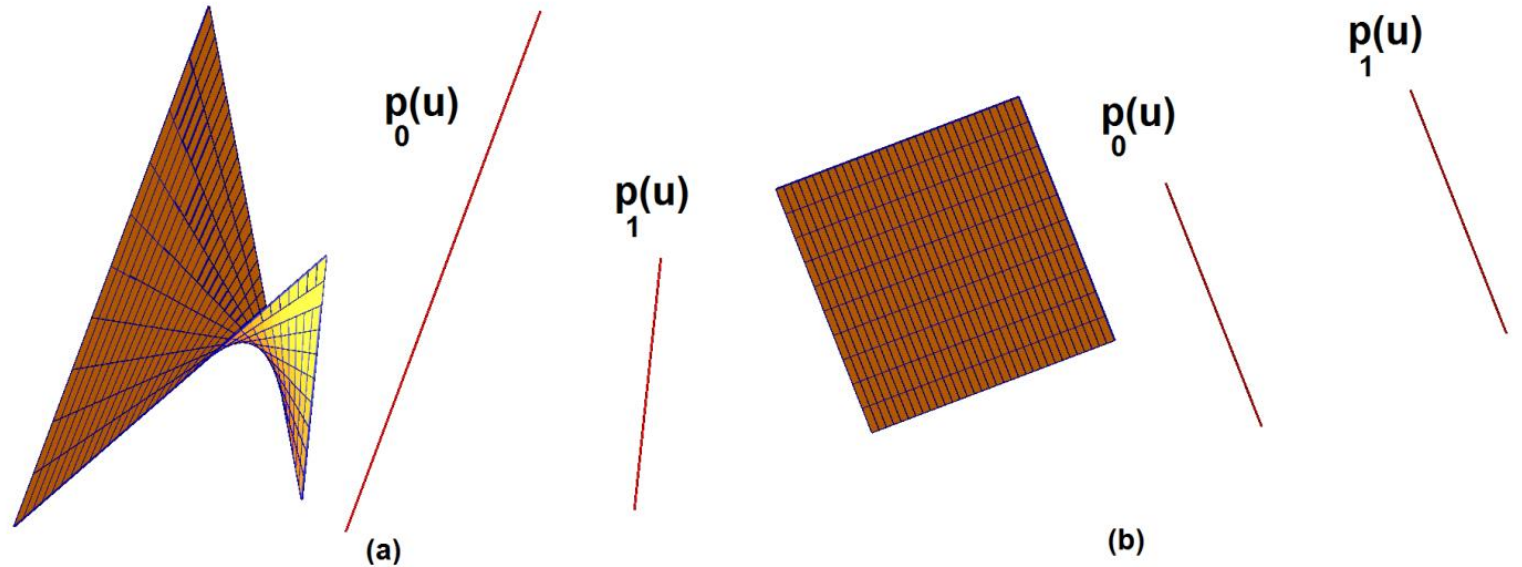
- One possibility is to sweep a line between two parametric curves.
 - Cylinder
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 - General Cylinder
 - Two translated same curves

$$\begin{aligned}Q(u, v) &= (1 - v)P_0(u) + v[P_0(u) + d] \\ &= P_0(u) + vd\end{aligned}$$



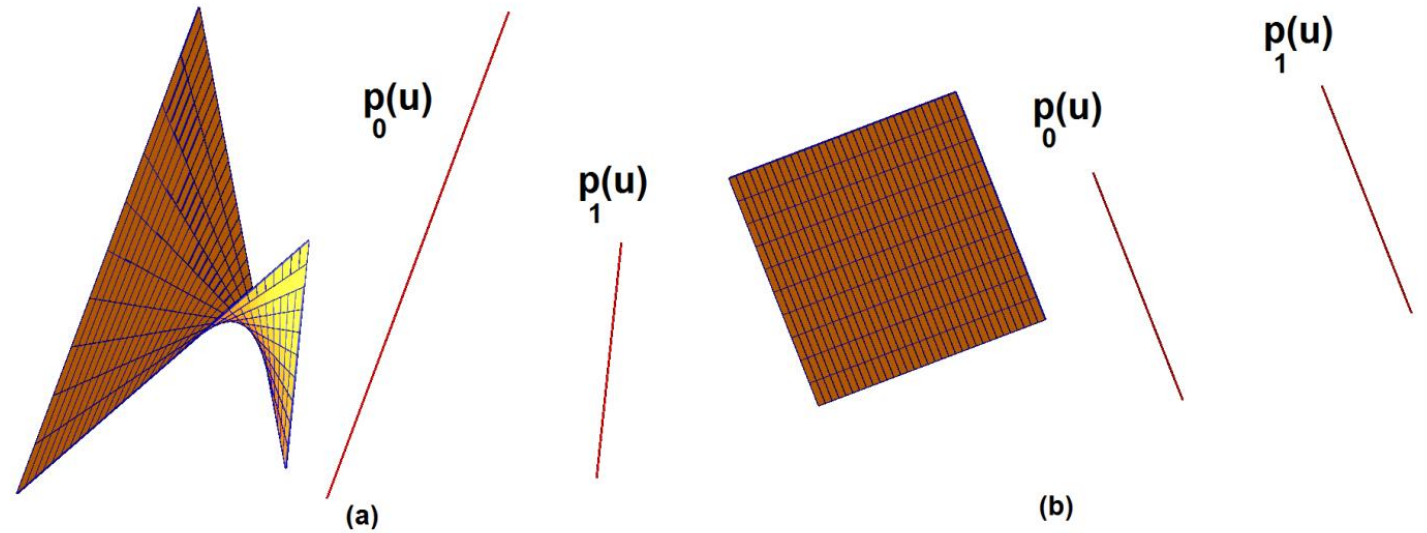
Ruled Surfaces

- One common way to find parametric surfaces is to combine curves along U and V.
 - Cylinder
 - Cone
 - General Cylinder
 - Bilinear Patch



Ruled Surfaces

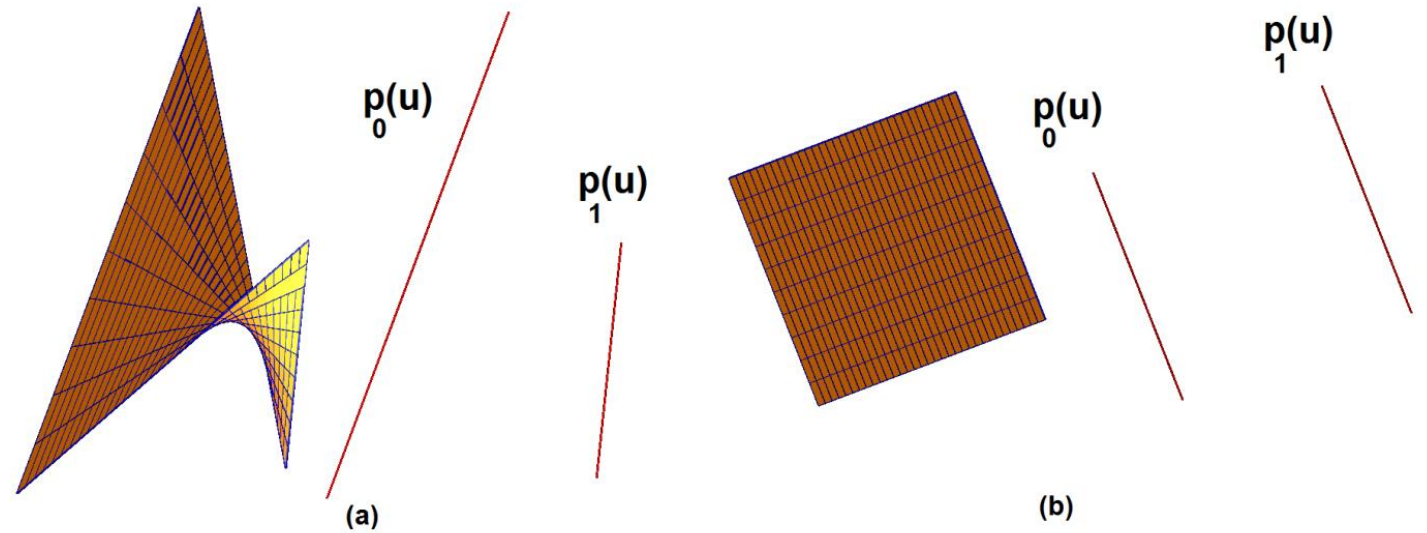
- Bilinear Patches



$$Q(u, v) = (1 - v)P_0(u) + vP_1(u)$$

Ruled Surfaces

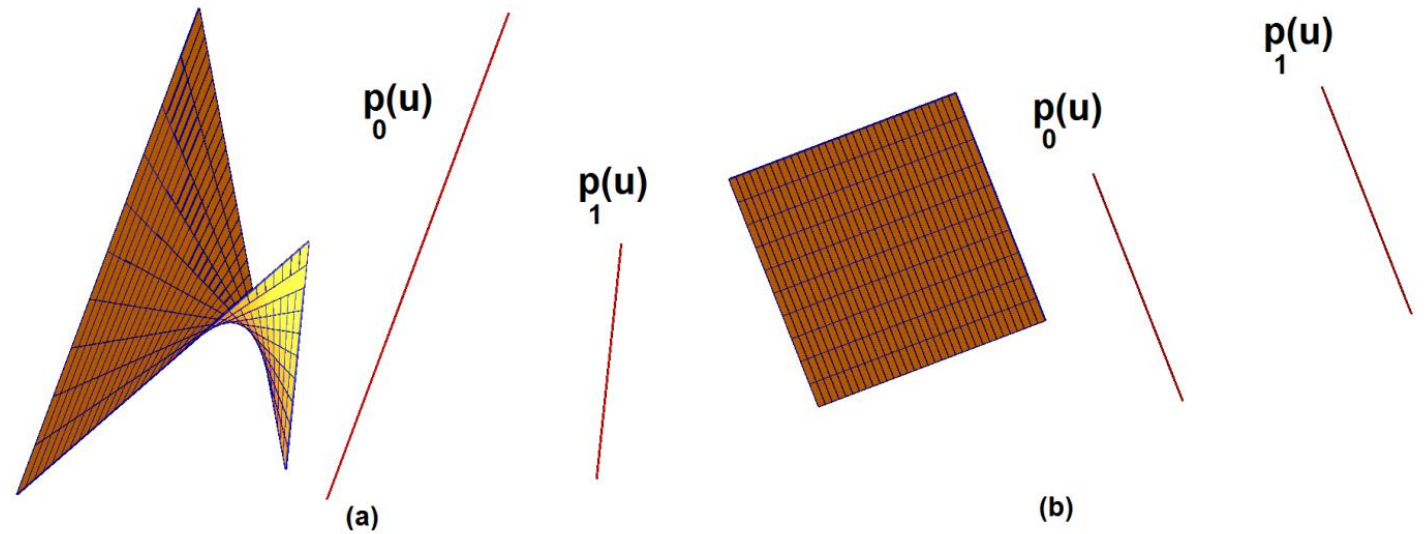
- Bilinear Patches



$$\begin{aligned} Q(u, v) &= (1 - v)P_0(u) + vP_1(u) \\ &= (1 - v)(1 - u)P_{00} + (1 - v)uP_{01} + vP_1(u) \end{aligned}$$

Ruled Surfaces

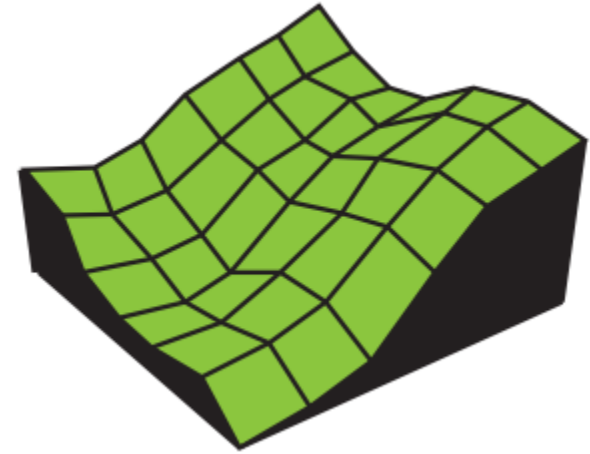
- Bilinear Patches



$$\begin{aligned} Q(u, v) &= (1 - v)P_0(u) + vP_1(u) \\ &= (1 - v)(1 - u)P_{00} + (1 - v)uP_{01} + vP_1(u) \\ &= (1 - v)(1 - u)P_{00} + (1 - v)uP_{01} + v(1 - u)P_{10} + v(u)P_{11} \\ 0 &\leq u, v \leq 1 \end{aligned}$$

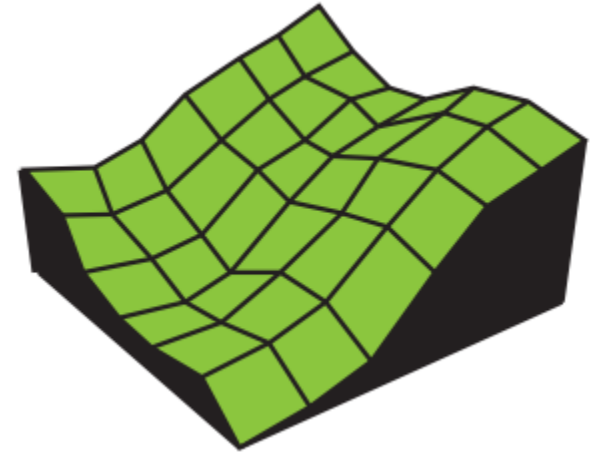
Coons Patches

- A useful generalization of ruled surfaces.



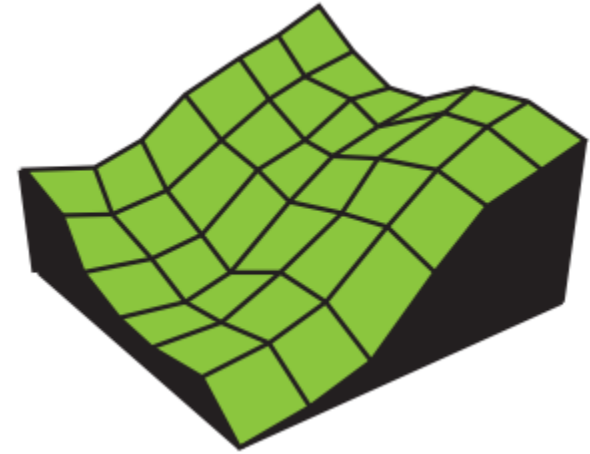
Coons Patches

- A useful generalization of ruled surfaces.
- This time, the input is four curves.



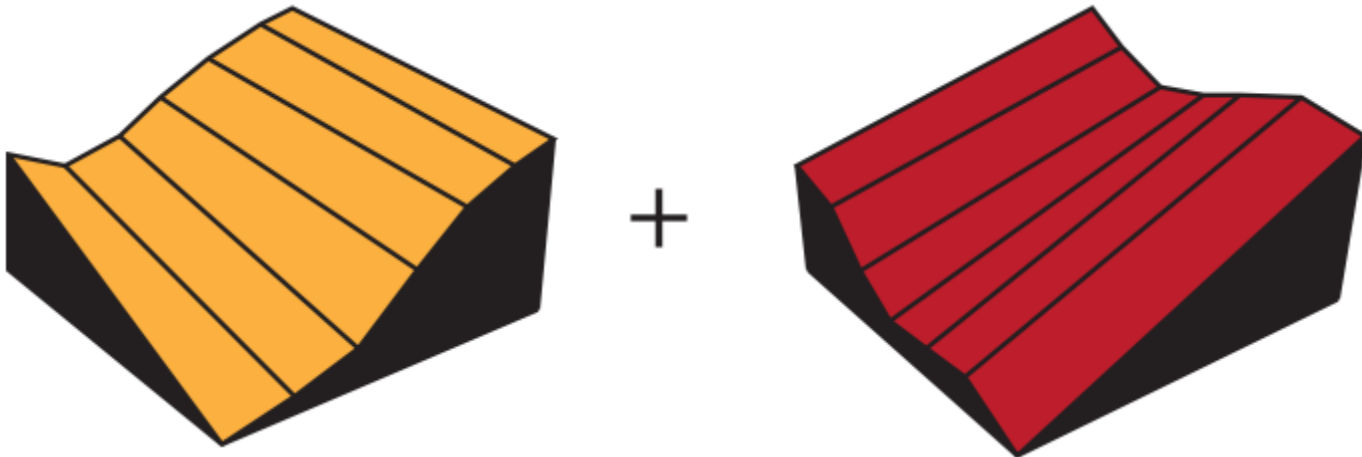
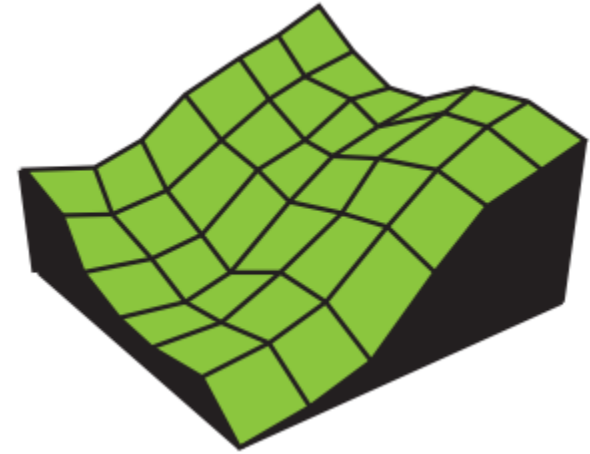
Coons Patches

- A useful generalization of ruled surfaces.
- This time, the input is four curves.
- How should we combine them?



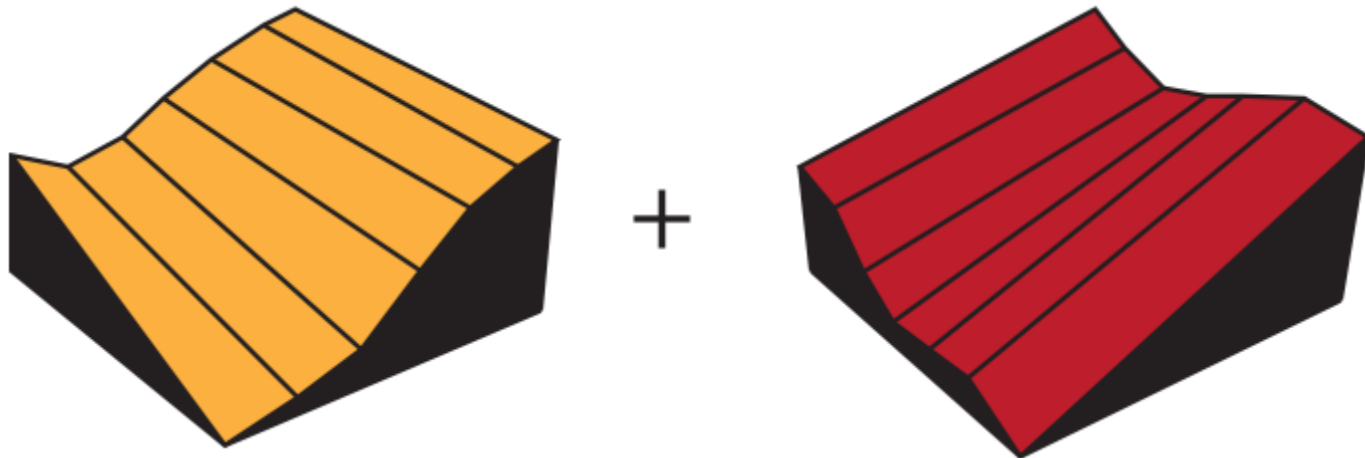
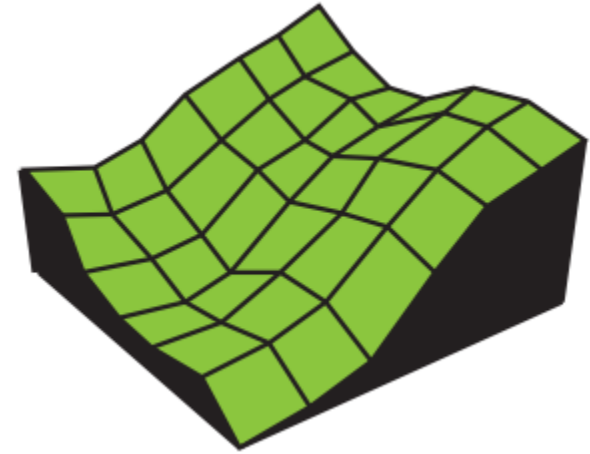
Coons Patches

- A useful generalization of ruled surfaces.
- This time, the input is four curves.
- How should we combine them? Add two ruled surface



Coons Patches

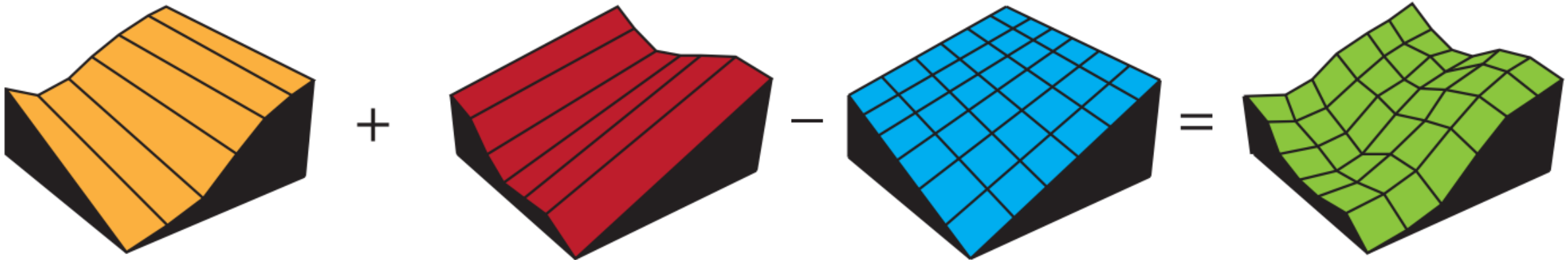
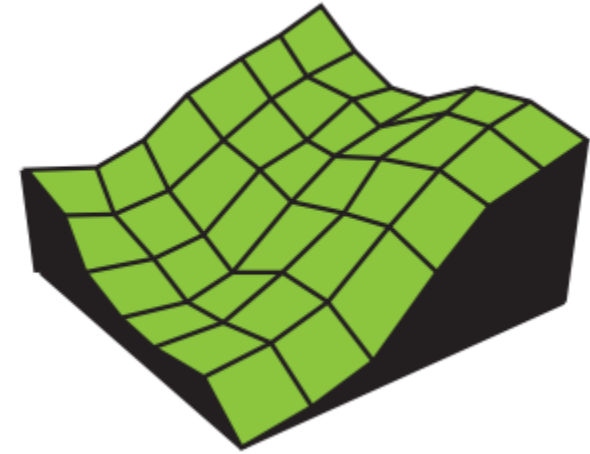
- A useful generalization of ruled surfaces.
- This time, the input is four curves.
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Problem?

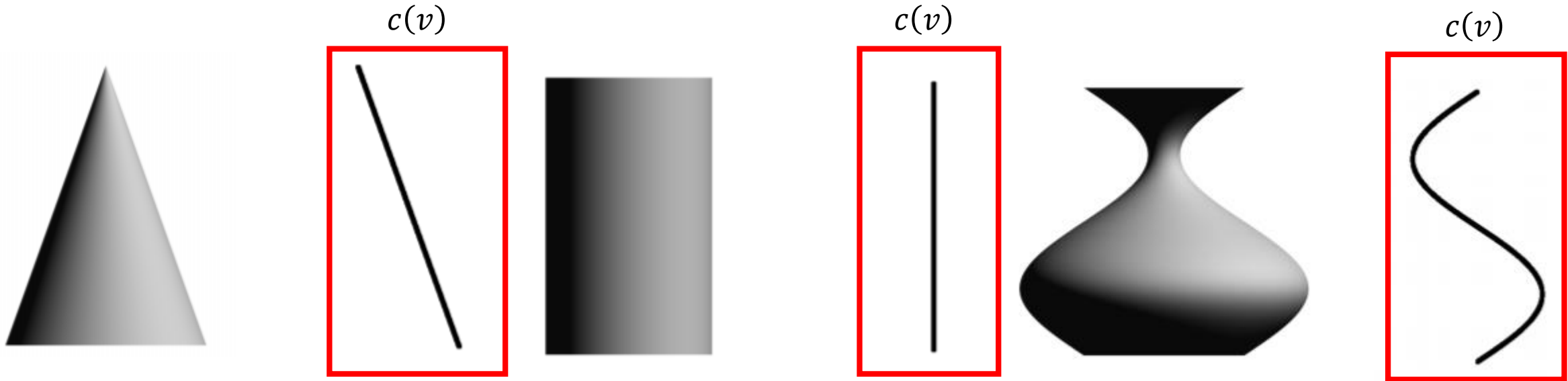
Coons Patches

- A useful generalization of ruled surfaces.
- This time, the input is four curves.
- How should we combine them? Add two ruled surface



Surface of Revolution

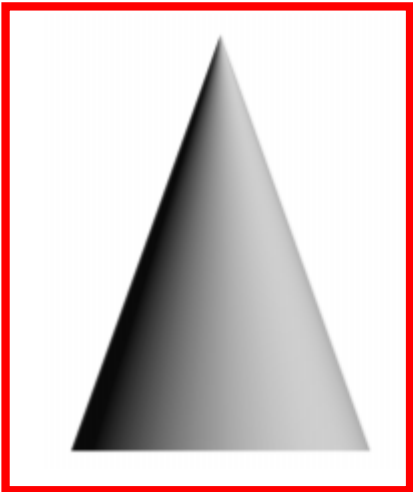
- A profile curve is given $c(v) = (X(v), Z(v))$



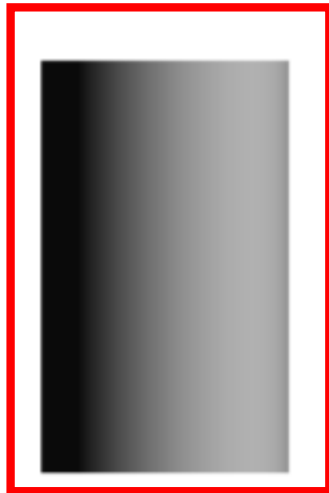
Surface of Revolution

- A profile curve is given $c(v) = (X(v), Z(v))$
- To obtain $Q(u, v)$ we sweep $c(v)$ along an axis.

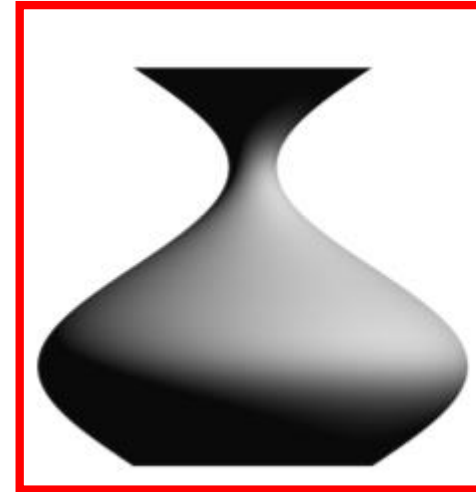
$Q(u, v)$



$Q(u, v)$



$Q(u, v)$



Surface of Revolution

- A profile curve is given $c(v) = (X(v), Z(v))$
- To obtain $Q(u, v)$ we sweep $c(v)$ along an axis.

$$Q(u, v) = (X(v) \cdot \cos u, X(v) \cdot \sin u, Z(v))$$

$$0 \leq u \leq 2\pi$$

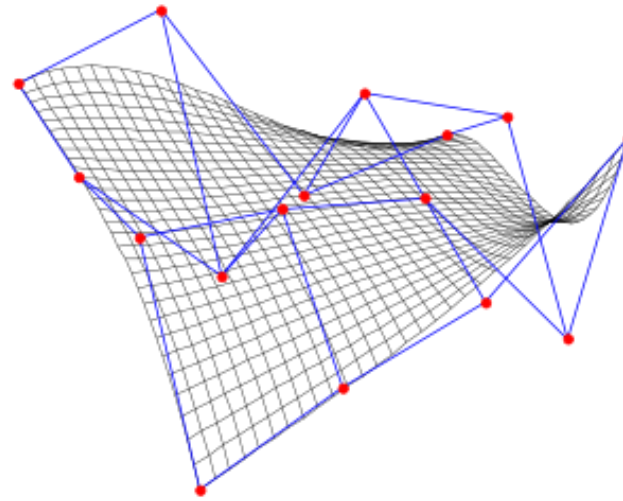
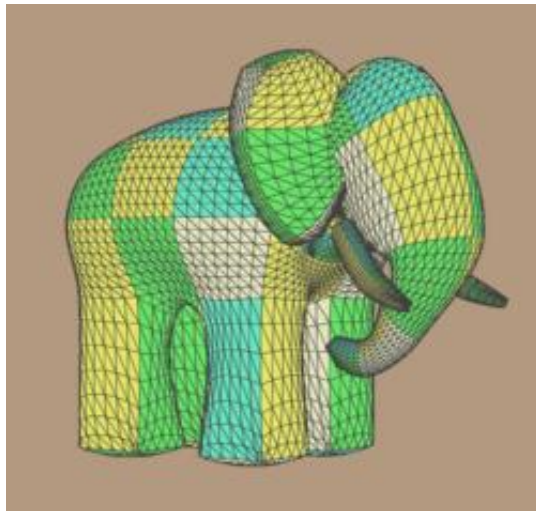
$$v \in I$$



Bezier Patches

- We have a grid of control points with two Bernstein basis functions along u and v .

$$Q(u, v) = \sum_{i=0}^d \sum_{j=0}^{\hat{d}} P_{i,j} B_{i,d}(u) B_{j,\hat{d}}(v), \quad \begin{array}{l} 0 \leq u \leq 1 \\ 0 \leq v \leq 1 \end{array}$$

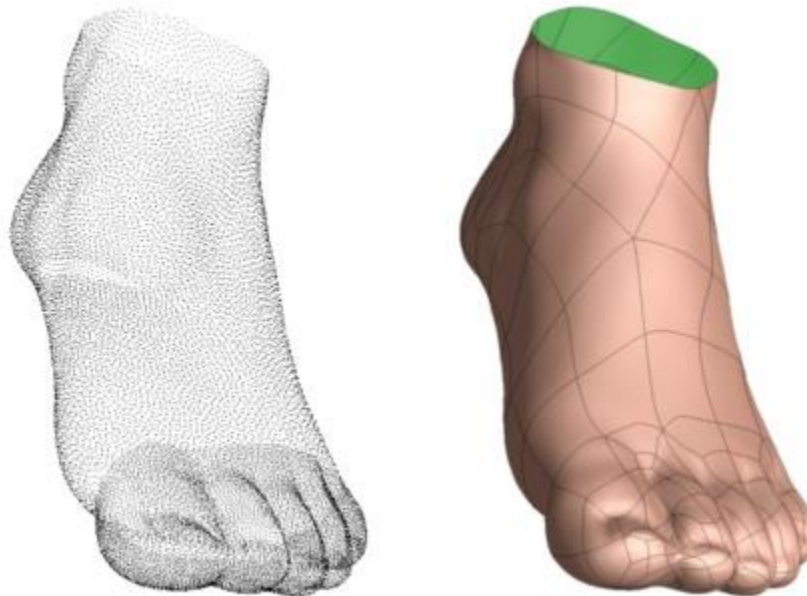


Bezier Patches

- You can replace Bernstein basis functions with BSpline or NURBS basis functions

$$Q(u, v) = \sum_{i=0}^d \sum_{j=0}^{\acute{d}} P_{i,j} B_{i,d}(u) \boxed{B_{j,\acute{d}}(v)}, \quad \begin{matrix} 0 \leq u \leq 1 \\ 0 \leq v \leq 1 \end{matrix}$$

$N_{i,k}(v)$

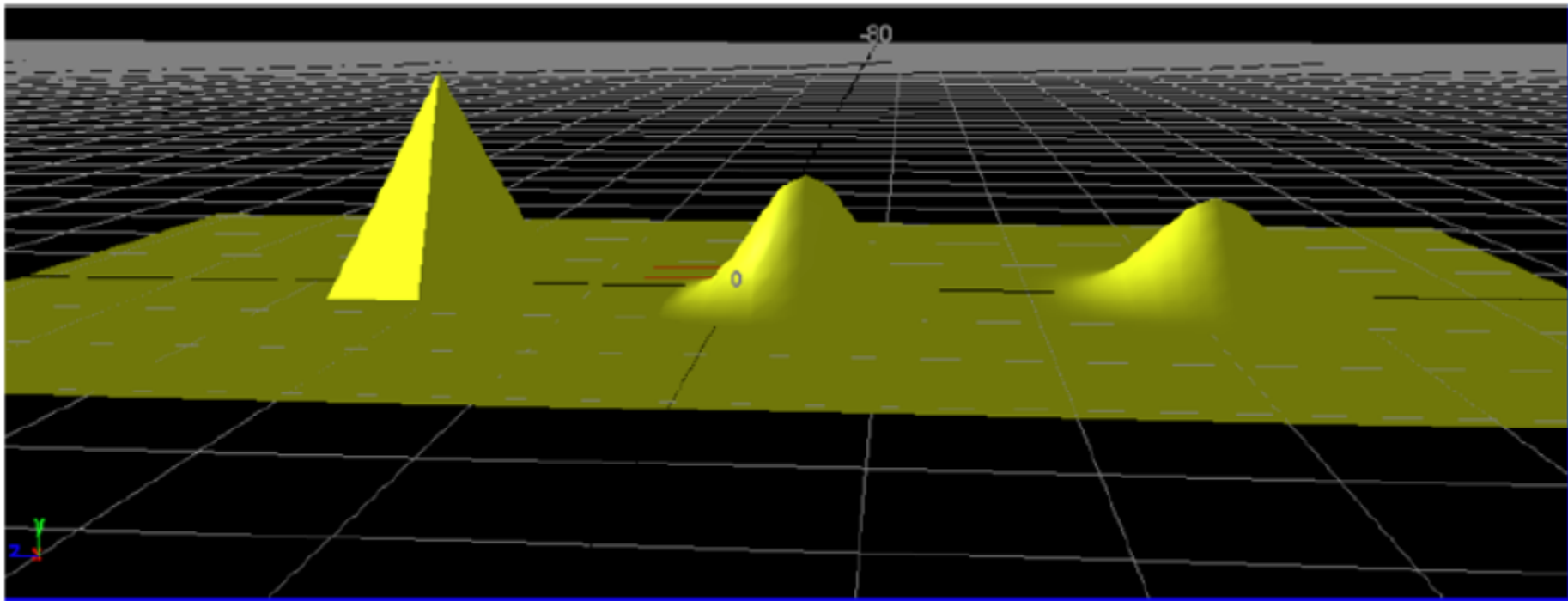


BSpline Patches

$$Q(u, v) = \sum_{i=0}^m \sum_{j=0}^{m'} P_{i,j} N_{i,k}(u) N_{j,k'}(v) \quad u_{k-1} \leq u \leq u_{m+1} \quad v_{k'-1} \leq v \leq v_{m'+1}$$

BSpline Patches

$$Q(u, v) = \sum_{i=0}^m \sum_{j=0}^{m'} P_{i,j} N_{i,k}(u) N_{j,k'}(v) \quad u_{k-1} \leq u \leq u_{m+1} \quad v_{k'-1} \leq v \leq v_{m'+1}$$



Applications

- Surface Reconstruction

Automatic Reconstruction of B-Spline Surfaces of Arbitrary Topological Type

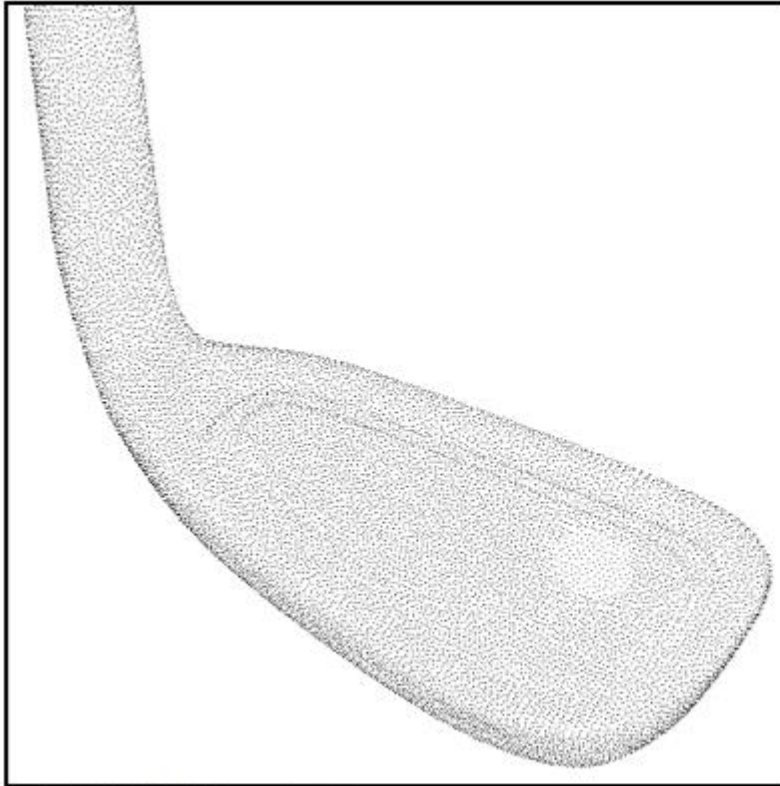
Matthias Eck
University of Darmstadt

Hugues Hoppe
Microsoft Research

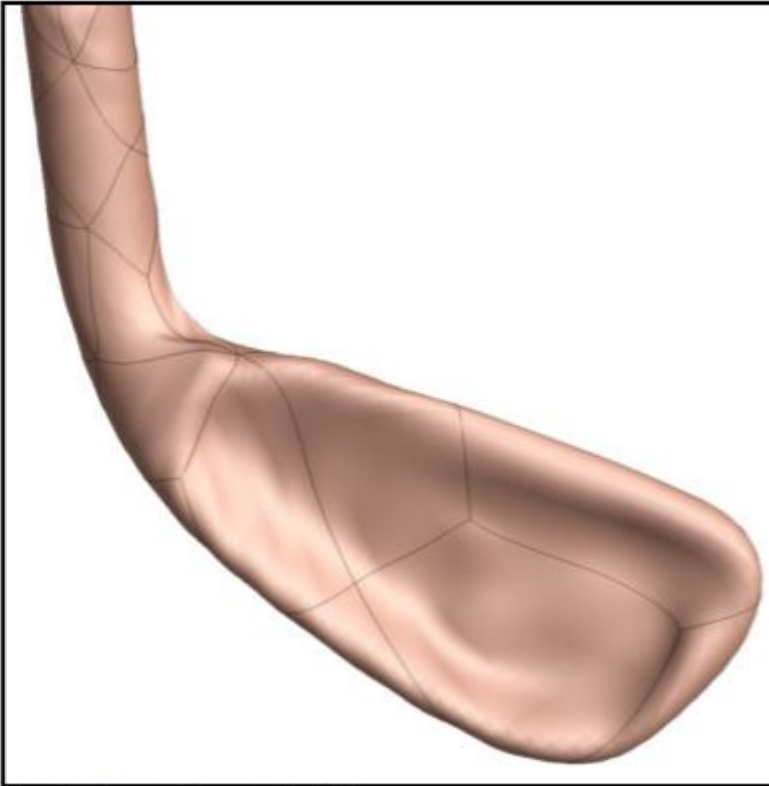
SIGGRAPH 1996

Applications

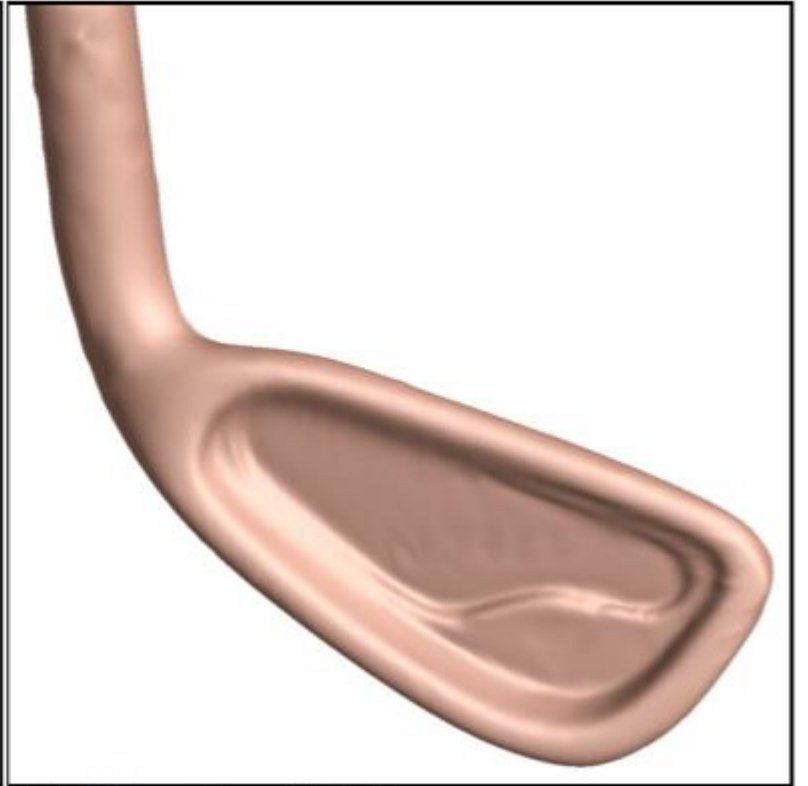
- Surface Reconstruction



(a) P (16,585 points)



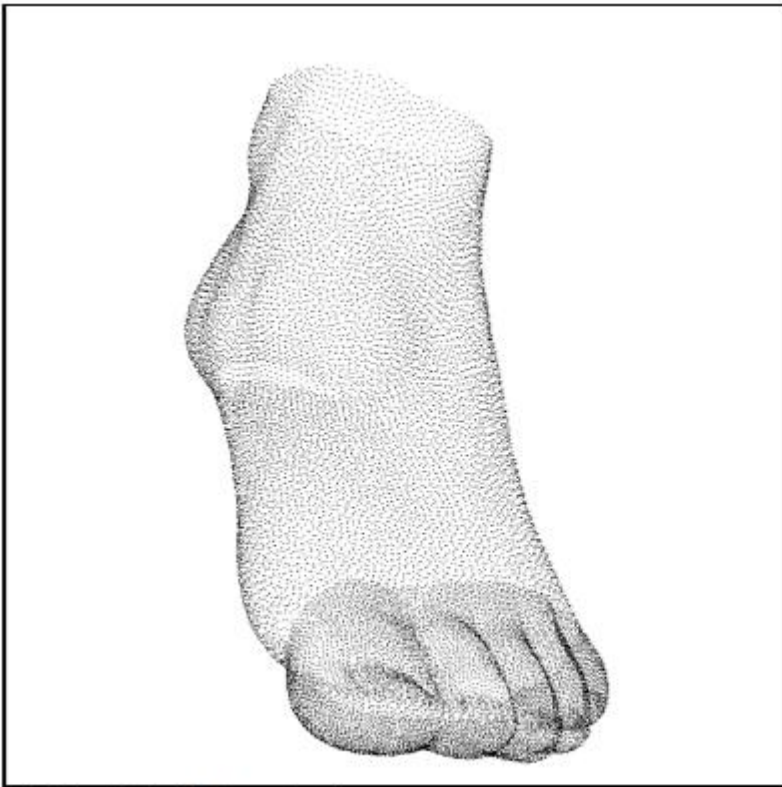
(b) 35 patches, 1.36% max error



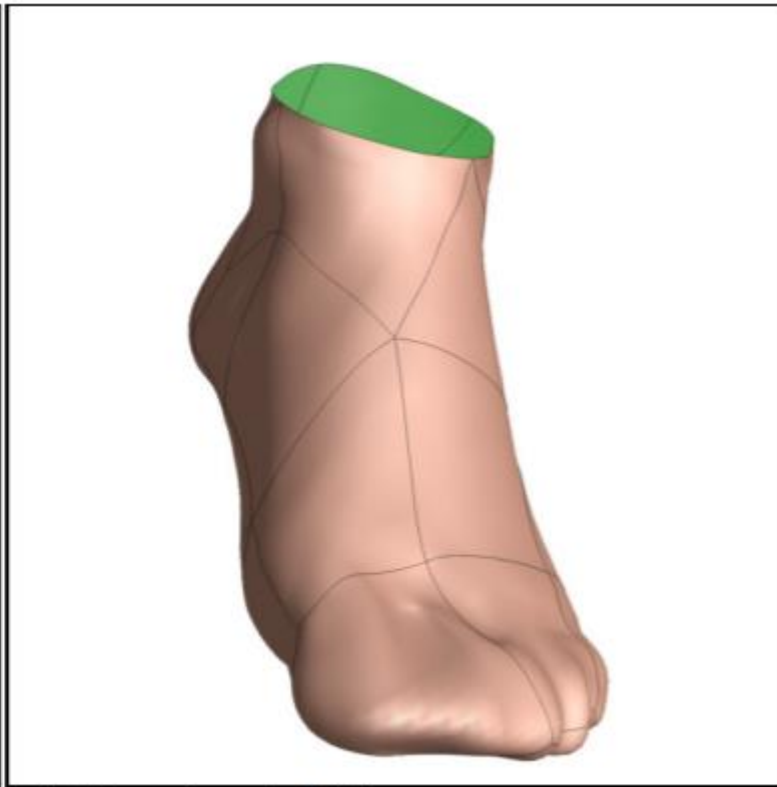
(c) 285 patches, 0.41% max error

Applications

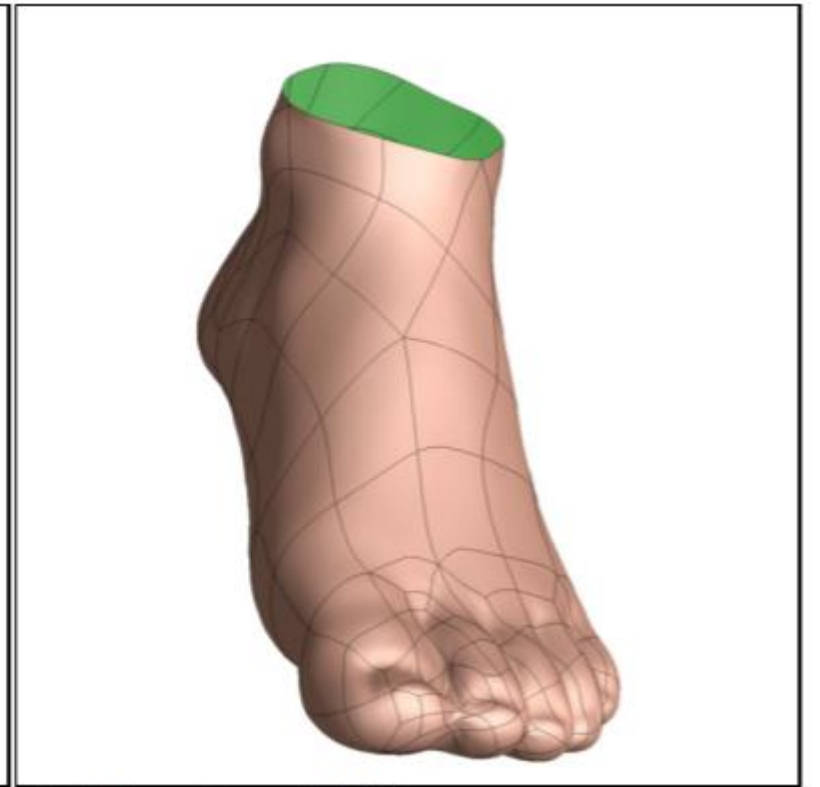
- Surface Reconstruction



(d) P (20,021 points)



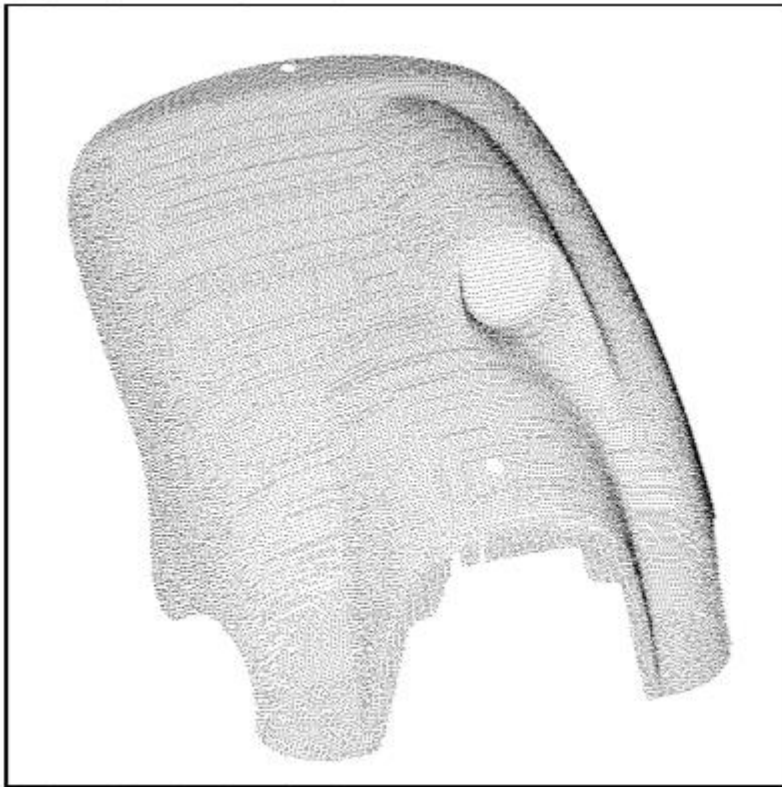
(e) 29 patches, 1.20% max error



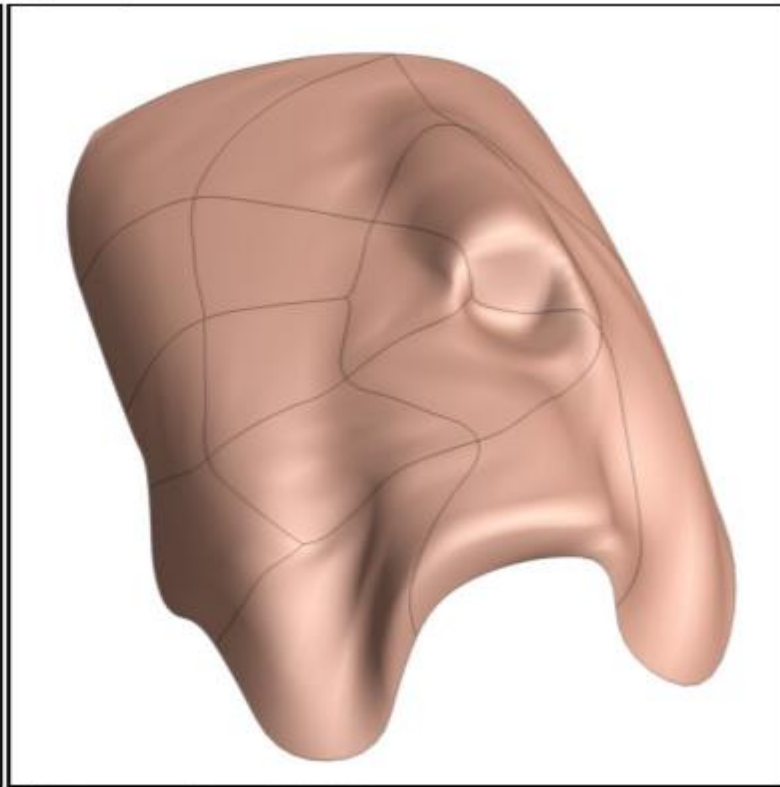
(f) 156 patches, 0.27% max error

Applications

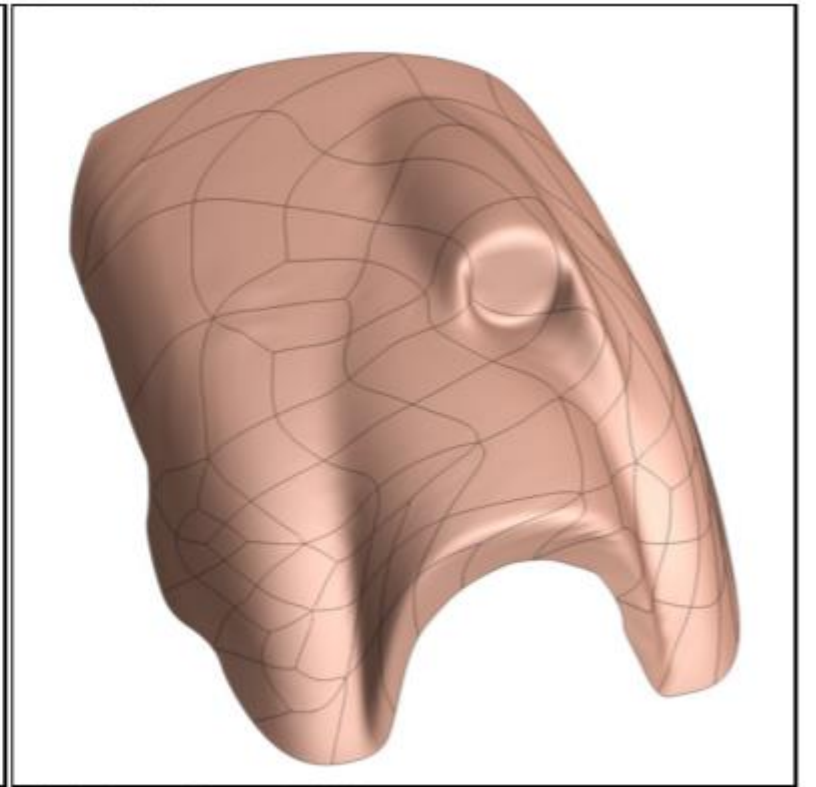
- Surface Reconstruction



(g) P (37,974 points)



(h) 15 patches, 3.00% max error



(i) 94 patches, 0.69% max error

Applications

- Sketch-based modeling

Sketch-based Modeling with Few Strokes

Joseph Jacob Cherlin [†]

Faramarz Samavati [†]

Mario Costa Sousa [†]

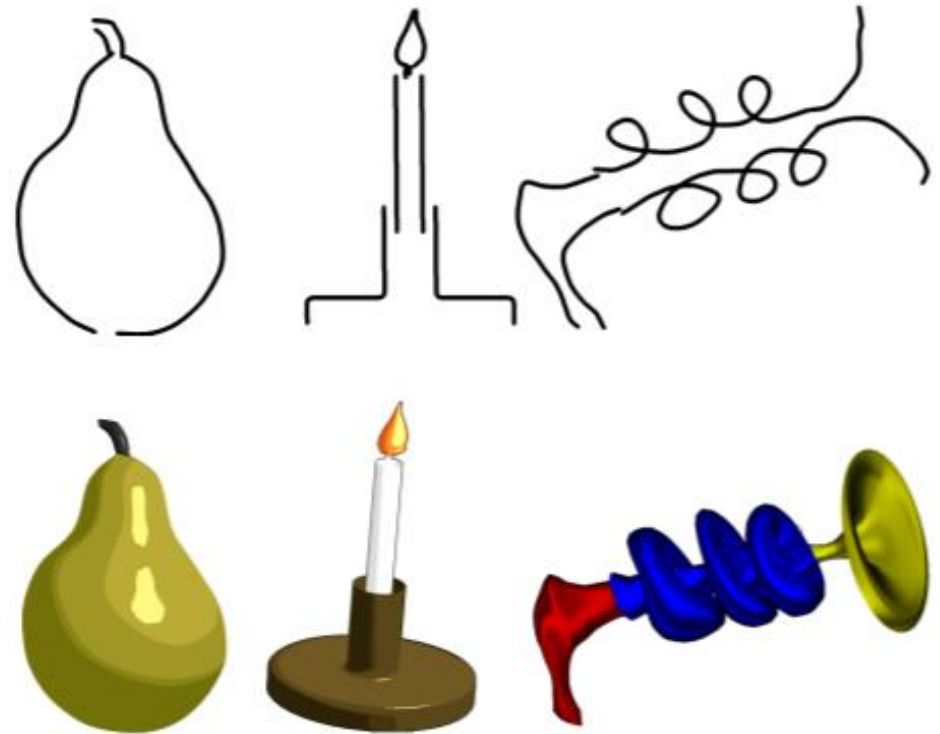
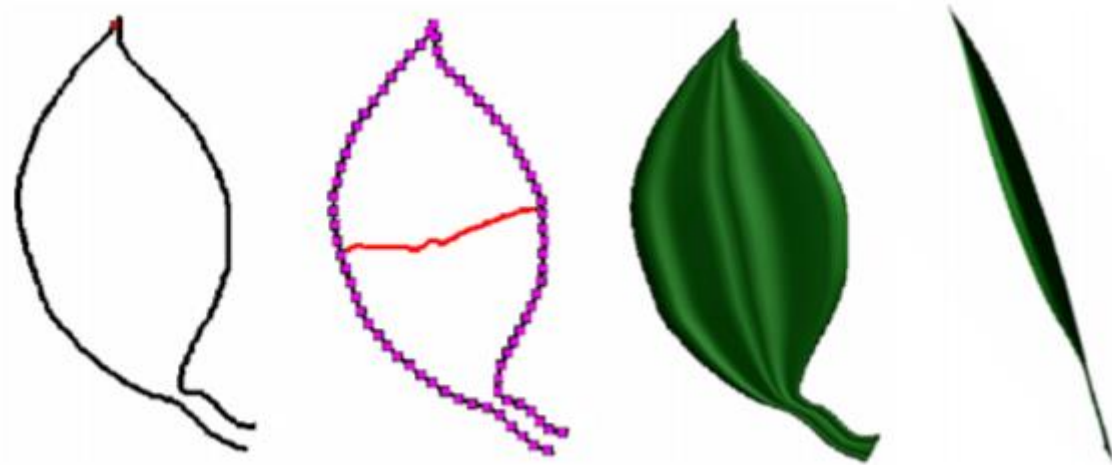
Joaquim A. Jorge [‡]

[†] University of Calgary

[‡] Technical University of Lisbon

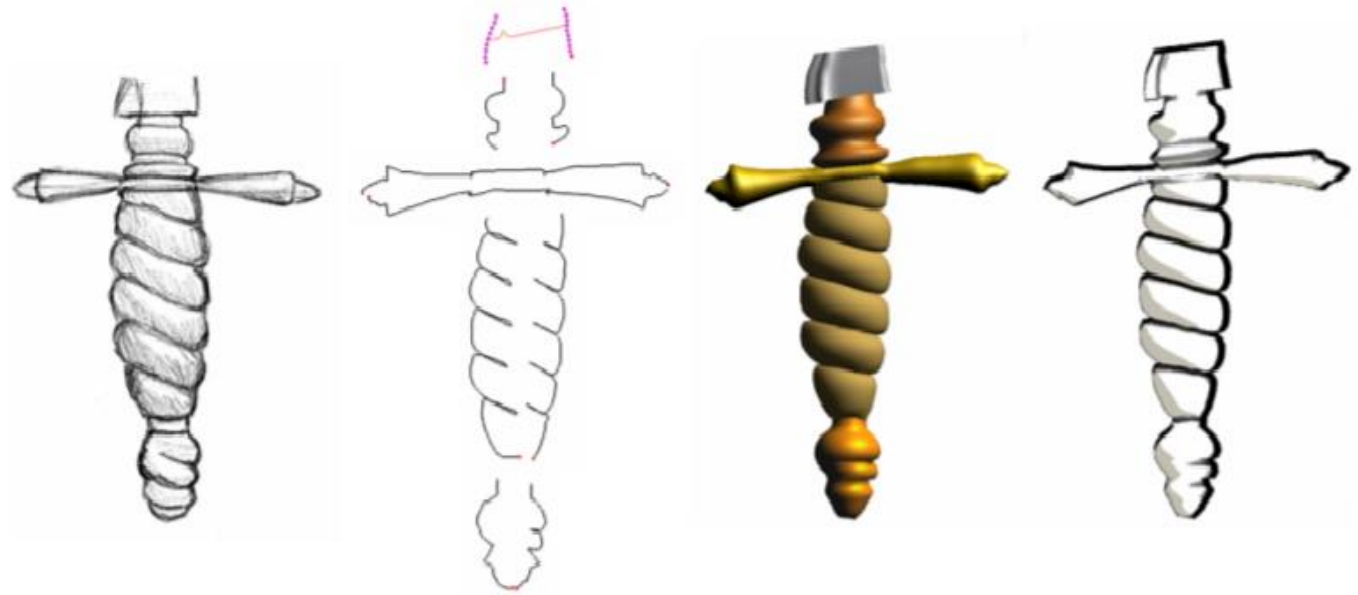
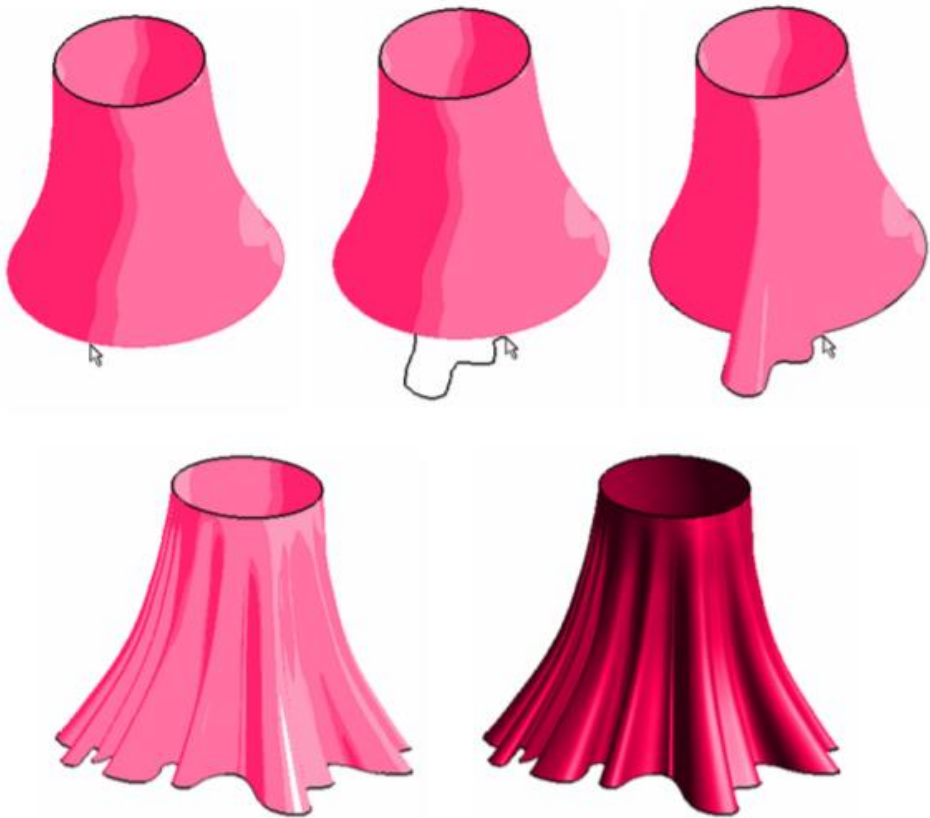
Applications

- Sketch-based modeling



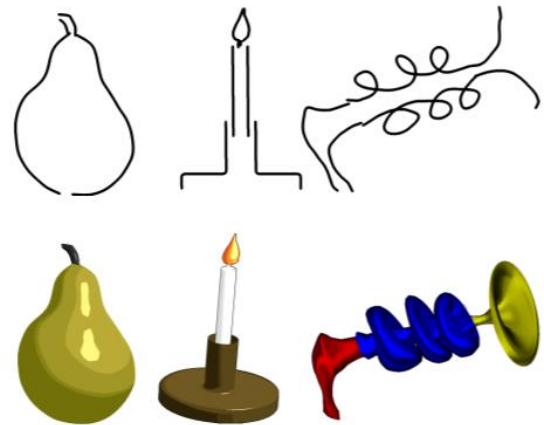
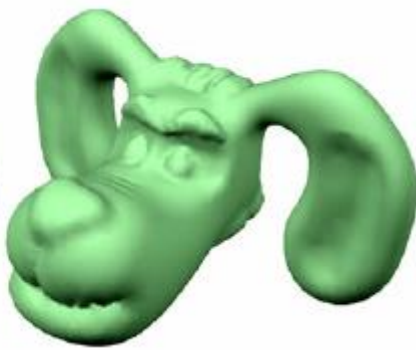
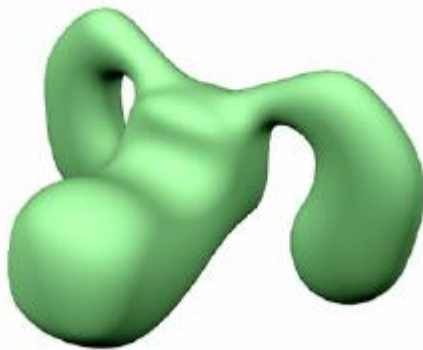
Applications

- Sketch-based modeling



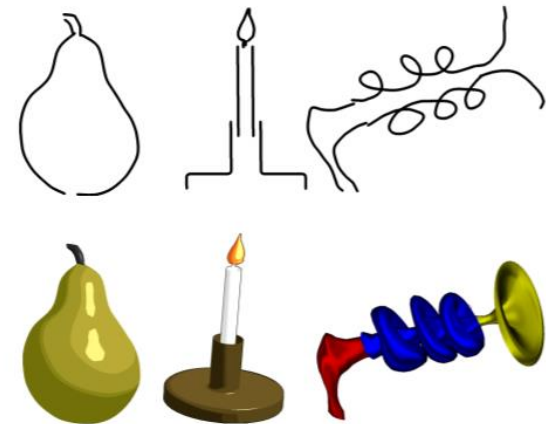
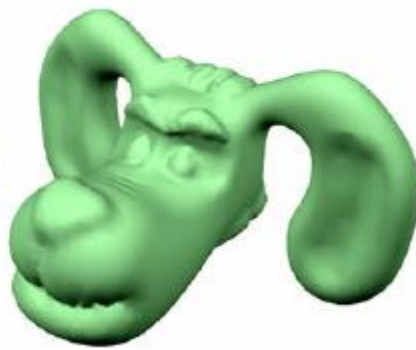
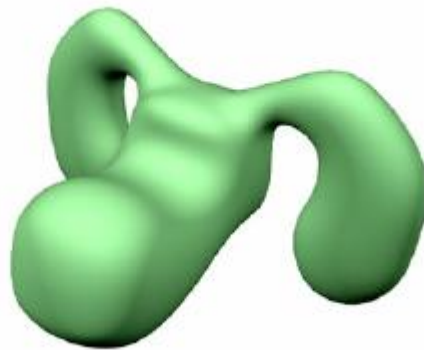
Polygonal Mesh

- So far what we have discussed is a bit limited.



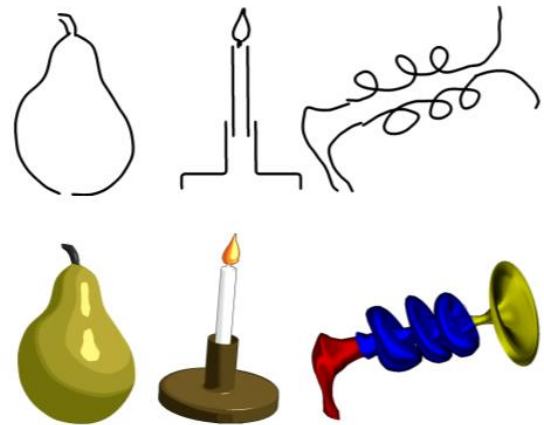
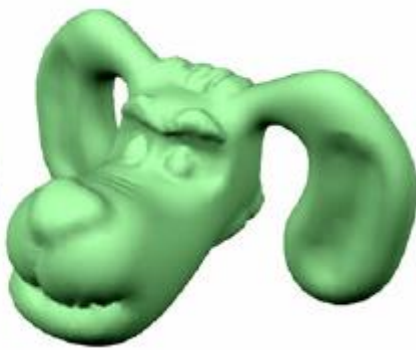
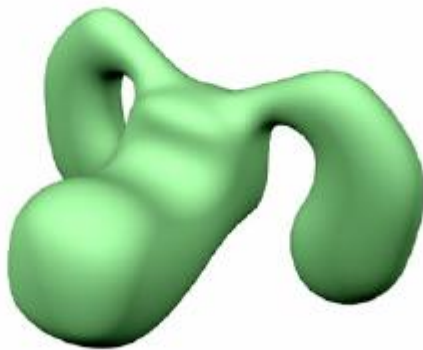
Polygonal Mesh

- So far what we have discussed is a bit limited.
- We have limited control on the topology of shapes.



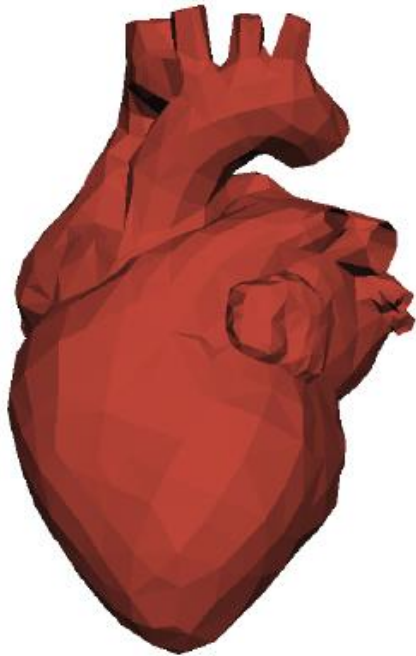
Polygonal Mesh

- So far what we have discussed is a bit limited.
- We have limited control on the topology of shapes.
- We have to have many patches or complicated functions.



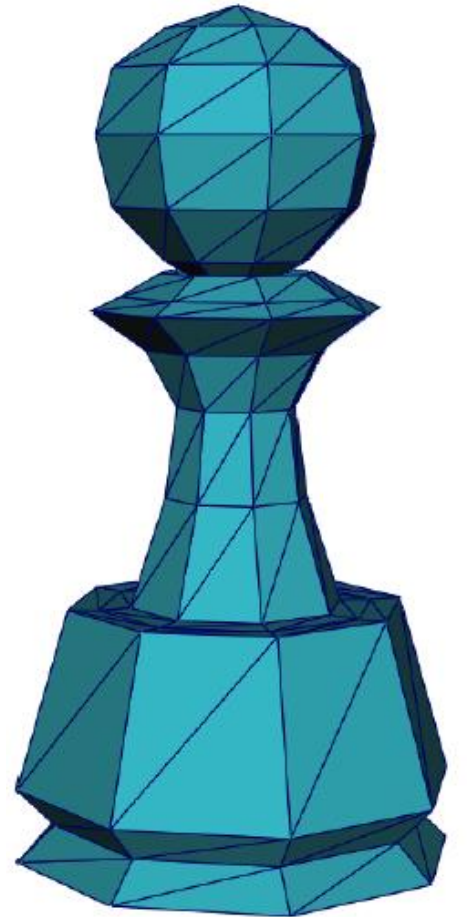
Polygonal Mesh

- How can we have complex shapes with handles, holes, sharp features, etc?



Polygonal Mesh

- Make a surface by attaching very small polygonal faces.



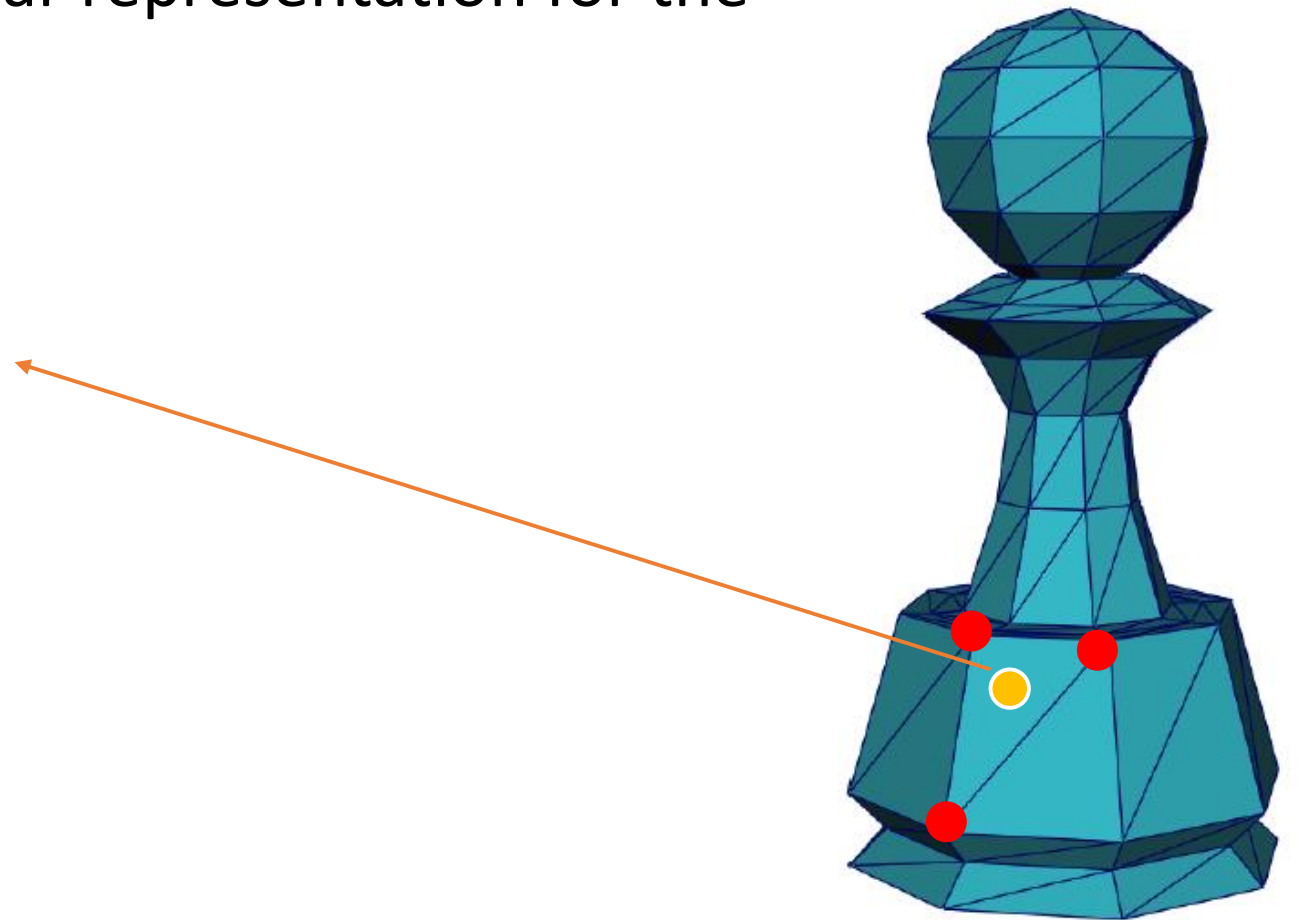
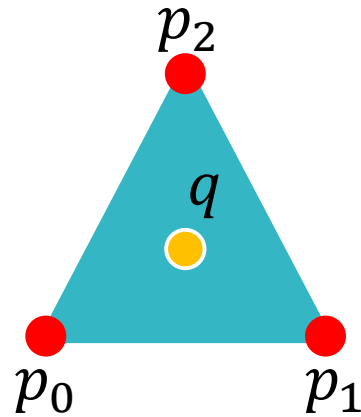
Polygonal Mesh

- Make a surface by attaching very small polygonal faces.
- In fact, mesh M is composed of a set of vertices V and faces F
 - $M = (F, V)$.



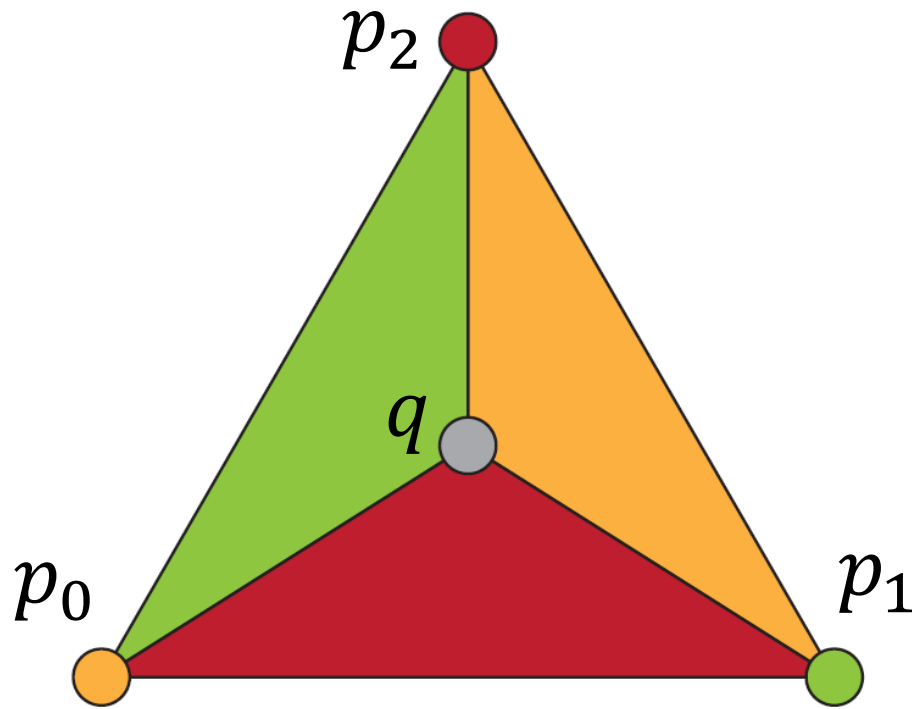
Polygonal Mesh

- This way, we can define a linear representation for the shape.



Polygonal Mesh

- Barycentric Coordinate



$$q = \alpha p_0 + \beta p_1 + \gamma p_2$$

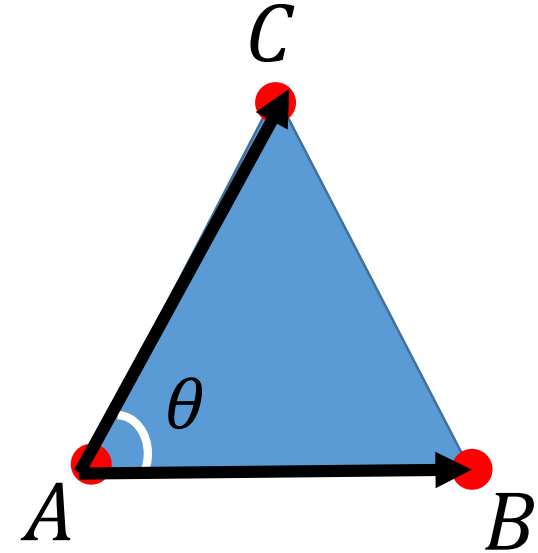
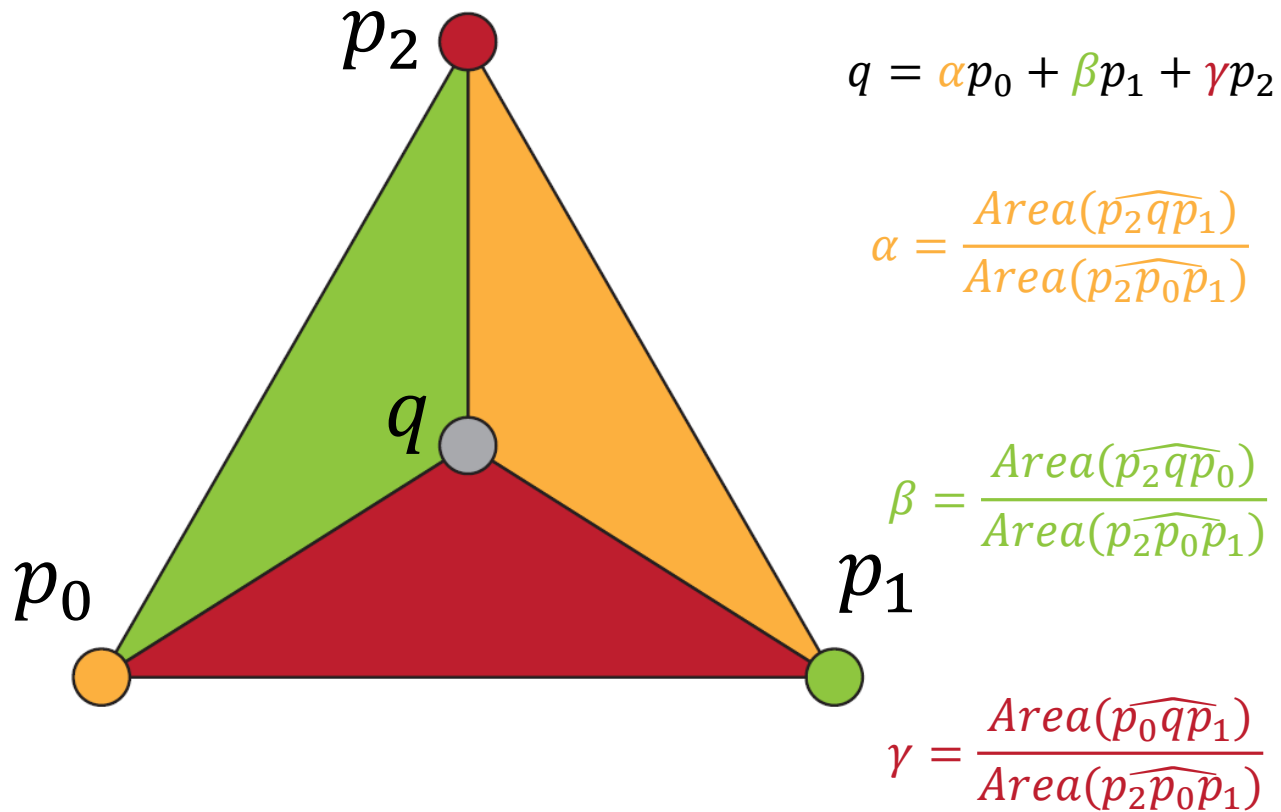
$$\alpha = \frac{\text{Area}(\widehat{p_2 q p_1})}{\text{Area}(\widehat{p_2 p_0 p_1})}$$

$$\beta = \frac{\text{Area}(\widehat{p_2 q p_0})}{\text{Area}(\widehat{p_2 p_0 p_1})}$$

$$\gamma = \frac{\text{Area}(\widehat{p_0 q p_1})}{\text{Area}(\widehat{p_2 p_0 p_1})}$$

Polygonal Mesh

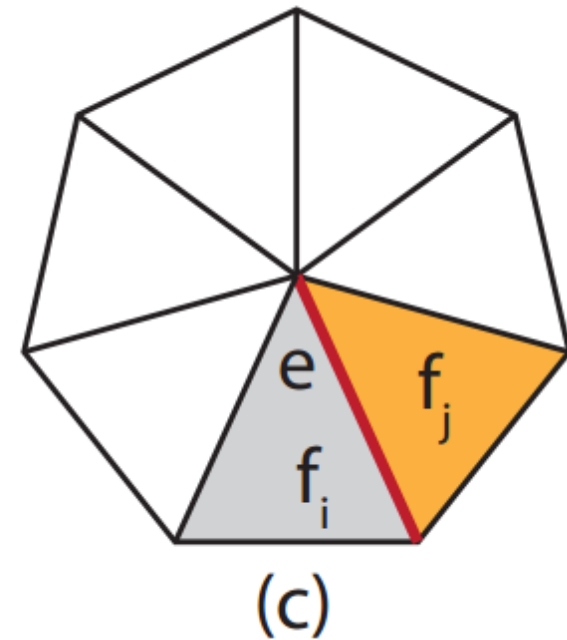
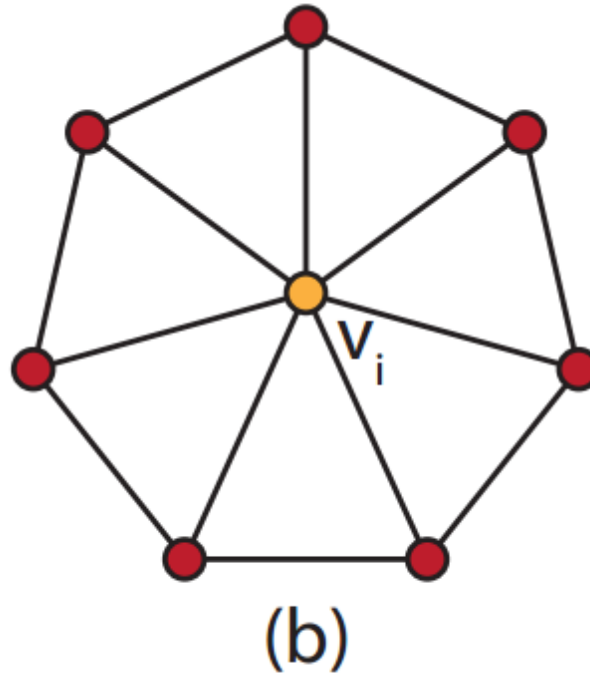
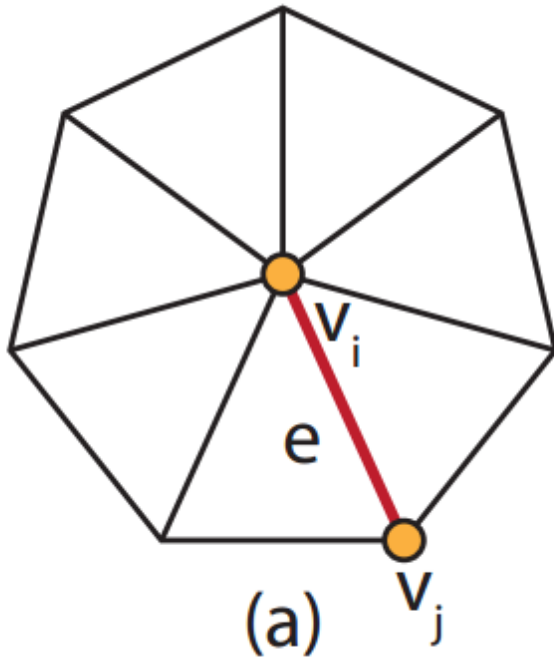
- Barycentric Coordinate



$$\text{Area}(\widehat{ABC}) = \frac{\|B - A\| * \|C - A\| * \sin \theta}{2}$$

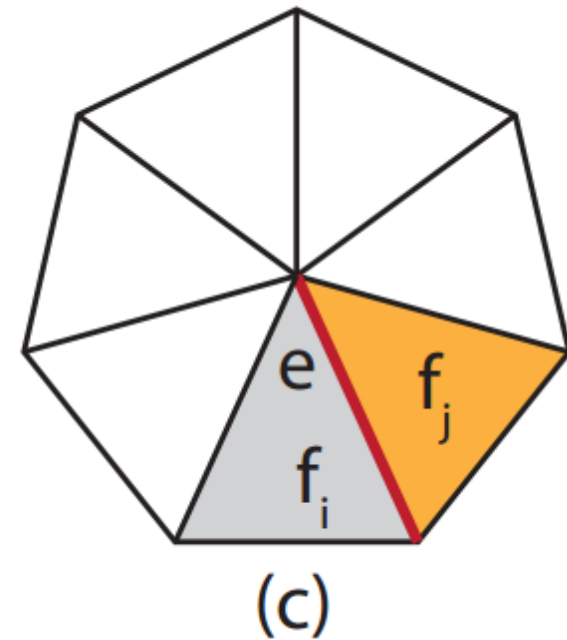
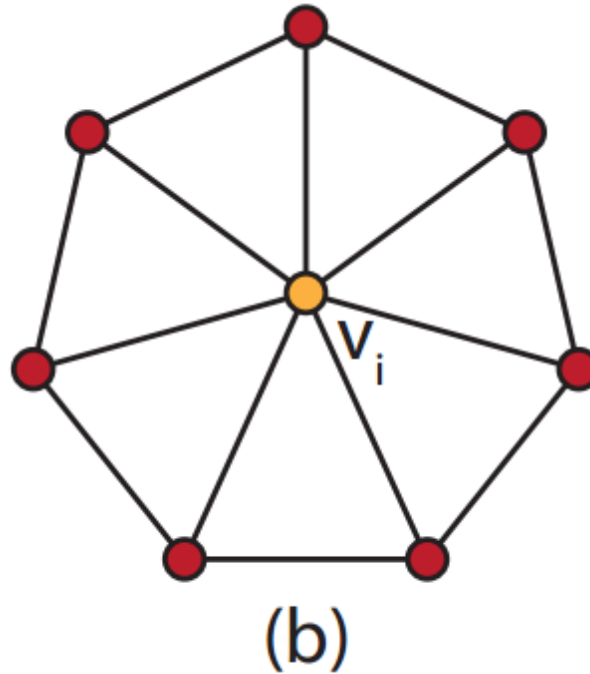
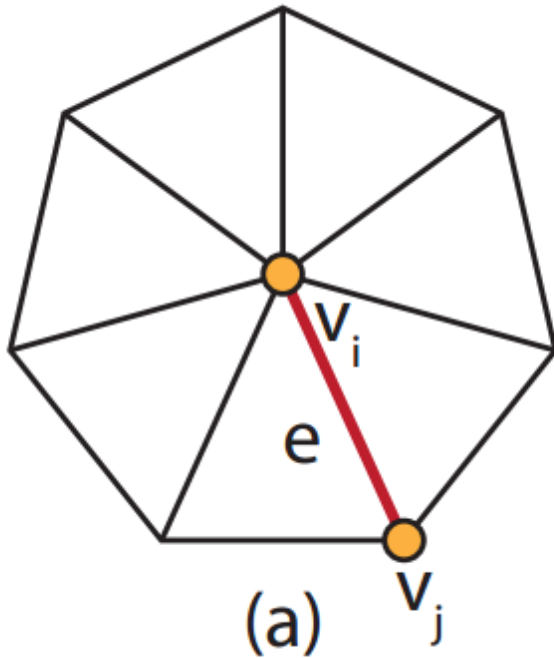
Polygonal Mesh

- Edge e connects v_i to v_j .



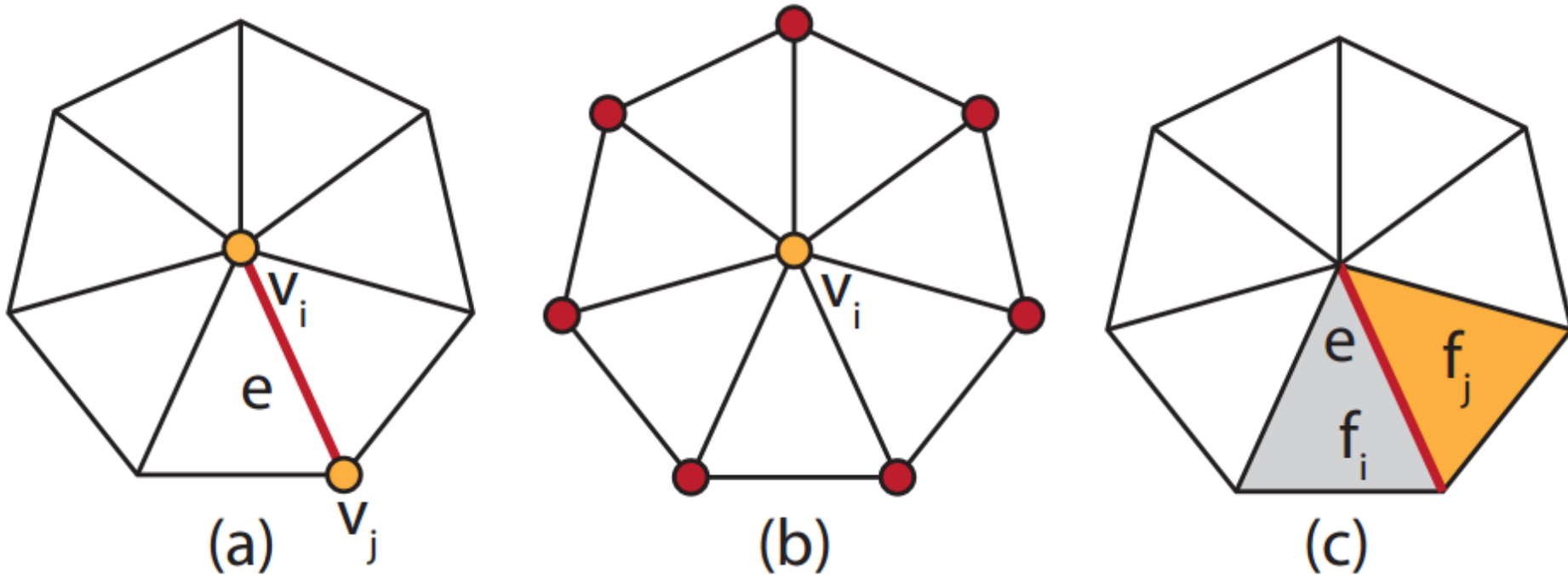
Polygonal Mesh

- Edge e connects v_i to v_j .
- v_i and v_j are neighbors.



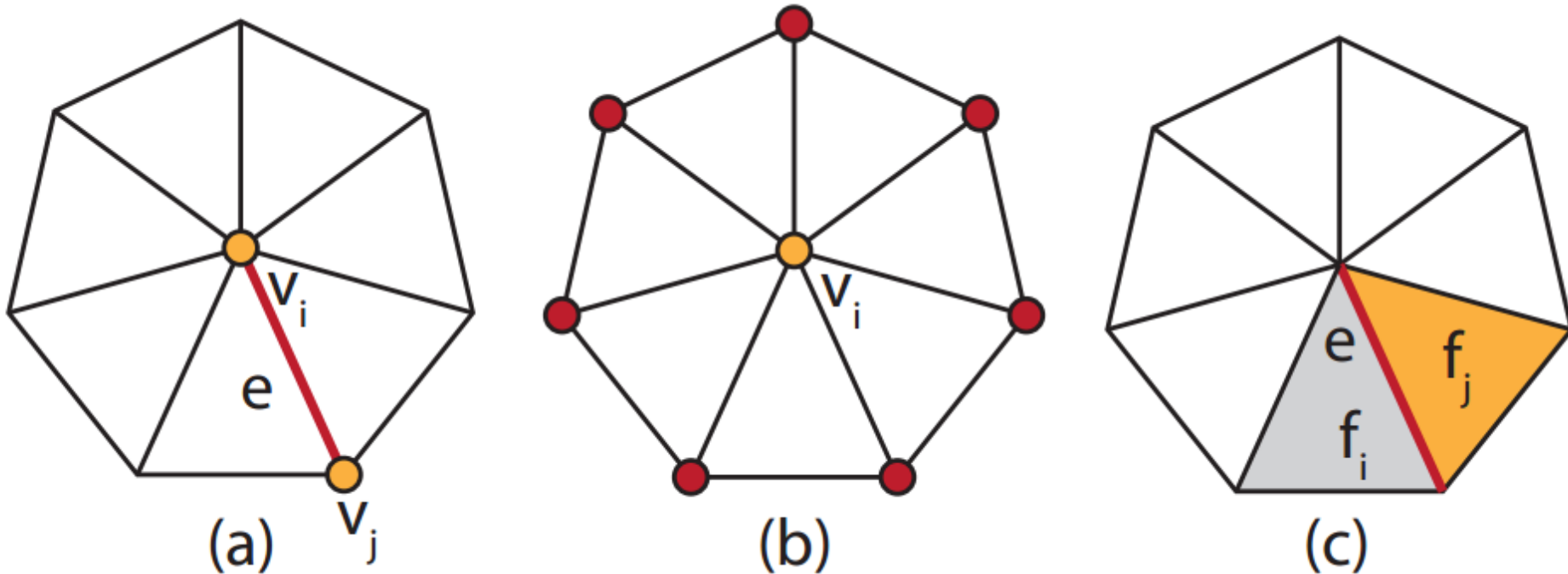
Polygonal Mesh

- Number of vertices connected to v_i is the **valence** of v_i .



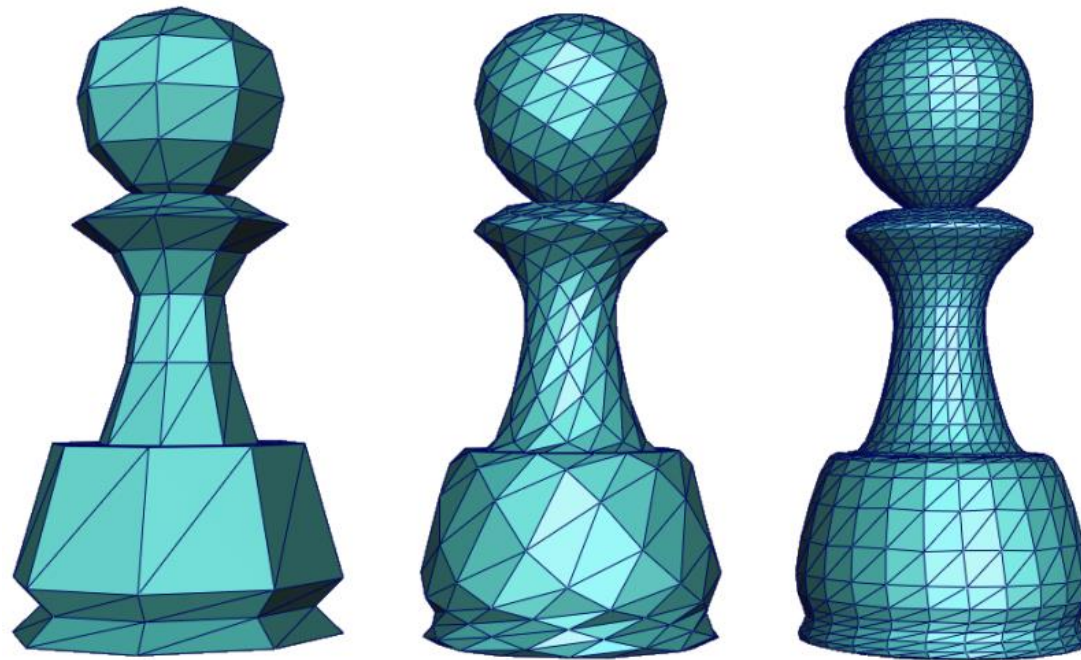
Polygonal Mesh

- Number of vertices connected to v_i is the **valence** of v_i .
- In a triangular mesh if valence of a vertex is six, it is a **regular** vertex.



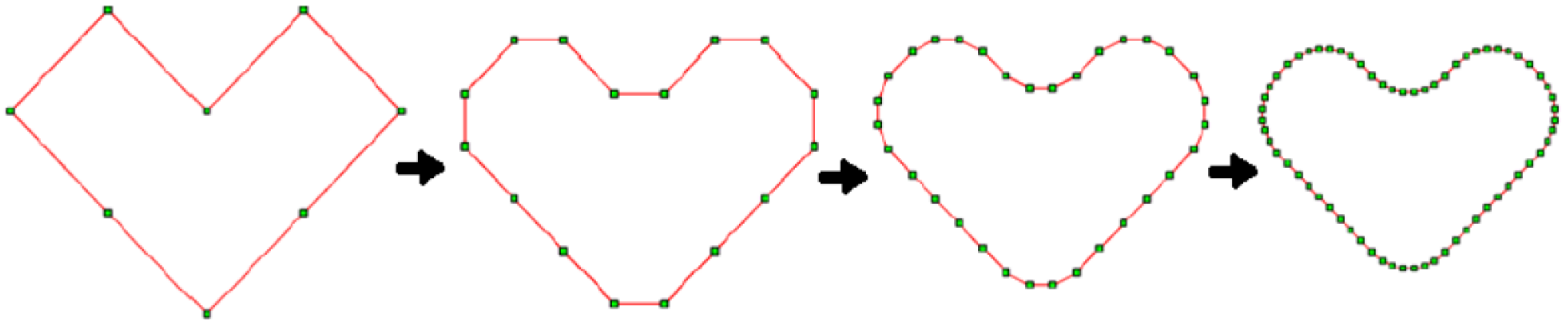
Polygonal Mesh

- How to model a complex and smooth shape?
 - Make a coarse model, subdivide it



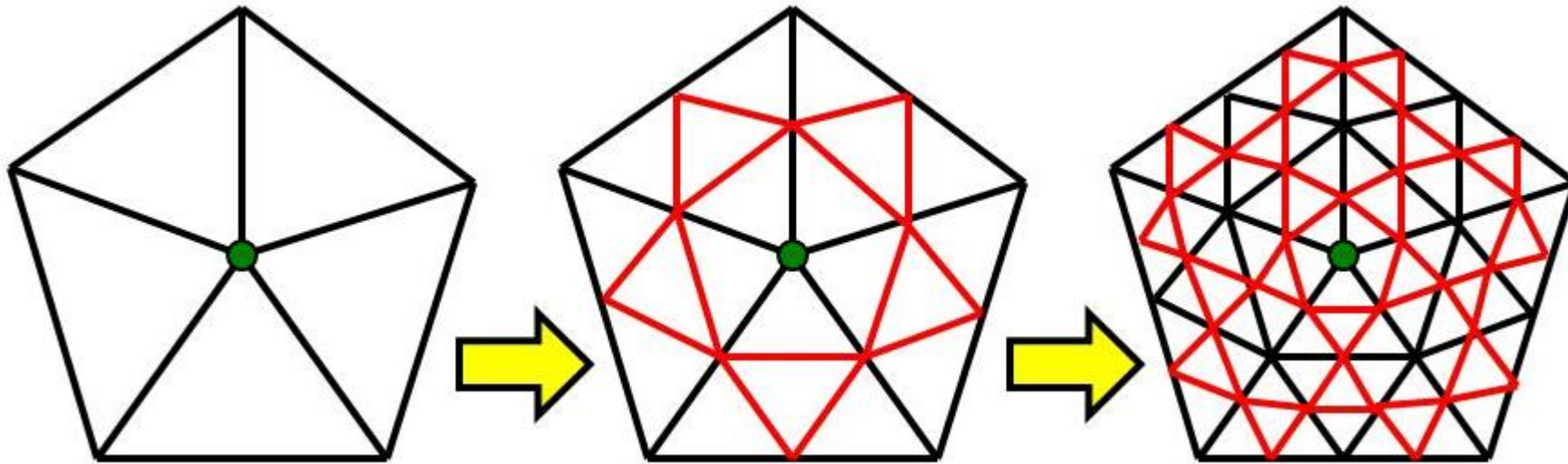
Polygonal Mesh

- How to model a complex and smooth shape?
 - Make a coarse model, subdivide it
 - You remember subdivision for curves?
 - We had a set of coarse vertices and using a combination of neighboring vertices, we could generate a smooth curve.



Polygonal Mesh

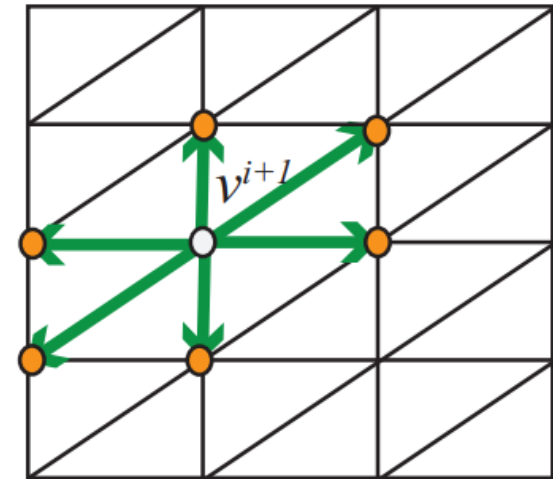
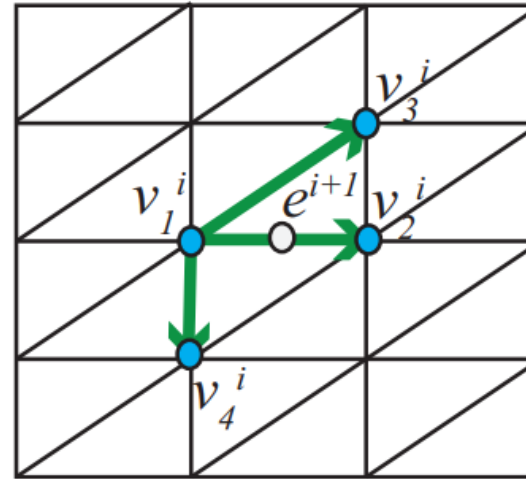
- Loop Subdivision
 - Only for triangular meshes



Polygonal Mesh

- Loop Subdivision
 - Only for triangular meshes

$$e^{i+1} = \frac{3}{8}v_1^i + \frac{3}{8}v_2^i + \frac{1}{8}v_3^i + \frac{1}{8}v_4^i$$



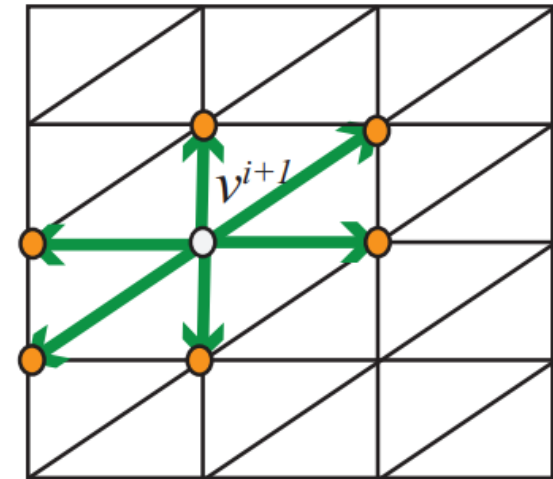
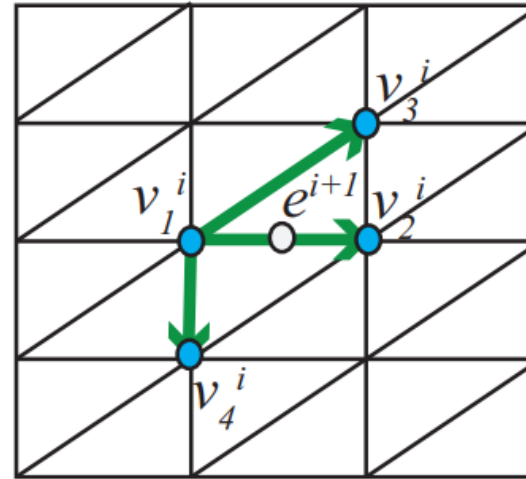
Polygonal Mesh

- Loop Subdivision
 - Only for triangular meshes

$$e^{i+1} = \frac{3}{8}v_1^i + \frac{3}{8}v_2^i + \frac{1}{8}v_3^i + \frac{1}{8}v_4^i$$

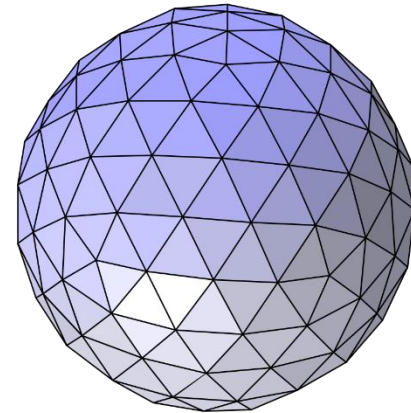
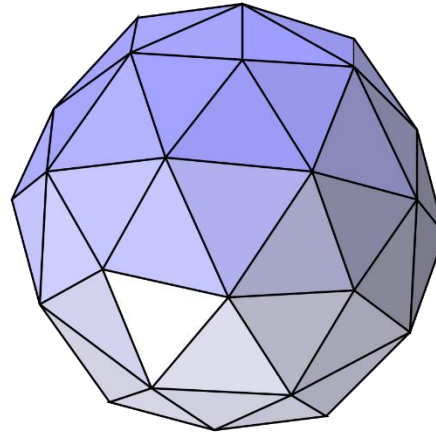
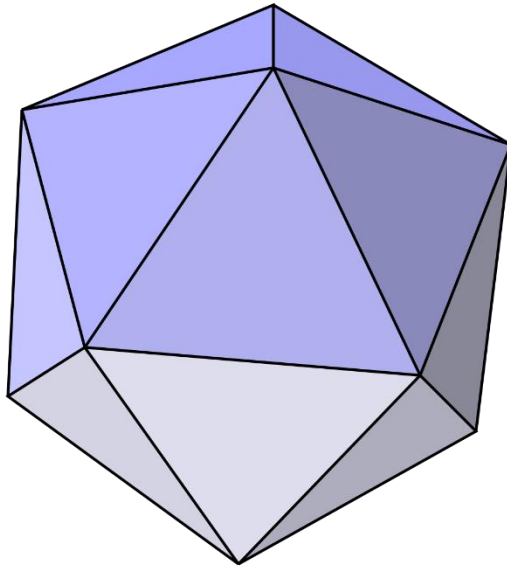
$$v^{i+1} = (1 - n\alpha)v^i + \alpha \sum v_j^i$$

$$\alpha = \frac{1}{n} \left(\frac{5}{8} - \left(\frac{3}{8} + \frac{1}{4} \cos \frac{2\pi}{n} \right)^2 \right)$$



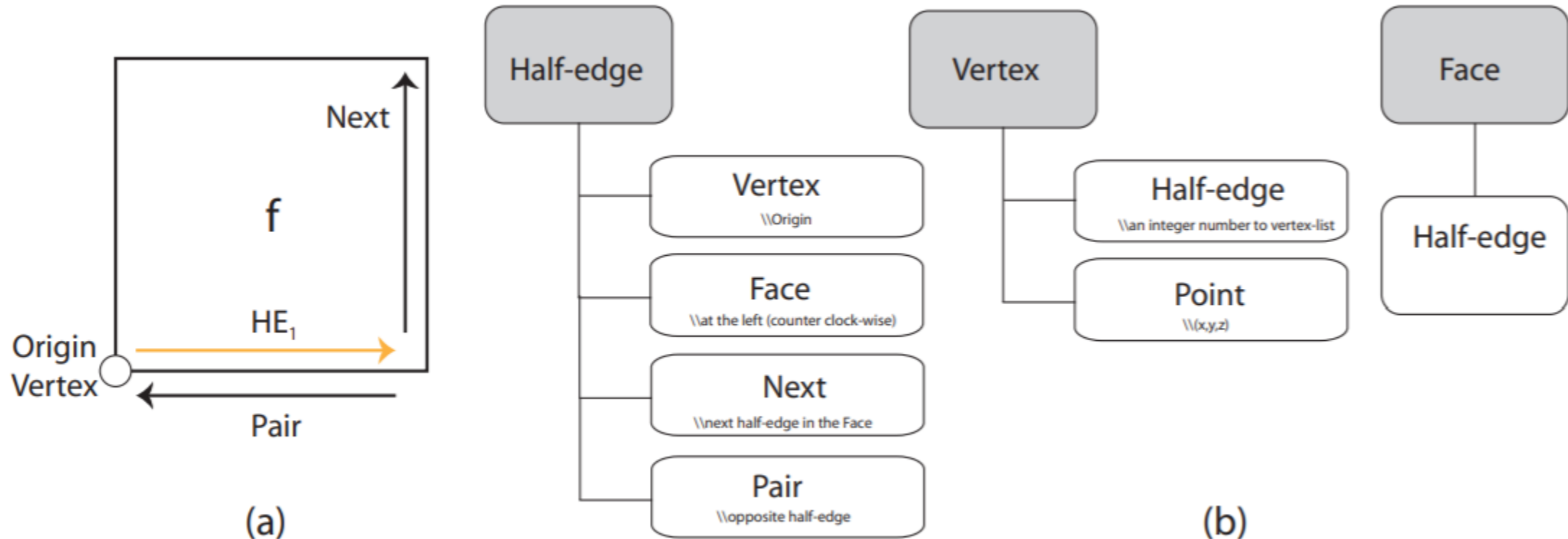
Polygonal Mesh

- Loop Subdivision
 - Only for triangular meshes



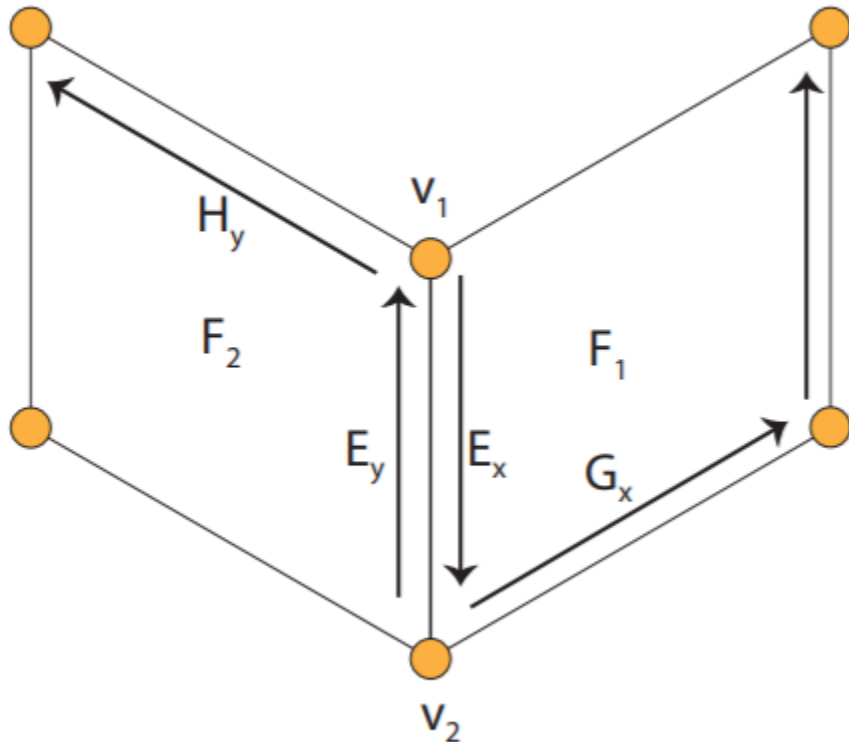
Polygonal Mesh

- How to access neighbors of a vertex
 - Half-edge data structure



Polygonal Mesh

- How to access neighbors of a vertex?



Half-edge-List

Half-edge	Vertex (origin)	face	next	pair
E_x	v_1	F_1	G_x	E_y
E_y	v_2	F_2	H_y	E_x
...	...	...	...	...

Face	Half-edge
F_1	E_x or G_x
F_2	H_y
...	...

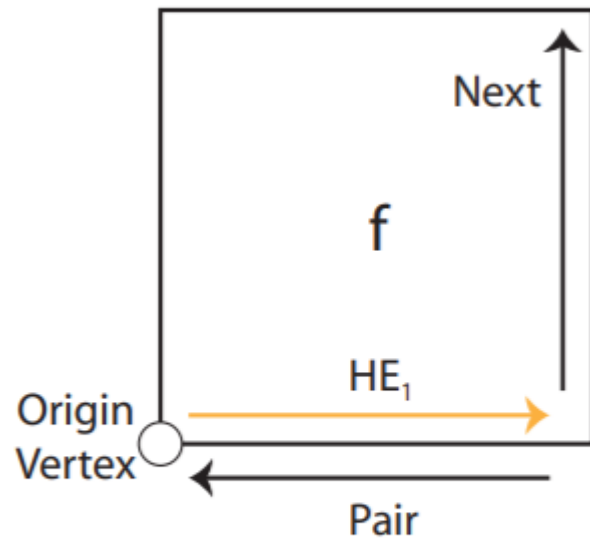
Face-List

Vertex	Half-edge	X	Y	Z
v_1	E_x or H_y			
v_2	G_x			
...	...			

Vertex-List

Polygonal Mesh

- How to access neighbors of a vertex
 - Half-edge data structure



(a)

How do you find all
the neighbors of a vertex?

Polygonal Mesh

- Laplacian for mesh?



Polygonal Mesh

- Laplacian
 - Review for image

0	1	0
1	-4	1
0	1	0

Polygonal Mesh

- Laplacian
 - Review for image

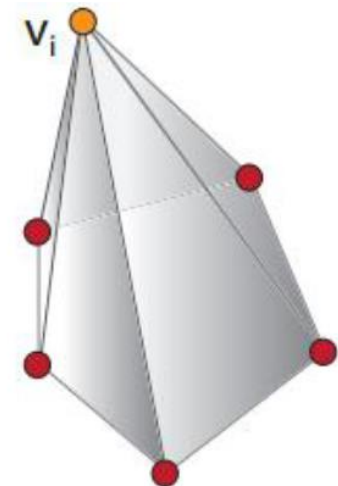
0	1	0
1	-4	1
0	1	0

- How can we extend it to meshes?

Polygonal Mesh

- Laplacian
 - Review for image
- How can we extend it to meshes?
 - Summation of all vertices in the neighborhood over its valence

0	1	0
1	-4	1
0	1	0



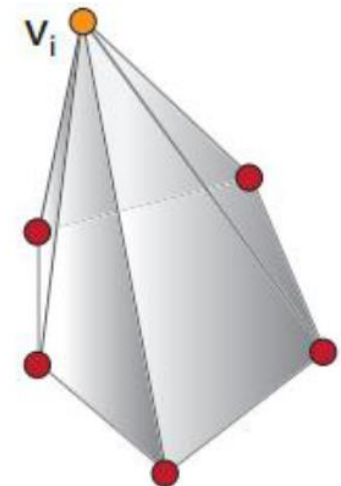
Polygonal Mesh

- Laplacian
 - Review for image

0	1	0
1	-4	1
0	1	0

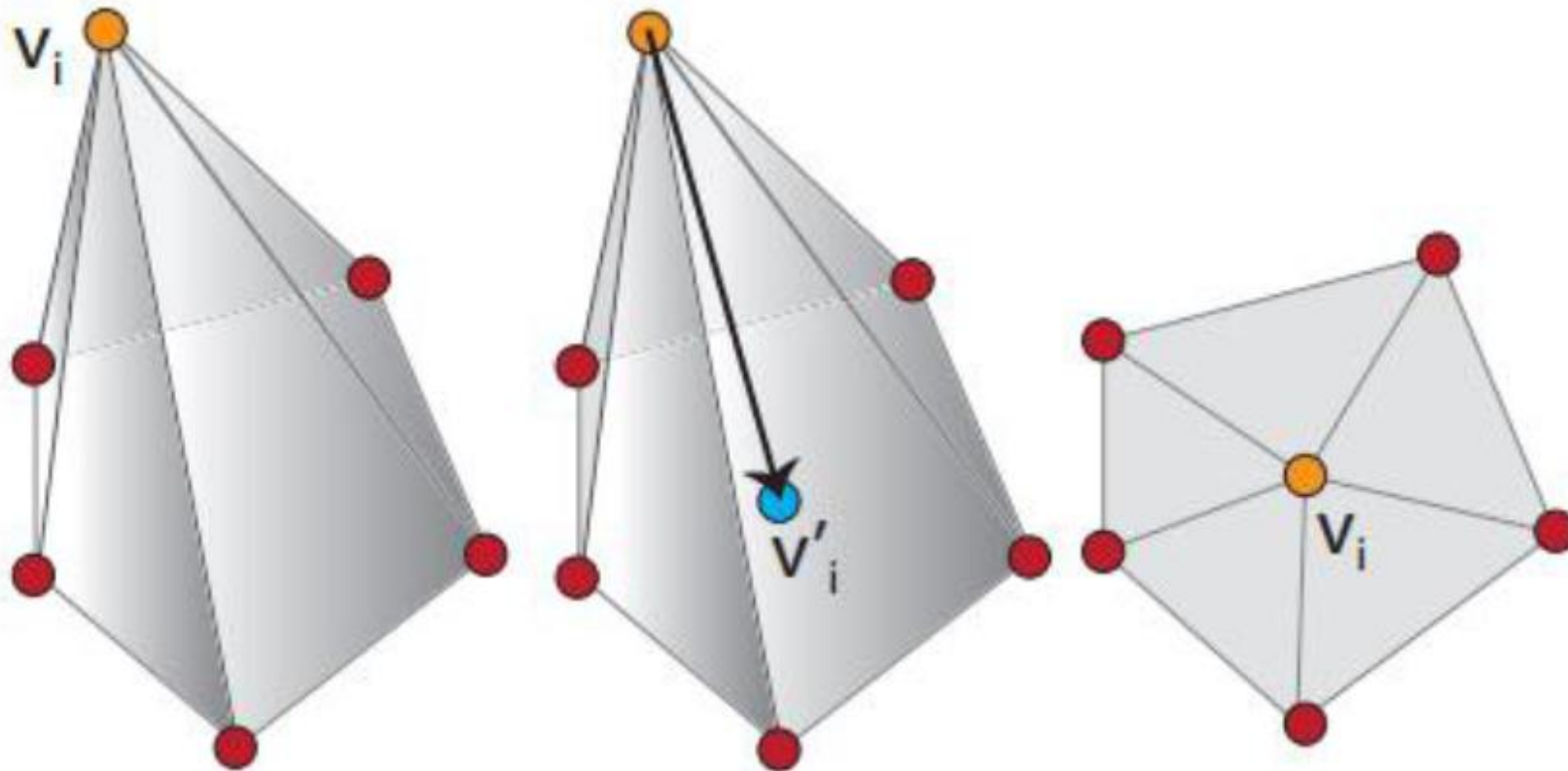
- How can we extend it to meshes?
 - Summation of all vertices in the neighborhood over its valence

$$\hat{v}_i = \frac{\sum_{v_j \in N(v_i)} v_j}{|N(v_i)|}$$



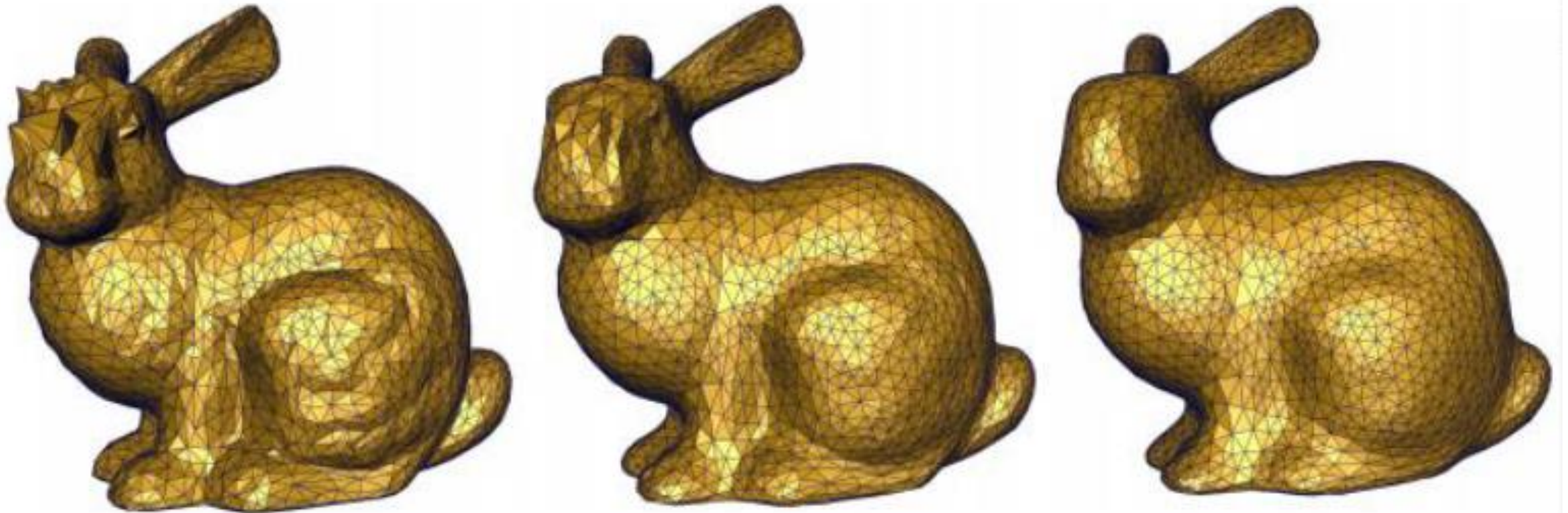
Polygonal Mesh

- We can smooth meshes using Laplacian



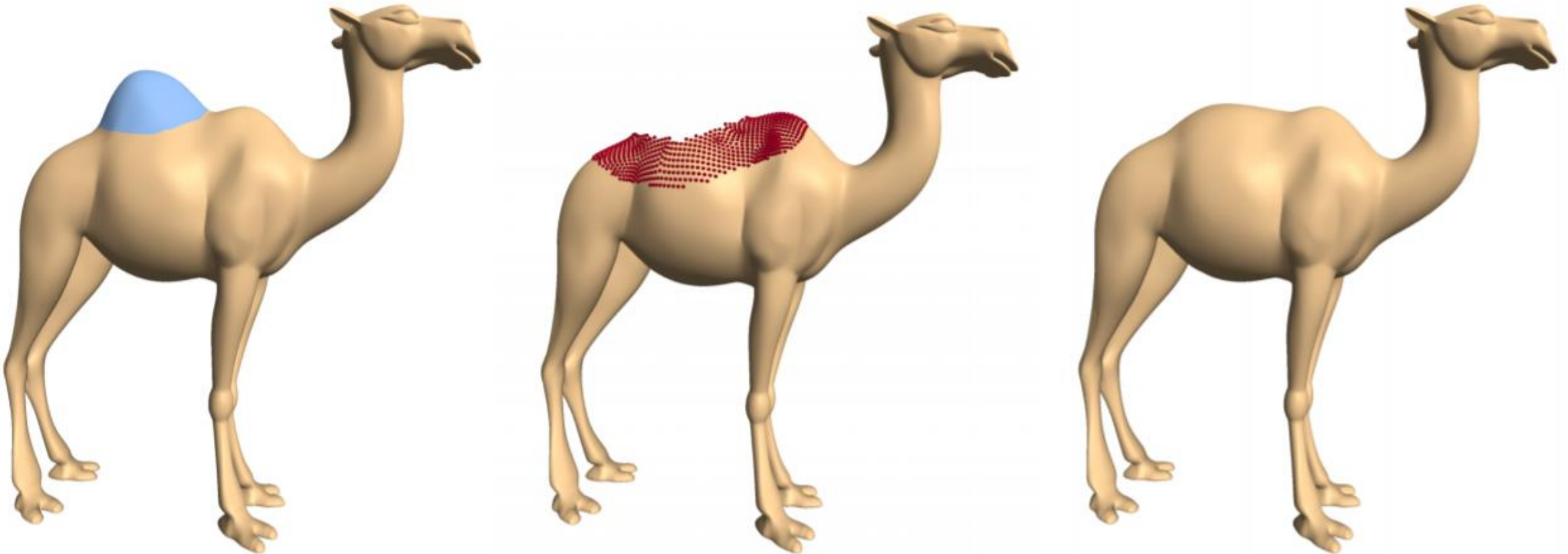
Polygonal Mesh

- We can smooth meshes using Laplacian



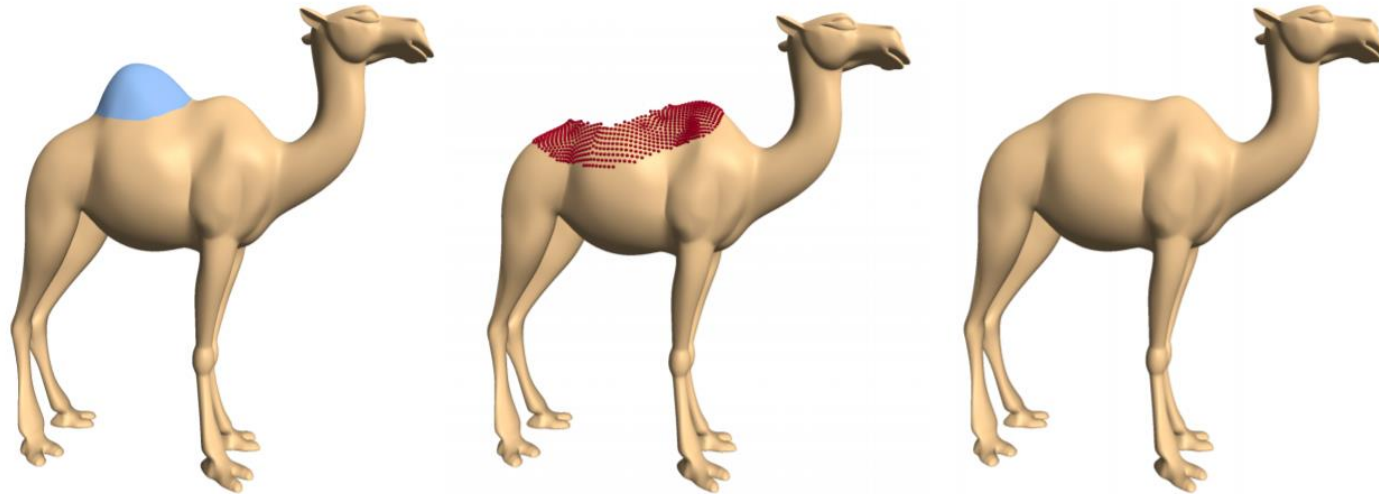
Polygonal Mesh

- Similarly we can use it for (sort of) mesh inpainting



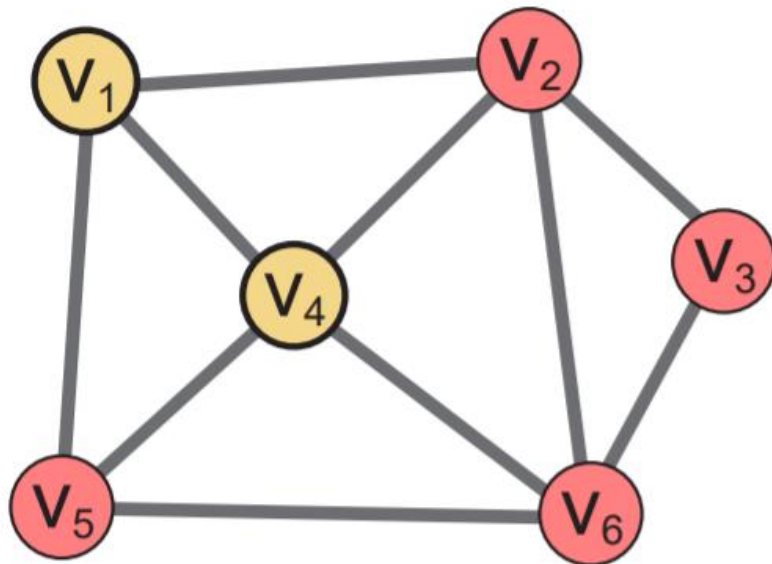
Polygonal Mesh

- Very Similar formulation to image inpainting
 - We know the location of some vertices, we want to smoothly find the location of other vertices.



Polygonal Mesh

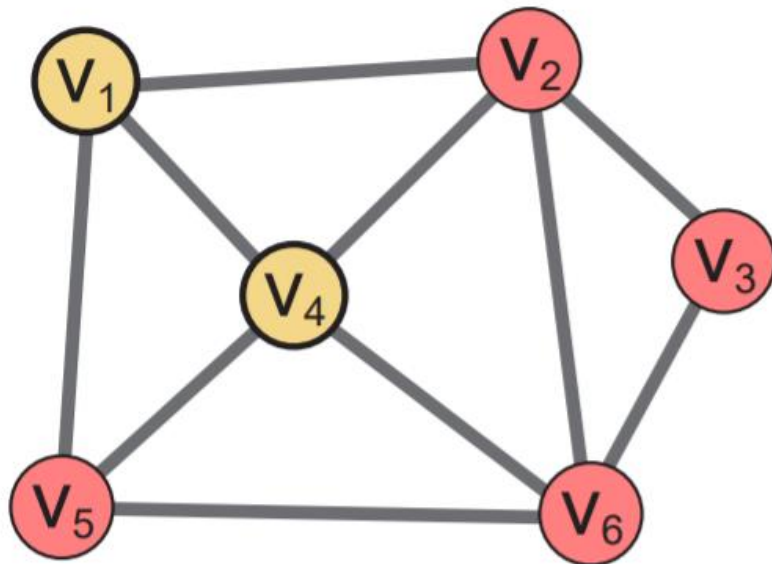
- Very Similar formulation to image inpainting
 - We know the location of some vertices, we want to smoothly find the location of other vertices.



1	$-\frac{1}{3}$	0	$-\frac{1}{3}$	$-\frac{1}{3}$	0
$-\frac{1}{4}$	1	$-\frac{1}{4}$	$-\frac{1}{4}$	0	$-\frac{1}{4}$
0	$-\frac{1}{2}$	1	0	0	$-\frac{1}{2}$
$-\frac{1}{4}$	$-\frac{1}{4}$	0	1	$-\frac{1}{4}$	$-\frac{1}{4}$
$-\frac{1}{3}$	0	0	$-\frac{1}{3}$	1	$-\frac{1}{3}$
0	$-\frac{1}{4}$	$-\frac{1}{4}$	$-\frac{1}{4}$	$-\frac{1}{4}$	1
1	0	0	0	0	0
0	0	0	1	0	0

Polygonal Mesh

- Very Similar formulation to image inpainting
 - We know the location of some vertices, we want to smoothly find the location of other vertices.



1	$-\frac{1}{3}$	0	$-\frac{1}{3}$	$-\frac{1}{3}$	0
$-\frac{1}{4}$	1	$-\frac{1}{4}$	$-\frac{1}{4}$	0	$-\frac{1}{4}$
0	$-\frac{1}{2}$	1	0	0	$-\frac{1}{2}$
$-\frac{1}{4}$	$-\frac{1}{4}$	0	1	$-\frac{1}{4}$	$-\frac{1}{4}$
$-\frac{1}{3}$	0	0	$-\frac{1}{3}$	1	$-\frac{1}{3}$
0	$-\frac{1}{4}$	$-\frac{1}{4}$	$-\frac{1}{4}$	$-\frac{1}{4}$	1
1	0	0	0	0	0
0	0	0	1	0	0

Matrix A

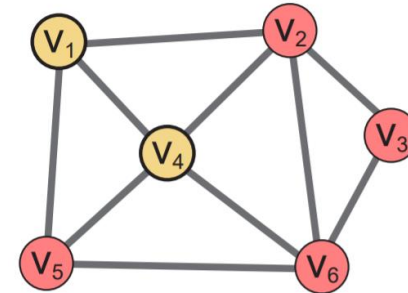
Polygonal Mesh

- Similar System

$$Ax = b$$

$$A = \begin{pmatrix} L \\ F \end{pmatrix}, \quad F_{ij} = \begin{cases} 1 & \text{fixed vertices} \\ 0 & \text{free vertices} \end{cases}$$

$$b_k = \begin{cases} 0 & \text{fixed vertices} \\ x_{\text{fixed}} & \text{free vertices} \end{cases}$$



1	$-\frac{1}{3}$	0	$-\frac{1}{3}$	$-\frac{1}{3}$	0
$-\frac{1}{4}$	1	$-\frac{1}{4}$	$-\frac{1}{4}$	0	$-\frac{1}{4}$
0	$-\frac{1}{2}$	1	0	0	$-\frac{1}{2}$
$-\frac{1}{4}$	$-\frac{1}{4}$	0	1	$-\frac{1}{4}$	$-\frac{1}{4}$
$-\frac{1}{3}$	0	0	$-\frac{1}{3}$	1	$-\frac{1}{3}$
0	$-\frac{1}{4}$	$-\frac{1}{4}$	$-\frac{1}{4}$	$-\frac{1}{4}$	1
1	0	0	0	0	0
0	0	0	1	0	0

Polygonal Mesh

- More on Laplacian for meshes

Least-squares Meshes

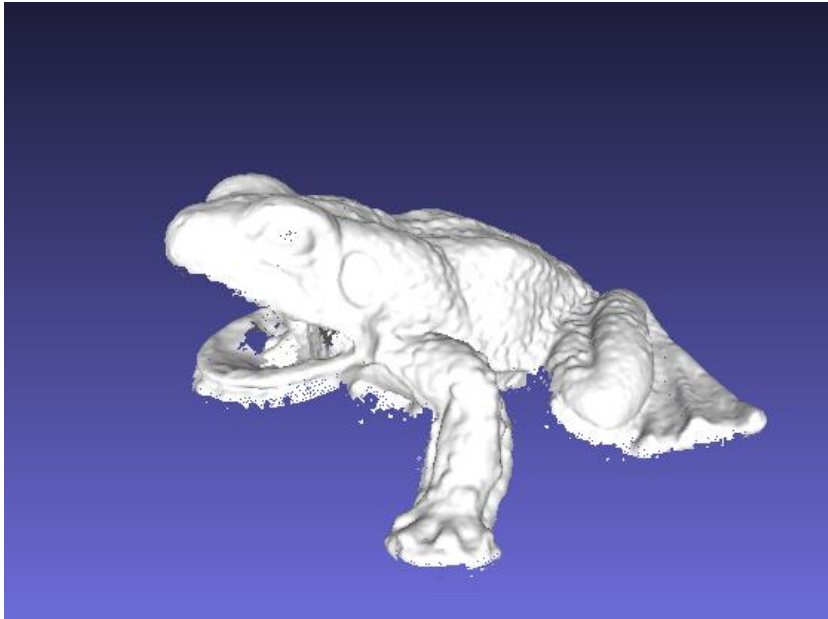
Olga Sorkine

Daniel Cohen-Or

SMI, 2004

Polygonal Mesh

- How to attach textures on a mesh?



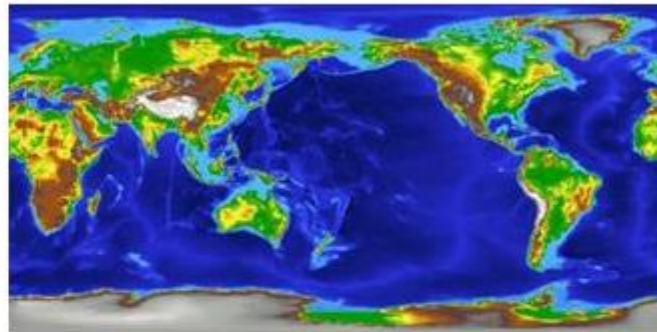
Polygonal Mesh

- How to attach textures on a mesh?
 - Texture is 2D, mesh is 3D (coordinates)



Object

+



Texture

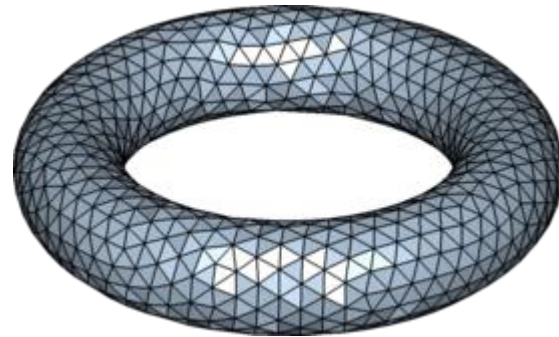
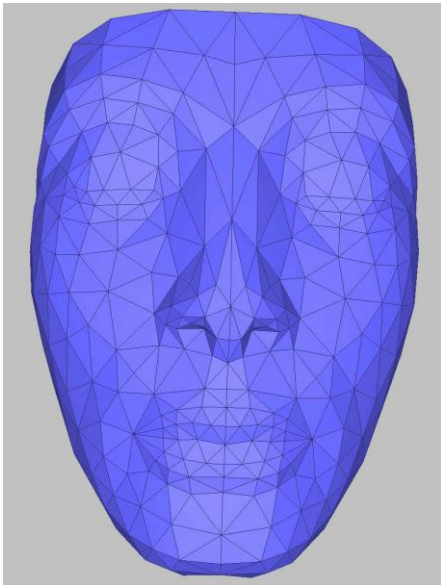
=



Texture
Mapped
Object

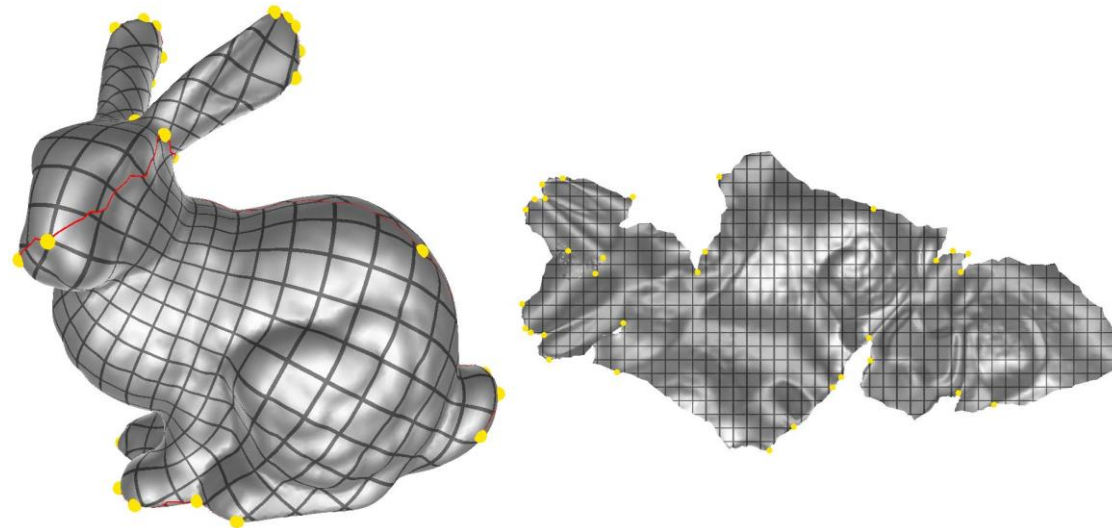
Polygonal Mesh

- How to attach textures on a mesh?
 - Texture is 2D, mesh is 3D (coordinates)
 - Texture is a disk, mesh can be disk, sphere, torus,... (topology)



Polygonal Mesh

- How to attach textures on a mesh?
 - Texture is 2D, mesh is 3D (coordinates)
 - Texture is a disk, mesh can be disk, sphere, torus,... (topology)
- We need to map all meshes to one intermediate domain with a disk topology (parameterization).



Polygonal Mesh

- One possible recipe:

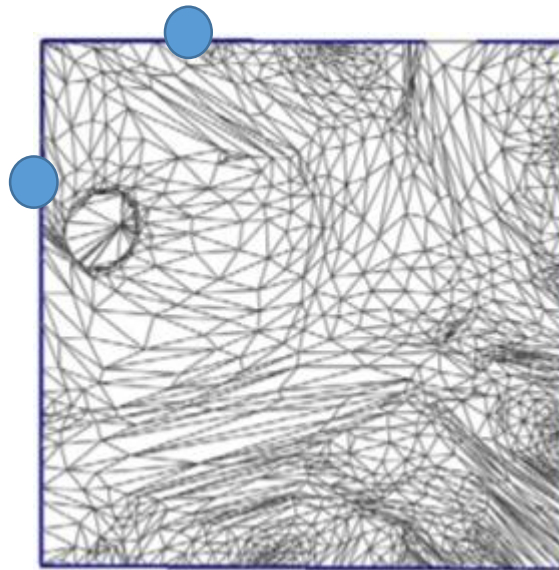
Polygonal Mesh

- One possible recipe:
 - Make a cut if needed (topology not disk)



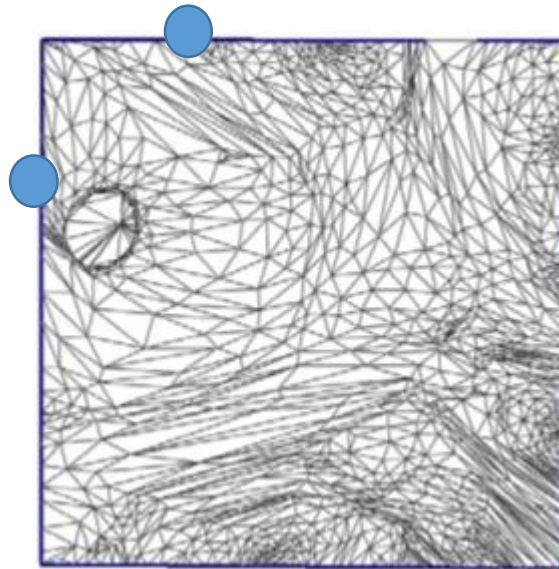
Polygonal Mesh

- One possible recipe:
 - Make a cut if needed (topology not disk)
 - Fix the position of the vertices on the edges on a disk. Note that when you open up the mesh, each vertex appears twice on the disk's boundary.



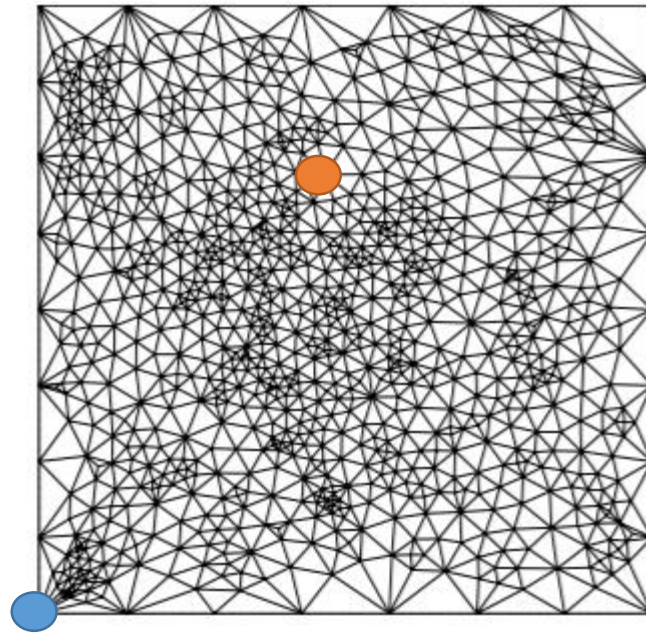
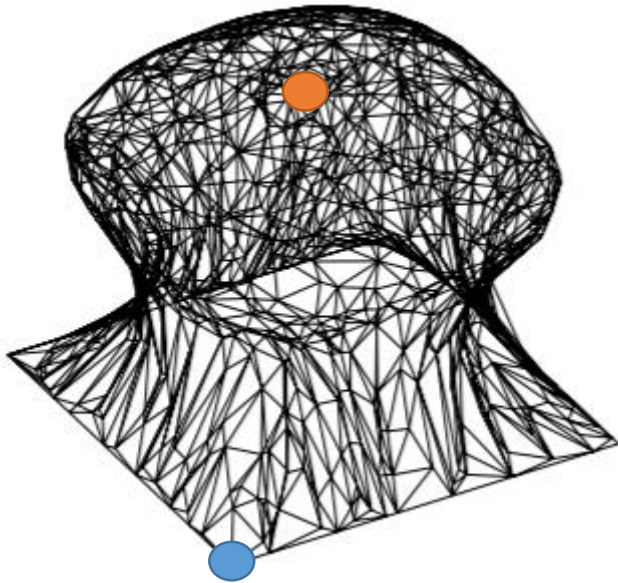
Polygonal Mesh

- One possible recipe:
 - Now fix the location of the vertices along the boundary



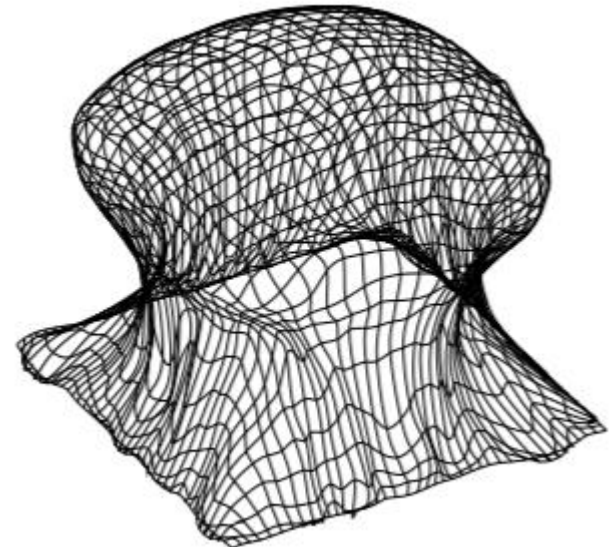
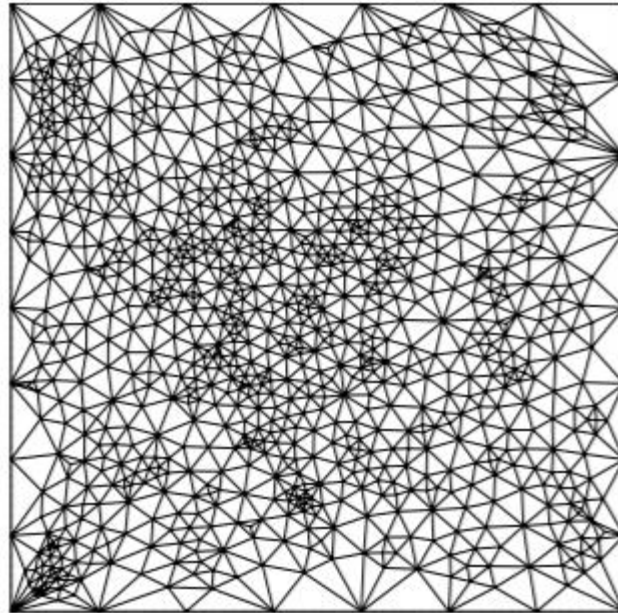
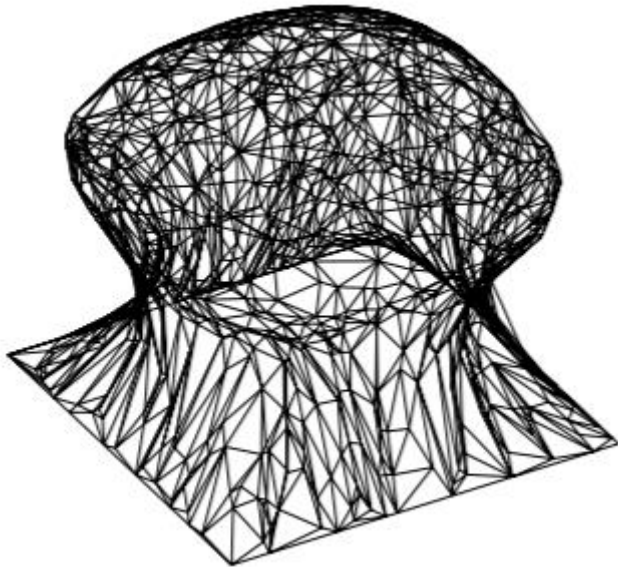
Polygonal Mesh

- One possible recipe:
 - Now fix the location of the vertices along the boundary and use Laplacian matrix to find the location of other vertices.



Polygonal Mesh

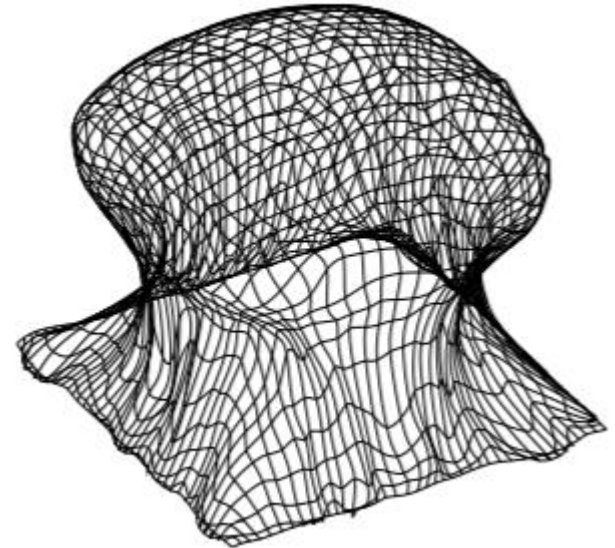
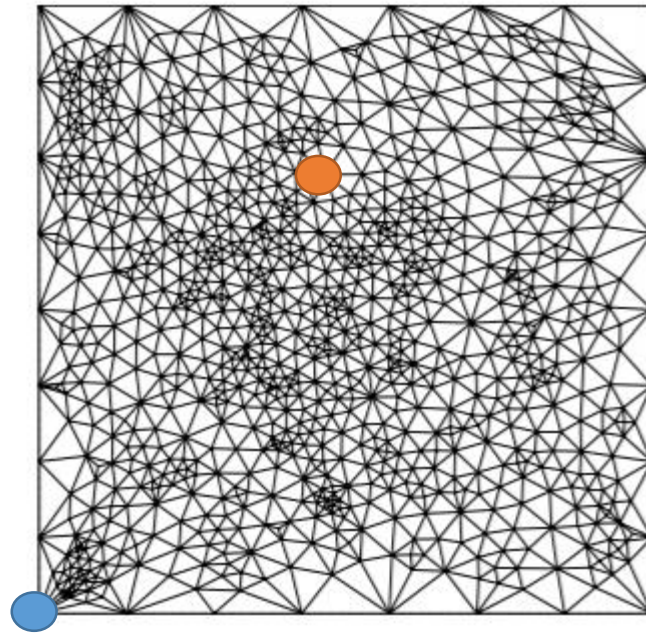
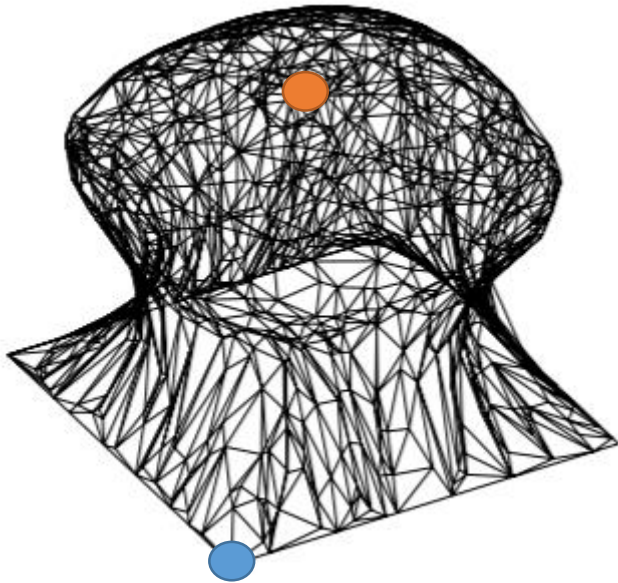
- Now you can either map the texture from the disk or sample the disk regularly and create a regular quad mesh.



Polygonal Mesh

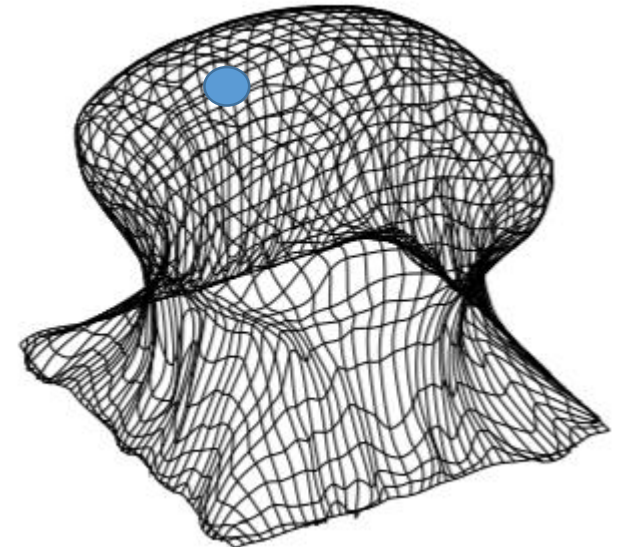
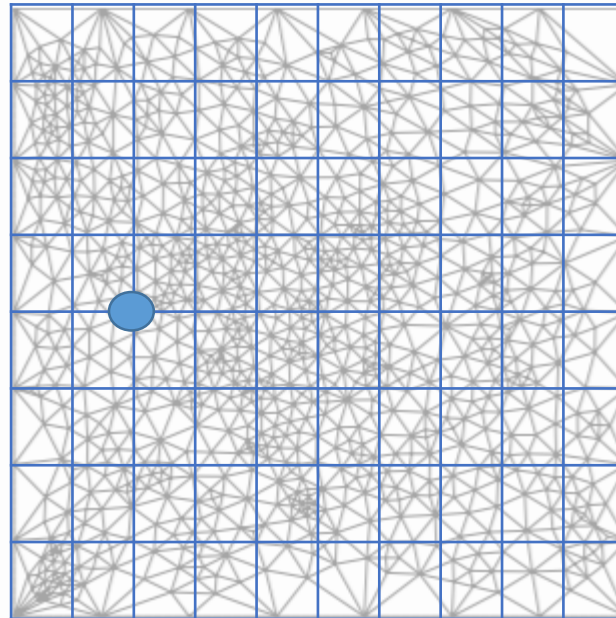
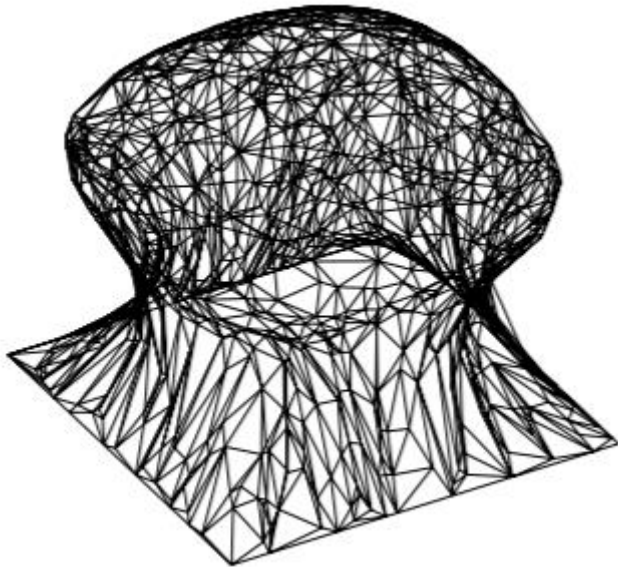
- Now you can either map the texture from the disk or sample the disk regularly and create a regular quad mesh.

How?



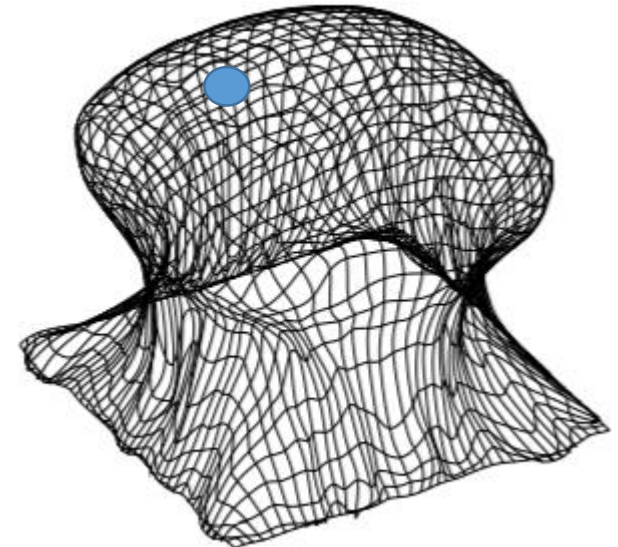
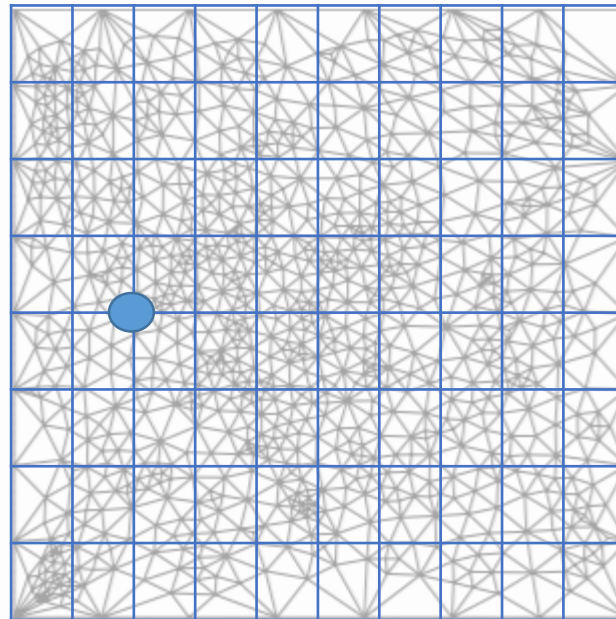
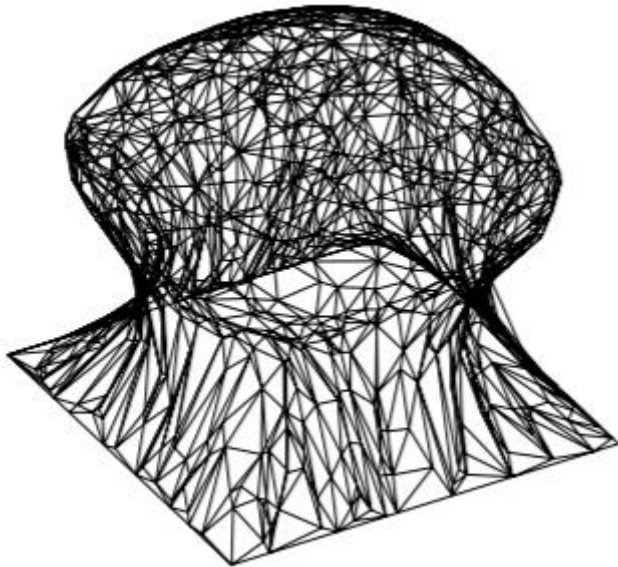
Polygonal Mesh

- Sample using a grid, find the location using barycentric coordinate.



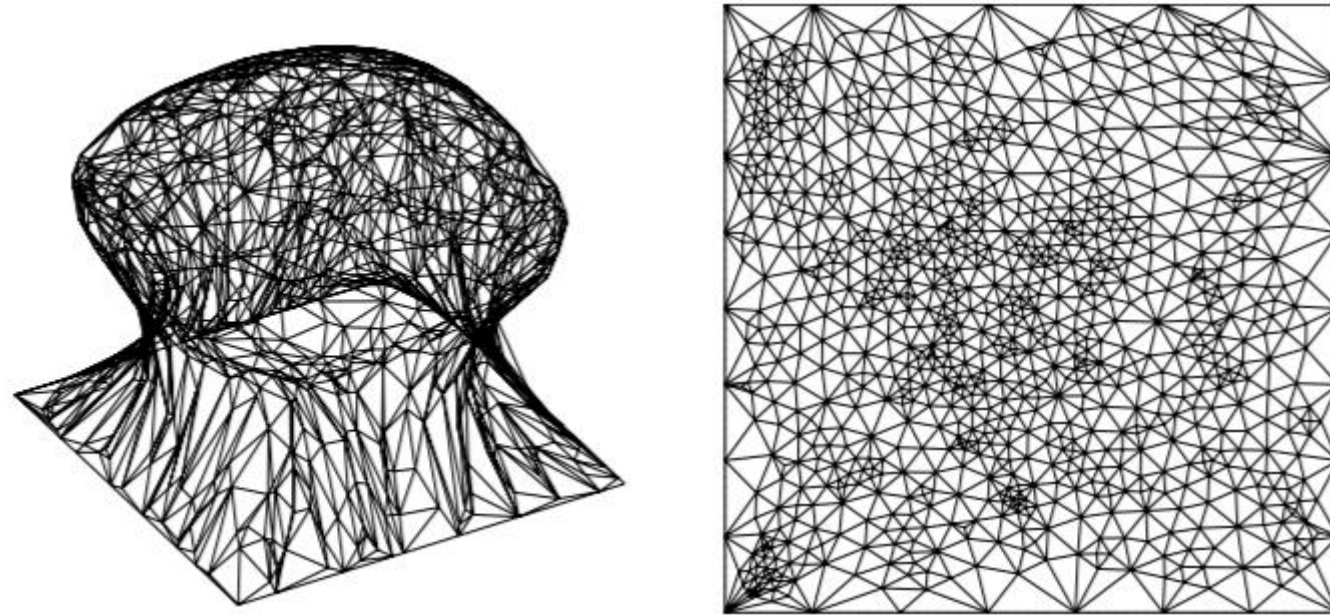
Polygonal Mesh

- Sample using a grid, find the location using barycentric coordinate.
- Sample points fall in one of the triangles in 2D domain and we can find the location of the sample on 3D mesh using the same barycentric coordinate.



Polygonal Mesh

- This approach is called Floater's parameterization

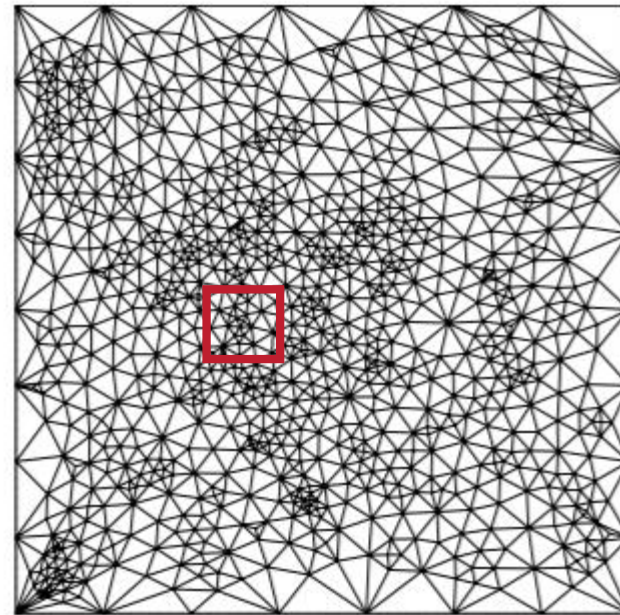
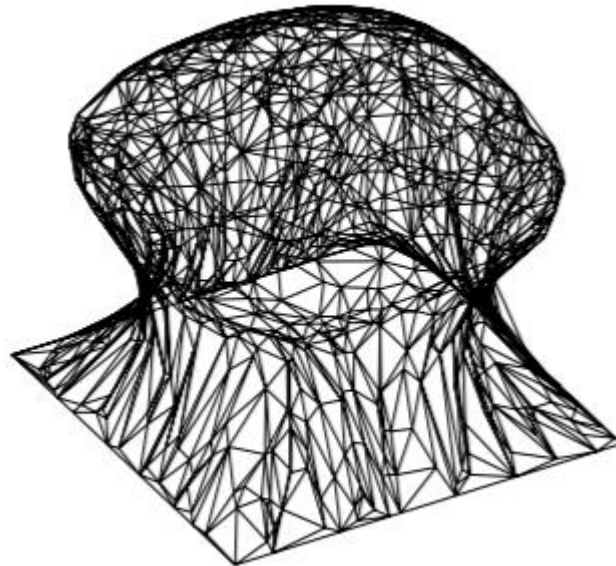


Parametrization and smooth approximation
of surface triangulations

Michael S. Floater

Polygonal Mesh

- The problem with this approach is that it makes a lot of distortions as large triangles may be mapped to small triangles



Polygonal Mesh

- Geometry Images

Geometry Images

Xianfeng Gu

Steven J. Gortler

Hugues Hoppe

TOG 2002

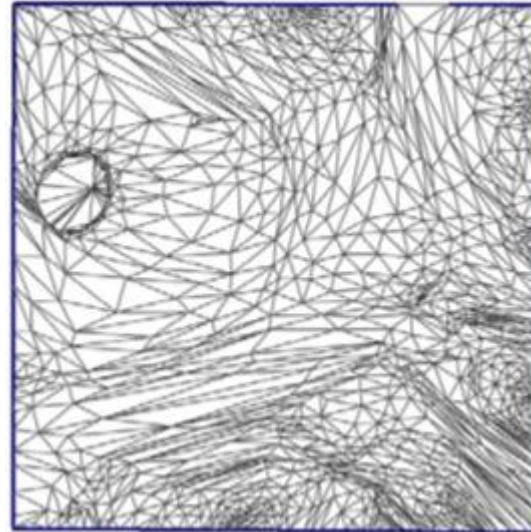
Polygonal Mesh

- Geometry Images
 - Find a cut



Polygonal Mesh

- Geometry Images
 - Find a cut
 - Project to a 2D domain using an initial parameterization (Floater)

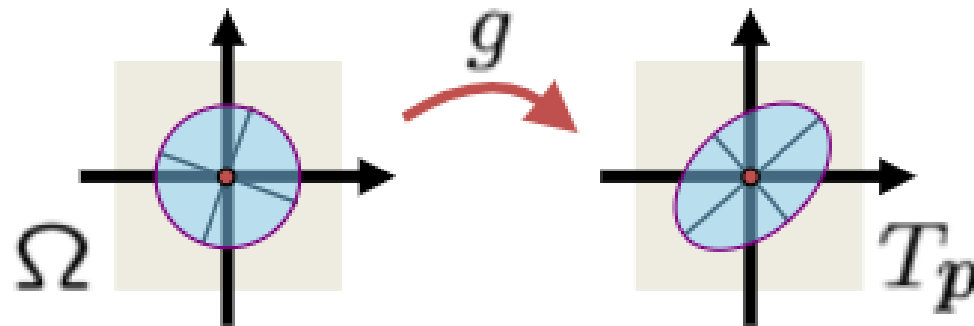


Polygonal Mesh

- Geometry Images

- Find a cut
- Project to a 2D domain using an initial parameterization (Floater)
- Minimize stretch of triangles defined by **singular values** of the local mapping

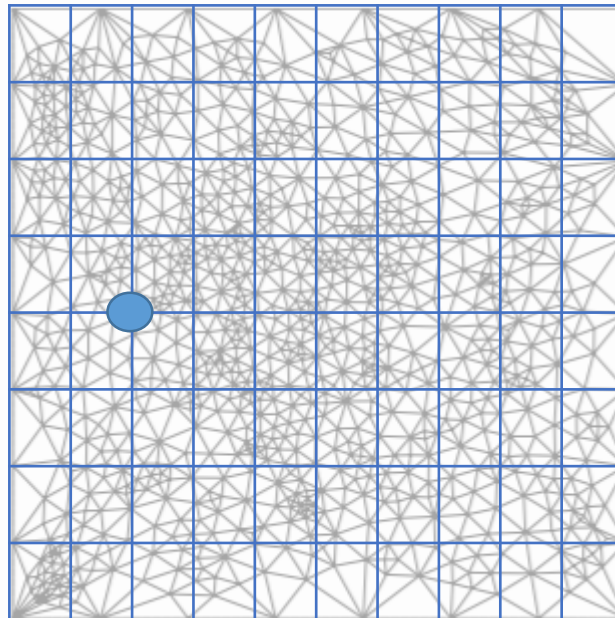
Note that it is impossible to map a mesh in spherical topology to a disk with no distortion. (circle to circle with the same area is impossible).



Polygonal Mesh

- Geometry Images

- Find a cut
- Project to a 2D domain using an initial parameterization (Floater)
- Minimize stretch of triangles defined by singular values of the local mapping
- Sample the 2D points



Polygonal Mesh

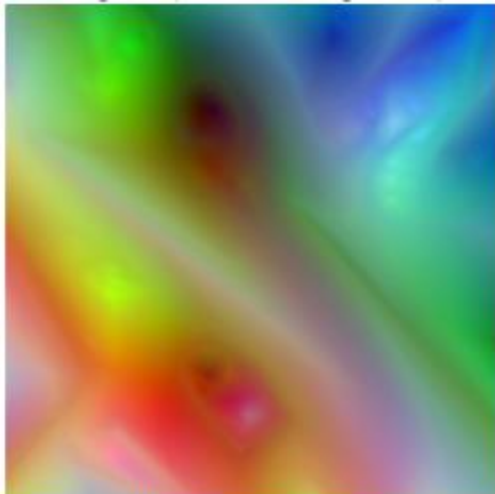
- Geometry Images
 - Find a cut
 - Project to a 2D domain using an initial parameterization (Floater)
 - Minimize stretch of triangles defined by singular values of the local mapping
 - Sample the 2D points
 - Use RGB for x,y,z and make an image



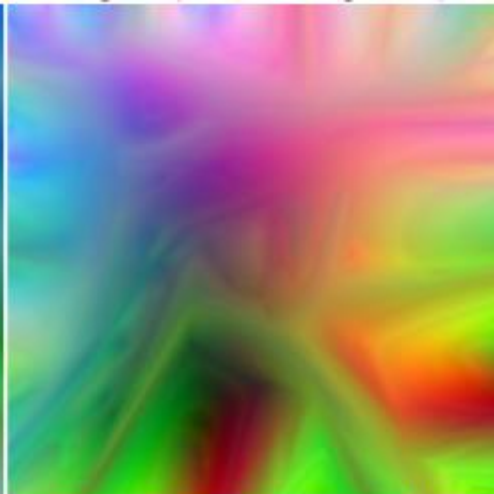
Geometry image (257x257)

Polygonal Mesh

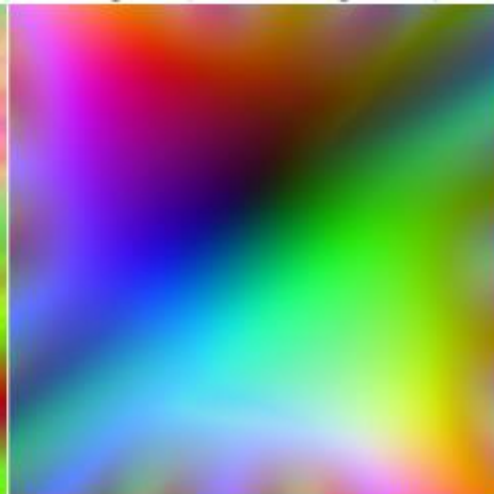
- What are these images?



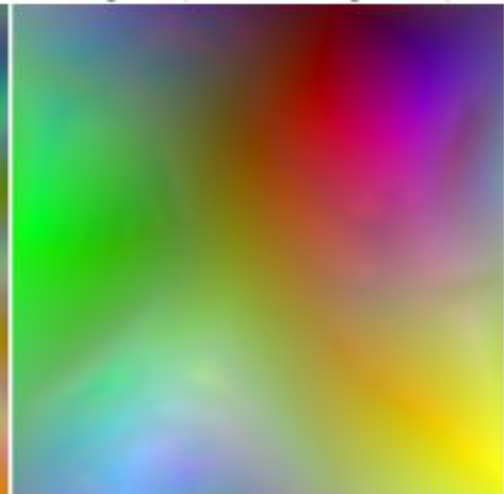
Geometry image (257x257)



Geometry image (257x257)



Geometry image (129x129)



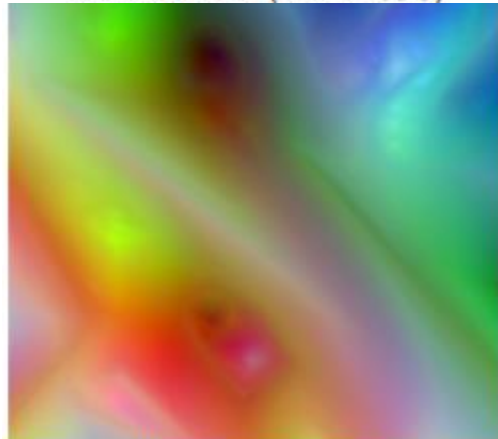
Geometry image (257x257)

Polygonal Mesh

- What are these images?



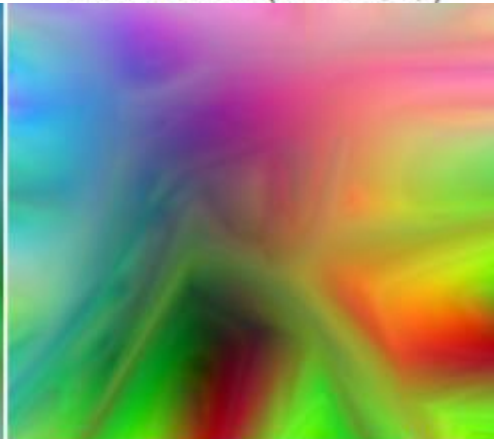
Reconstruction (PSNR=66.8)



Geometry image (257x257)



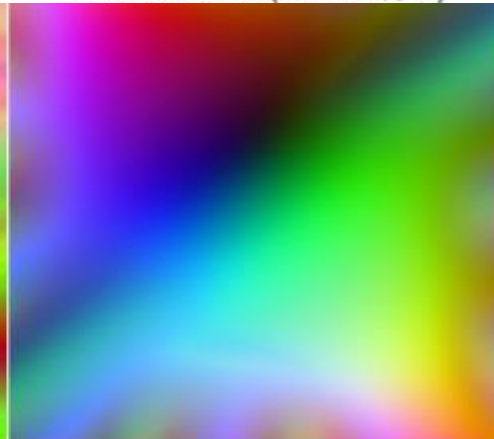
Reconstruction (PSNR=64.9)



Geometry image (257x257)



Reconstruction (PSNR=75.2)



Geometry image (129x129)



Reconstruction (PSNR=78.6)



Geometry image (257x257)

Polygonal Mesh

- What are the main benefits of geometry images?

